

Quantum Mechanics

Lectures 1–19 — Prof. V. Balakrishnan

Handwritten lecture notes, transcribed

Contents

1	Lecture 1: Quantum Mechanics (Introduction)	3
2	Lecture 2	5
3	Lecture 3	6
4	Lecture 4	11
5	Lecture 5	15
6	Lecture 6	18
7	Lecture 7 :-	22
8	Lecture 8	28
9	Lecture 10	40
10	Lecture 11	46
11	Lecture 12	60
12	Lecture 13	69
13	Lecture 14	76
14	Lecture 15	80
15	Lecture 17	89
16	Lecture 16	98
17	Lecture 18	103

1 Lecture 1: Quantum Mechanics (Introduction)

(in Q.M)

- * Math is simpler than classical physics.
- * Every thing at ultimate level turns out to obey Q.M.
- * Q.M applies to everything i.e. photons, electrons, humans, galaxy. It is just that its manifestations become dramatic when we observe small object, & become more dramatic if small objects are moves with very high speed.
- * The fact that two solid object does not penetrate one another is due to Q.M. principle (pauli exclusion principle).
- * magnetism, superconductor, electric conduction, propogation of sounds in solids, these are all quantum phenomenas.
- 'The phenomena of diamagnetism, paramagnetism depends on quantum mechanical principales, as they are not explainable from classical physics.'
- * Black body radiation is also explained from Q.M.
- * uncertainty principle.

The position and canonical conjugate momentum of any particle can not be measured simultaneously and accuratly (to arbitrary precision).

$$\Delta x \cdot \Delta p \gtrsim \hbar$$

uncertainty principle is intrinsic in nature :-

It is not lack of resolution / experimental lack, it is that the position and momentum are not defined to be measured to with arbitary accuracy (when measured simultaneously).

$$\Delta r \cdot \Delta p_r \geq \hbar/2$$

$$\Delta y \cdot \Delta p_y \geq \frac{\hbar}{2}$$

- * we can see that hamiltonian formalism is translated to Quantum mechanics.

The whole assumption is that we have

- Hamiltonian and
- Hamiltonian framework then we translate to Q.M.

Q.) It is very deep and harder question to ask how to quantize or how do we do Q.M. for a system which are not classical hamiltonian in classical limit.

e.g. friction in system.

(Q) *"Problem of dissipation in Q.M is still an open problem."*

classically, we know that

$$\{x, p_x\} = 1$$

Q.M

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2} \quad \text{where } \hbar = h/2\pi$$

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\text{variance}}$$

in Q.M $\delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$ is calculated w.r.t quantum mechanical probability distribution. and that distribution will be specified through the Shrödinger's equation.

For any two arbitrary canonically conjugate dynamical variable $\exists$ uncertainty principle.

$$(\Delta A) \cdot (\Delta B) \geq \frac{1}{2} |\langle [A, B] \rangle|$$

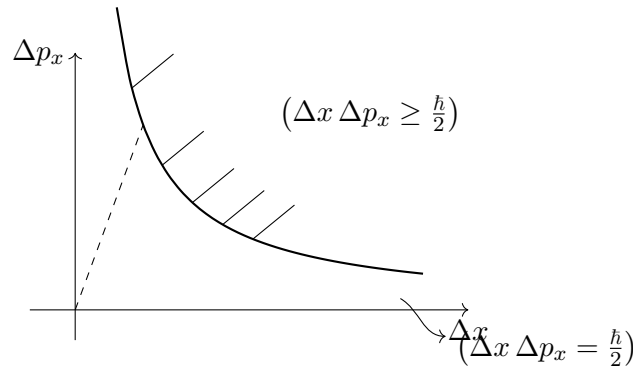
$$[A, B] = AB - BA$$

$\langle [A, B] \rangle$ = expectation value of $[A, B]$.

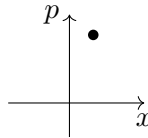
we know that

$$\{x, p_y\} = 0 \quad \text{— classical}$$

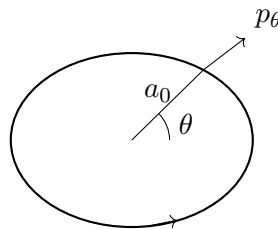
$$(\Delta x \cdot \Delta p_y \geq 0) \quad \text{— Quantum Physics}$$



Now, the idea of phase space point and trajectory is no more in Q.M.



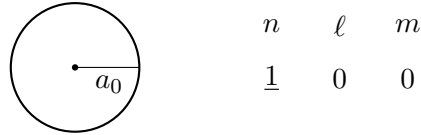
The idea of orbit :-



Here, θ, p_θ can be measured with arbitrary precision

so, we can not use orbits to describe motion of $\bar{e}$ in an atom.

Also, Orbital theory is not completely correct.



If any e^- is orbiting with radius a_0 , its angular momentum can not be zero. but it is told that

$$\frac{\sqrt{\ell(\ell+1)} h}{2\pi} = 0$$

All objects in Q.M are treated with state vector which is generalization of wave function.

ex.

- e^- is a state vector
- photon is a state vector.
- Even the molecular orbital theory is not correct (s, p, d, f) [n orbitals also] are not correct.

”we should not extrapolate this categorisation of objects into waves & particles to the microscopic domain”. It is conceivable that we have objects which have properties of both waves & particles, & it is also conceivable that $\exists$ some objects which has properties of either of them, depending upon how we measure them.”

2 Lecture 2

The state of a quantum system is not specified by a point in phase-space but rather it is specified by an element of linear vector space. i.e. vector ”state vector”. $(|\psi\rangle) / |\Psi(t)\rangle$.

→ This state vector is an element of LVS, and all the required information can be extracted from just state vector. and this state vector evolves in time.

→ we have to throw the idea of representing system as variation of many dynamical variables (independent), $x_1, x_2, x_3 - x_n$ along with idea of phase space.

Digression for LVS

Linear vector space :-

Contains set of elements (”vectors”). $|\psi\rangle, |\phi\rangle, |\chi\rangle \in V$ with following properties / definition.

- $|1\rangle + |2\rangle = |3\rangle \in V$; closure
- $|1\rangle + (|2\rangle + |3\rangle) = (|1\rangle + |2\rangle) + |3\rangle$; Addition is associative
- $|1\rangle + |2\rangle = |2\rangle + |1\rangle$; Addⁿ is commutative
- $a|\psi\rangle \in V \hookrightarrow$ fields can be scalar Real/complex ($a, b, c, \dots \in F(C \text{ or } R)$)
- $a(b|1\rangle) = (ab)|1\rangle \in V$
- $a(|1\rangle + |2\rangle) = a|1\rangle + a|2\rangle$
- $\exists$ Null vector such that $|1\rangle + |0\rangle = |1\rangle$

- $\exists$ an additive inverse such that $|1\rangle + |-1\rangle = |0\rangle$

$$\begin{aligned}\langle\phi|\psi\rangle^* &= \langle\psi|\phi\rangle \\ (\langle\phi|\psi\rangle^T)^\dagger &= \langle\psi|\phi\rangle^* \Rightarrow \langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*\end{aligned}$$

The Norm of a vector :-

$$\begin{aligned}\|\psi\| &\stackrel{\text{def}}{=} \langle\psi|\psi\rangle^{1/2} > 0; \quad = 0 \text{ iff } |\psi\rangle = |0\rangle \\ \|\psi + \chi\| &\leq \|\psi\| + \|\chi\| \quad (\text{triangle inequality}) \\ |\langle\phi|\psi\rangle|^2 &\leq \langle\phi|\phi\rangle \langle\psi|\psi\rangle \quad (\text{cauchy-Schwarz inequality}) \\ &\begin{cases} \bar{a} \cdot \bar{b} = |a||b| \cos \theta \\ \cos \theta \leq 1 \Rightarrow \bar{a} \cdot \bar{b} \leq |a||b| \end{cases}\end{aligned}$$

$\hookrightarrow$ equals only when

$$|\phi\rangle = a|\psi\rangle$$

overview — next $\hookrightarrow$

3 Lecture 3

Linear independence :-

if $|\phi_1\rangle, |\phi_2\rangle, \dots |\phi_n\rangle$ are linearly independent

then if

$$\sum_i a_i |\phi_i\rangle = 0$$

so solⁿ becomes $a_i = 0 = a_1 = a_2$

Span :-

A set of vectors $|\phi_1\rangle, |\phi_2\rangle, \dots |\phi_n\rangle$ are said to span the LVS, iff any vector can be written as linear combination of these vectors. ($|\phi_1\rangle, \dots, |\phi_n\rangle$).

example

(1) $\hat{e}_x$ & $\hat{e}_y \longrightarrow$ L.I, do not span $\mathbb{R}^3$

(2) $\hat{e}_x, \hat{e}_y, \hat{e}_z \rightarrow$ L.I, span $\mathbb{R}^3$ (orthogonal set) (unit vectors)

(3) $\hat{e}_x, \hat{e}_y, \hat{e}_z, \hat{e}_x + \hat{y} \longrightarrow$ span, not L.I

(4) $\hat{e}_x, \hat{e}_y + \hat{e}_x, \hat{e}_z + \hat{e}_y + \hat{e}_x \longrightarrow$ L.I, span $\mathbb{R}^3$ (oblique set of coordinate axis vectors) (not-unit vector)

Basis :-

A set of vectors which are linearly independent & span the LVS forms a basis set in the LVS.

$$\langle \phi_i | \phi_j \rangle = \delta_{ij}$$

For orthogonal basis $\delta_{ij} = 0$ if $i \neq j$

Dimententionality :-

The no. of basis vectors defines the dimententionality of LVS.

$\therefore \dim V = \#$ of vectors in basis set.

$\rightarrow$ Infinite dimententional LVS does not have finite basis

$\mathbb{R}^2$ is 2-d ; $\mathbb{R}^3$ is 3-D LVS, $\dots$ $\mathbb{R}^n$ is n -dimensional LVS.

Here is what can go wrong for infinite-dimentional LVS.

for $\mathbb{R}^n$:

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{pmatrix} = |\phi\rangle$$
$$\langle \phi | \phi \rangle = \sum_{i=1}^N |x_i|^2 = \|\phi\|^2$$

[we could have defined $\|\phi\|$ as $\left(\sum_{i=1}^N |x_i|^p\right)^{1/p}$, but for $p = 2$ only l_p is self dual. dual of $l_p = l_q$ such that $\frac{1}{p} + \frac{1}{q} = 1$ (for $p = q = 2$, $l_p = l_q$)

for $N \rightarrow \infty$ there is no gaurentee that $\langle \phi | \phi \rangle$ converges / have finite value.

so we have to make this piece (length of vector) to be finite.

$$\sum_{i=1}^N |x_i|^2 < \infty$$

$[|x_i|^2 \text{ should vanish before } n \rightarrow \infty \text{ (i.e. } n^{-1} \rightarrow 0) \Rightarrow |x_i| \text{ should vanish faster than } \frac{1}{\sqrt{n}}]$

$$|f\rangle = \begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix} = \sum_{i=1}^N f_i |i\rangle \Rightarrow \langle f | f \rangle < \infty \Rightarrow \sum_{i=1}^N |f_i|^2 < \infty$$

linear vector space of square summable sequences. , l_2

ex.,

(1)

$$(x_1, x_2, x_3 \dots x_\infty) = \left(1, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{4}} \dots\right)$$

$\hookrightarrow$ diverges. not an element of l_2 .

(2)

$$x_r = \frac{1}{r^\epsilon} \quad \text{for, } \epsilon > \frac{1}{2} \text{ it is in } l_2$$

$[|\phi\rangle = (x_1, x_2 \dots x_r)$ is in l_2 if $x_r = \frac{1}{r^\epsilon}$

$\langle\phi|\phi\rangle = (x_1^2 + x_2^2 + \dots x_r^2)$, $\sum_{r=1}^{\infty} (x_r)^2 \rightarrow \text{converges}$, $\sum_r \frac{1}{r^{2\epsilon}} \rightarrow \text{converges}$, $2\epsilon > 1 \Rightarrow (\epsilon > \frac{1}{2})$

(3)

$$x_n = \frac{\ln n}{n^{0.6}} \quad \text{in } l_2$$

$$|x_n|^2 = \frac{(\ln n)^2}{n^{1.2}} \in l_2$$

also $\frac{(\ln n)^{100}}{n^{0.6}} \in l_2$

(log is weaker than power & power is weaker than exponential)

Normalization :-

$$\frac{|\psi\rangle}{\sqrt{\langle\psi|\psi\rangle}} = \frac{|\psi\rangle}{\|\psi\|}$$

Gram – Schmidt orthonormalization :-

finding two orthonormal basis vectors.

let us say we have any $|\psi_1\rangle, |\psi_2\rangle$ two basis vectors

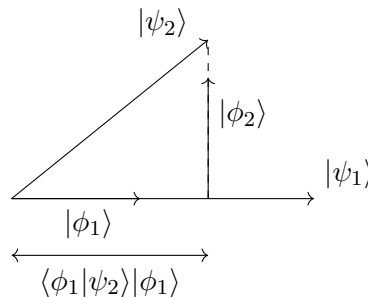
we can produce $|\phi_1\rangle$ & $|\phi_2\rangle$ such that $\langle\phi_i|\phi_j\rangle = \delta_{ij}$.

$|\phi_1\rangle$ = one of orthonormal vector would be simply $\frac{|\psi_1\rangle}{\|\psi_1\|}$

$$|\phi_2\rangle = \frac{|\psi_2\rangle - \langle\phi_1|\psi_2\rangle|\phi_1\rangle}{\|\xi\|} = |\phi_2\rangle$$

This will guarantee that

$$\langle\phi_1|\phi_2\rangle = \boxed{0}$$



$\hookrightarrow$ or

$$\left\{ |\phi_1\rangle \langle\phi_1| \right\} |\psi\rangle = \langle\phi_1|\psi\rangle |\phi_1\rangle$$

$$(n \times 1)(1 \times n) \longrightarrow (n \times n)$$

(projection operator)

let us say

$$|\psi\rangle = v_1|\phi_1\rangle + v_2|\phi_2\rangle + v_3|\phi_3\rangle$$

magnitude projection along $|\phi_1\rangle \hookrightarrow (\langle\phi_1|\psi\rangle = v_1)$.

projection of $|\psi\rangle$ along $|\phi_1\rangle =$

$$\left(|\phi_1\rangle\langle\phi_1|\right)|\psi\rangle = |\phi_1\rangle v_1 = v_1|\phi_1\rangle$$

$\hookrightarrow$ projection operator. (of $(n \times n)$ matrix identification)

[In LVS in addition to linear vectors we have linear operators & these operators acts upon these vectors to produce another vector.]

$$\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k} \quad \boxed{P = \hat{i}\hat{i} \ (\hat{j}\hat{j}) \ \hat{k}\hat{k}}$$

$$(\hat{i}\hat{i})\vec{v} = \vec{v} \cdot \hat{i} \hat{i} = v_1\hat{i} \quad \text{or} \quad \hat{i}v_1$$

$\hookrightarrow$ dyadic in tensor algebra / or tensor product

(projection operator)

$$P_n = |\phi_n\rangle\langle\phi_n|$$

$$P_n^2 = |\phi_n\rangle\langle\phi_n|\phi_n\rangle\langle\phi_n|$$

(① unity iff $\|\phi_n\| = 1$)

$$= |\phi_n\rangle\langle\phi_n|$$

$$\left(\sum_n |\phi_n\rangle\langle\phi_n|\right)|\psi\rangle = 1 \cdot |\psi\rangle$$

so,

$$\sum_n |\phi_n\rangle\langle\phi_n| = \text{unit operator.}$$

we know that,

$$P_n^2 = P_n \quad (\text{Idempote matrix}) \text{ i.e. for matrix } M; \ M^\alpha = M \text{ for any } \alpha \in \mathbb{Z}$$

$$P_n^2 - P_n = 0$$

$$P_n(P_n - \mathbb{I}) = 0 \quad \text{eigen value} = (0, 1)$$

$\hookrightarrow$ cauchy relation $\lambda(\lambda - 1) = 0$ will be satisfied by matrix itself.

These are two very important properties of LVS.

1) orthonormality

$$\langle\phi_i|\phi_j\rangle = \delta_{ij}$$

2) completeness.

$$\sum_{n=1}^n |\phi_n\rangle\langle\phi_n| = \mathbb{I}$$

ex.

$$|\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |\phi_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$[|\phi_1\rangle \text{ \& } |\phi_2\rangle]$ also provides basis for operators]

$$\langle\phi_i|\phi_j\rangle = \delta_{ij}$$

$$|\phi_1\rangle\langle\phi_1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$|\phi_2\rangle\langle\phi_2| = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\sum_{i=1}^2 |\phi_i\rangle\langle\phi_i| = |\phi_1\rangle\langle\phi_1| + |\phi_2\rangle\langle\phi_2| = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$|\phi_1\rangle\langle\phi_2| = \text{operator} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$|\phi_2\rangle\langle\phi_1| = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

we know that,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{cases} a = a_{11} \\ b = a_{12} \\ c = a_{21} \\ d = a_{22} \end{cases}$$

$$= a_{11}|\phi_1\rangle\langle\phi_1| + a_{12}|\phi_1\rangle\langle\phi_2| + a_{21}|\phi_2\rangle\langle\phi_1| + a_{22}|\phi_2\rangle\langle\phi_2|$$

$$a_{11} = \langle\phi_1|A|\phi_1\rangle$$

$$a_{12} = \langle\phi_1|A|\phi_2\rangle$$

$$a_{21} = \langle\phi_2|A|\phi_1\rangle$$

$$a_{22} = \langle\phi_2|A|\phi_2\rangle$$

$$A_{nm} = \langle\phi_n|A|\phi_m\rangle$$

Any vector $|\psi\rangle$ can be uniqlly expanded in form

$$|\psi\rangle = \sum_n c_n |\phi_n\rangle$$

Any operator A can be expanded in form ; where $\{|\phi_n\rangle\langle\phi_m|\}$ forms **basis**.

$$A = \sum_{n,m} A_{nm} |\phi_n\rangle\langle\phi_m|$$

4 Lecture 4

l_2 : space of square summable sequence

$$\sum_{i=1}^{\infty} |x_i|^2 < \infty$$

$$\left(l_2(a, b) \right) = \int_a^b |f(x)|^2 dx \quad (< \infty)$$

necessary condⁿ for $f(x)$ to be l_2 is that

$$\left\{ \lim_{x \rightarrow \pm \infty} f(x) \rightarrow 0 \right\}$$

why not $(x \rightarrow b, a)$?

$\hookrightarrow$ it should die to zero sufficiently rapidly

$\lim_{|x| \rightarrow \infty} f(x) \rightarrow 0$ should go faster than $\frac{1}{\sqrt{|x|}}$ (?) **How.**

$\rightarrow$ we will require wave function to be bounded.

Let's look at space of square integrable functions. $\rightarrow l_2(-1, 1)$

where,

$L_1 \rightarrow$ space of integrable functions.

$l_2 \rightarrow$ space of square integrable fun. (self dual)

$$l_2(-1, 1) = \int_{-1}^1 dx |f(x)|^2 < \infty$$

[let us find basis set for any function. so that we can expand it in the given basis]

let $|\psi_0\rangle, |\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle \dots \rightarrow x^0, x^1, x^2, \dots$ forms basis. (orthonormal)

$$f_0(x) = c_0$$

whats

$$\int_{-1}^1 dx c_0^2 = 1 \quad \Rightarrow \quad c_0 = \frac{1}{\sqrt{2}}$$
$$f_1(x) = ax + b \quad (\text{in general}).$$

find a, b such that

$$\begin{cases} \langle \phi_0 | \phi_0 \rangle = 1 \\ \langle \phi_1 | \phi_1 \rangle = 1 \\ \langle \phi_0 | \phi_1 \rangle = 0 \end{cases}$$
$$\Rightarrow \int_{-1}^1 dx f_1(x) f_0(x) = 0$$
$$\Rightarrow \int_{-1}^1 (ax + b) \frac{1}{\sqrt{2}} dx = 0$$
$$\Rightarrow \left[\frac{ax^2}{2} + bx \right]_{-1}^1 = 0$$

$$\left(\frac{a}{2} + b\right) - \left(\frac{a}{2} - b\right) = 0$$

$$2b = 0 \quad \Rightarrow \quad \boxed{b = 0}$$

So

$$f_1(x) = ax$$

so normalizing

$$\int_{-1}^1 f_1(x)^2 dx = 1$$

$$\int a^2 x^2 dx = 1$$

$$\frac{a^2}{3} (x^3) \Big|_{-1}^1 = 1$$

$$\frac{a^2}{3} (1 + 1) = 1$$

$$\boxed{a = \left(\frac{3}{2}\right)^{1/2}}$$

$$f_2(x) = ax^2 + bx + c$$

$$\int_{-1}^1 dx f_2(x) f_0(x) = 0$$

$$\int dx f_2(x) f_1(x) = 0$$

$$\int dx (f_2(x))^2 = 1$$

three condⁿ are sufficient to find a, b, c .

Let us Redefine Normalization condition to get rid of weird constants.

lets call these functions as $P_n(x)$ (Legendre polynomials)

we have,

$$\int_{-1}^1 dx P_n(x) P_m(x) = \frac{2}{2n+1} \delta_{nm} = \frac{2}{2m+1} \delta_{mn} = \frac{2 \delta_{mn}}{\sqrt{(2n+1)(2m+1)}}$$

if

$$P_0(x) = 1 \quad P_1(x) = x, \quad P_2(x) = \frac{3x^2 - 1}{2}$$

Redefining normalization condition for legendre polynomials.

Even the fourier transform (series) is really writing a function in some basis set. Any periodic fun. can be written as (expanded in fourier series). where ' $\cos nx$ ' and ' $\sin nx$ ' are basis set for state vector $f(x)$.

$$f(x + \beta) = f(x)$$

$$f(x) = \sum_{n=0}^{\infty} (a_n \cos nx + b_n \sin nx) \quad (n = 0, 1, 2, 3 \dots)$$

a_n, b_n are coefficients of expansion

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}$$

[Generalized normalized condition

$$\int_{-\infty}^{\infty} d\mu(x) \phi_n^*(x) \phi_m(x) = A_n \delta_{nm} \quad (A_n \rightarrow \text{weight factor})$$

$d\mu(x) \rightarrow$ measure

ex.

$$\int_{-\infty}^{\infty} dx e^{-x^2} \phi_n^*(x) \phi_m(x) = A_n \delta_{nm}$$

$\hookrightarrow \phi_n(x) =$ Hermite polynomials

$d\mu(x) \rightarrow$ measure is used to make sure that $\int_{-\infty}^{\infty} d\mu(x) \phi_n^*(x) \phi_n(x) < \infty$ remains finite.

for $d\mu(x) = dx e^{-x}$, $\phi(x)$ forms Laguerre polynomial.]

If $f(x)$ is not periodic, we can expand $f(x)$ in fourier integral

$$f(x) = \int_{-\infty}^{\infty} dk e^{ikx} \tilde{f}(k)$$

for orthonormality

$$\langle \phi_n | \phi_m \rangle = 0, \quad \Longleftrightarrow \quad \int_a^b f_n^*(x) f_m(x) dx = \delta_{mn}$$

for completeness

$$\begin{aligned} \sum_{n=1}^{\infty} |\phi_n\rangle \langle \phi_n| &= 1 \\ \sum_n f_n^*(x) f_n(x') &= \delta(x - x') \\ \int dx' \delta(x - x') \psi(x') &= \psi(x) \end{aligned}$$

$\rightarrow$ unit operator

[so in this way we can define our own polynomials. let, $d\mu(x) = dx \cdot x^2$, $\phi_n(x)$ forms Vishal's polynomial. $\smile$]

If $\{|\phi_n\rangle\}$ is an orthonormal basis in any LVS, any ket $|\psi\rangle \in V$ can be uniquely expanded as

$$|\psi\rangle = \sum_{n=1}^N c_n |\phi_n\rangle \quad (n \rightarrow \infty) \text{ for LVS of infinite dimensions.}$$

where,

$$c_m = \langle \phi_m | \psi \rangle$$

If we have different basis say $|\chi_n\rangle$

$$|\psi\rangle = \sum_{n=1}^N d_n |\chi_n\rangle = \sum_i d_i |\chi_i\rangle \quad \dots \textcircled{1}$$

where

$$d_j = \langle \chi_j | \psi \rangle \quad \dots \textcircled{2}$$

To find relation between c_m and d_i

we can expand $|\chi_i\rangle$ in $|\phi_n\rangle$ basis

$$\begin{aligned} |\chi_i\rangle &= \sum_m h_{mi} |\phi_m\rangle \\ \Rightarrow h_{mi} &= \langle \phi_m | \chi_i \rangle \end{aligned}$$

So, eqⁿ ① becomes

$$|\psi\rangle = \sum_i^N d_i |\chi_i\rangle = \sum_i \sum_m d_i h_{mi} |\phi_m\rangle$$

or

$$\sum_n \left(\sum_i h_{ni} d_i \right) |\phi_n\rangle$$

So

$$\begin{aligned} \boxed{c_n = \sum_i h_{ni} d_i} \\ c_n = \langle \phi_n | \psi \rangle = \sum_i \langle \phi_n | \chi_i \rangle \langle \chi_i | \psi \rangle \\ = \langle \phi_n | \left(\sum_i |\chi_i\rangle \langle \chi_i| \right) | \psi \rangle \end{aligned}$$

($\rightarrow$ unit operator)

$$= \langle \phi_n | \psi \rangle$$

So the whole point is any time we take new basis we automatically insert the identity (unit operator).

$$\begin{aligned} \langle \psi | &= \sum_{n=1}^N c_n^* \langle \phi_n |, \quad |\psi\rangle = \sum_{n=1}^N c_n |\phi_n\rangle \\ \langle \psi | \psi \rangle &= \sum_{n=1}^N c_n^* c_n \langle \phi_n | \phi_n \rangle = \sum_n |c_n|^2 < \infty \end{aligned}$$

as, $\langle \phi_n | \phi_n \rangle = 1$

we can expand any function $f(x)$ on basis set formed by polynomials (Legendre, Laguerre polynomials, hermite polynomials and fourier series (if $f(x)$ is periodic). If $f(x)$ is not periodic then we can expand it in continuous basis $\{e^{ikx}\}$ labelled by ' k '.

$$f(x) = \int_{-\infty}^{\infty} dk e^{ikx} \tilde{f}(k)$$

$\tilde{f}(k)$ = fourier transform of $f(x)$.

[$e^{ikx} \sim$ unit vectors / basis set. $\tilde{f}(k)$ are like components in e^{ikx} direction / basis.]

for $l_2(-1, 1)$

$$f(\theta) = \sum_{\ell=0}^{\infty} c_{\ell} P_{\ell}(\cos \theta)$$

$$c_{\ell} = \frac{1}{2} \int_{-1}^1 d(\cos \theta) P_{\ell}^*(\cos \theta) f(\theta)$$

$\{P_{\ell}(\cos \theta) \text{ is real so, } P_{\ell}^*(\cos \theta) = P_{\ell}(\cos \theta)$

→ comes from orthogonality relation

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} dk e^{ik(x-x')} = \delta(x-x')$$

Fourier inversion formula

$$f(x) = \int_{-\infty}^{\infty} dk \tilde{f}(k) e^{ikx}, \quad \tilde{f}(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dx e^{-ikx} f(x)$$

overview — Quantum Physics by Prof. Balakrishnan (YouTube playlist) — [ii](#) [prev](#) — [next](#) [iii](#)

5 Lecture 5

$l_2, L_2^{(a,b)}$ both are infinite dimensional LVS.

Idea of cauchy sequence :-

let us assume we have set of complex no.

$\{z_n\}$, then for convergence of sequence.

$$|z_n - z_m| \longrightarrow 0 \quad \text{as, } n, m \rightarrow \infty$$

$\{z_1, z_2, z_3 \cdots z_n \cdots\}$ formed according to some rule.]

This implies

also ($\Rightarrow \lim_{n \rightarrow \infty} z_n$ exists) $\lim_{n \rightarrow \infty} z_n = \text{some complex no.}$

If we include the limit point of sequence $z_n (n \rightarrow \infty)$, then $\{z_1, z_2 \cdots z_n\}$ is called as closed set. (or closed sequence).

$\Rightarrow \{|\psi_n\rangle\}$ is a cauchy sequence if $(\lim_{n,m \rightarrow \infty} \|\psi_n - \psi_m\| \longrightarrow 0)$

$\Rightarrow$

$$\lim_{n \rightarrow \infty} |\psi_n\rangle = |\psi\rangle$$

limiting vector

if $|\psi\rangle \in V$ then it is called **complete vector space**

i.e. If the limit vector of every cauchy sequence in a LVS (V) is also in V , then V is a complete LVS.

A complete LVS with an inner product is a Hilbert space.

* The reason we need such sequence is to make sure that operations with infinite dimensional vectors do not lead to errors.

* A Hilbert space with a denumerable basis $\{|\phi_n\rangle\}$ ($n = 1, 2, 3 \dots \infty$) [$\rightarrow$ (able to count)] is called a *separable Hilbert space*.

[$\hookrightarrow$ we will be working in separable hilbert space in whole **Q.M**]

$$\Rightarrow |\psi\rangle = \sum_{n=1}^{\infty} c_n |\phi_n\rangle$$

$$c_n = \langle \phi_n | \psi \rangle \quad \text{if } \{|\phi_n\rangle\} \text{ forms orthonormal basis.}$$

(assuming basis are orthonormal)

$$\langle \phi_n | \phi_m \rangle = \delta_{mn}$$

$$\langle \psi | \psi \rangle = \sum_{n=1}^{\infty} |c_n|^2 = \|\psi\|^2 < \infty \quad (\text{finite}) \in l_2.$$

$$\begin{cases} \sum_{n=1}^{\infty} |\phi_n\rangle \langle \phi_n| = \mathbb{I} \\ \langle \phi_n | \phi_m \rangle = \delta_{mn} \end{cases}$$

Linear operators

operators acts on $|\psi\rangle$, where linear operator acts linearly

$$\begin{cases} \hat{A}(|\psi\rangle + |\chi\rangle) = \hat{A}|\psi\rangle + \hat{A}|\chi\rangle \\ \hat{A}(c|\psi\rangle) = c\hat{A}|\psi\rangle \\ (\hat{A} + \hat{B})|\psi\rangle = \hat{A}|\psi\rangle + \hat{B}|\psi\rangle \\ \hat{A}(\hat{B}|\psi\rangle) = (\hat{A}\hat{B})|\psi\rangle \end{cases} \quad (\hat{A}, \hat{B} \text{ are linear operators})$$

in general $\hat{A}\hat{B} \neq \hat{B}\hat{A}$ operators do not commute.

Postulates

(1) $\rightarrow$ Every dynamical system is described by a "state vector" $|\psi\rangle$ which is element of a Hilbert space.

where system can be

– 1 particle

– 2 particle

– 10 particle

– Human being $\dots$ galaxy

$\rightarrow$ every one of such system is described by their "own" state vector.

$\rightarrow$ Every information is burried in this vector $|\psi\rangle$.

(2) $\rightarrow |\psi\rangle$ is fun. of time generally $|\psi(t)\rangle$.

As system evolves $|\psi(t)\rangle$ changes with time and rule of evolution describes the change

(3) $\rightarrow$ Physical measurables are associated with "self adjoint" operators. i.e. $(\hat{L}, \hat{p}, x, \vec{v}, \dots$ every observable).

For every physical observable we have corresponding operator. not necessary independent with every; for ex. FOR k.E only velocity / momentum operator are will work.

To associate values with operator, we refer to eigen values of operators.

for operator

$$A|\psi\rangle = \lambda|\psi\rangle$$

then ' λ ' is eigen value of A with eigen vector $|\psi\rangle$.

For a given eigen value, if we have two or more linearly independent eigen vectors, then eigen values are repeated eigen values / degenerate eigen value.

It is also possible that no. of eigen values are infinite if LVS is infinite dimensional.

For $n \times n$ matrix it can have max of n - eigen values.

(4) Eigen values are in some sence "the only measurable values" for an operator cooresponding to physical observable should be real.

$\rightarrow$ If $A = A^\dagger$ (Hermitian conjugate) (All eigen values are real for operator A .)

i.e. if operator is self adjoint it's guaranteed that its all eigen-values are real.

Note :- self adjoint are not equally same as Hermitian conjugate except in case of matrices.

Note : we can have contineous set of eigen values in case of infinite dimensional LVS.

For $\underline{x}$, position of a particle can take any value for a definite momentum.

(4) The result of the measurement for any physical measurable observable is always *an* real eigen value for the corresponding operator.

{keep in mind every eigen value will be real for physical observable operators.}

Note :- We can not say which eigen value will appear on measurment because of quantum uncertainty laws.

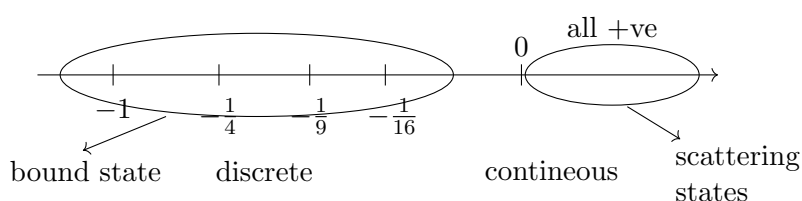
On the other hand we can repeatedly make measurments, take the average we call that mean value of the specific observable. or expectation value i.e. $\langle \underline{x} \rangle$. and we have rule in Q.M to calculate expectation value. It also gives rule to calculate scatter about mean.

This can be achieved by making infinite identical copies of system and make measurements on each of them and take mean. of $\langle x \rangle$.

Average may or may not be an eigen state of that operator.

* spectrum of an operator is the set of all eigen values of the operator. These eigen values can be contineous / discrete eigen values.

ex. the energy of e^- in H like atom goes like



For Any operator A , which acts upon $|\psi\rangle$ state of system

let $\lambda_1, \lambda_2 \dots$ be the eigen values of A with eigenvectors $|\phi_1\rangle, |\phi_2\rangle, \dots$

assuming $|\phi_i\rangle$ are orthonormal basis $\langle\phi_i|\phi_j\rangle = \delta_{ij}$

$$|\psi(t)\rangle = \sum_{n=1}^{\infty} c_n |\phi_n\rangle = \sum_{n=1}^{\infty} c_n(t) |\phi_n\rangle$$

$\hookrightarrow$ assuming operators are time independent (explicitly).

$$c_n(t) = \langle\phi_n|\psi\rangle$$

$\hookrightarrow$ probability amplitude saying the actual state is $|\phi_n\rangle$.

so c_n tells us how much of $|\psi\rangle$ points along $|\phi_n\rangle$.

$\{|c_n(t)|^2$ is the probability that the system is in the state $|\phi_n\rangle$ at time t .

if $\langle\psi|\psi\rangle = 1$

$$\Rightarrow \sum_n |c_n|^2 = 1$$

6 Lecture 6

$$\begin{aligned} \langle A \rangle &= \frac{\sum_n \lambda_n |c_n|^2}{\langle\psi|\psi\rangle} \bigg/ \sum_n |c_n|^2 \\ A|\phi_n\rangle &= \lambda_n |\phi_n\rangle \\ \langle\psi(t)|\psi(t)\rangle &= \sum_n |c_n|^2 = 1 \\ &= \frac{\sum_n \lambda_n \langle\psi|\phi_n\rangle \langle\phi_n|\psi\rangle}{\langle\psi|\psi\rangle} \\ &= \frac{\langle\psi|\sum_{n=1}^{\infty} \lambda_n |\phi_n\rangle \langle\phi_n| \psi\rangle}{\langle\psi|\psi\rangle} \\ &= \left\langle \psi \left| \sum_{n=1}^{\infty} A |\phi_n\rangle \langle\phi_n| \right| \psi \right\rangle \bigg/ \|\psi\|^2 \\ &= \frac{\langle\psi|A \sum_{n=1}^{\infty} |\phi_n\rangle \langle\phi_n| \psi\rangle}{\langle\psi|\psi\rangle} \\ &\quad \sum_{n=1}^{\infty} |\phi_n\rangle \langle\phi_n| = 1 \end{aligned}$$

$$\boxed{\langle A \rangle = \frac{\langle\psi|A|\psi\rangle}{\langle\psi|\psi\rangle}} \quad \text{or} \quad \boxed{\langle A \rangle = \frac{\langle\Psi(t)|A|\Psi(t)\rangle}{\langle\Psi|\Psi\rangle}}$$

[$\langle A \rangle$ is diagonal matrix element of $\hat{A}$ in basis formed from $\{|\Psi(t)\rangle\}$.]

for a normalized $|\psi\rangle$, $\langle\psi|\psi\rangle = 1$

So,

$$\boxed{\langle A \rangle(t) = \langle\psi|A|\psi\rangle}$$

where $|\psi\rangle$ is time dependent $|\Psi(t)\rangle$.

classical Physics / Quantum Physics

classical Physics

- Dyn. variables in **phase space** : $q_1, q_2 \dots ; p_1, p_2 \dots$
- PB relations

$$\begin{aligned}\{q_i, p_j\} &= \delta_{ij} \\ 0 &= \{q_i, q_j\} = \{p_i, p_j\}\end{aligned}$$

$H(q, p)$

$$\left. \begin{aligned}\dot{q}_i &= \frac{\partial H}{\partial p_i} \\ \dot{p}_i &= \frac{-\partial H}{\partial q_i}\end{aligned} \right\} \text{dynamics of variables}$$

such that,

$$\begin{aligned}\{q_i(t), p_j(t)\} &= \delta_{ij} \\ \{q_i(t), q_j(t)\} &= 0 \\ \{p_i(t), p_j(t)\} &= 0\end{aligned}$$

Quantum Physics

- Dynamical variables are replaced by self adjoint operators (acting upon the vectors in the **hilbert space**).
- Phase space $\xrightarrow{\text{replaced}}$ Hilbert space.
- PB's are replaced by commutator operators.

$$\begin{aligned}[A, B] &= AB - BA \\ \frac{[q_i, p_j]}{i\hbar} &= \frac{q_i p_j - p_j q_i}{i\hbar} = \delta_{ij} \\ [q_i, q_j] &= 0 = [p_i, p_j]\end{aligned}$$

Lec 6 (contd.) – Why we need it in the relation ?

Lec 6 > [Why we need $i\hbar$ in $\frac{[q_i, p_j]}{i\hbar} = \delta_{ij}$ relation ?]

◇ Commutator becomes anti-hermitian.

$$\begin{aligned}A &= A^\dagger \\ B &= B^\dagger \\ [A, B]^\dagger &= (AB - BA)^\dagger = B^\dagger A^\dagger - A^\dagger B^\dagger = -[A, B] = [B, A]\end{aligned}$$

So

$$[A, B]^\dagger = -[A, B]$$

since $AB - BA$ is also a physical measurable, so we have to make sure its eigen values are real, which suggests we have to make it hermitian.

$$\frac{[A, B]}{i\hbar} \text{ has real eigen values.}$$

' $\hbar$ ' is for diamentional reasons. such that $\frac{[q_i, p_j]}{i\hbar}$ becomes diamensionless.

Now we know that

$$\langle A \rangle(t) = \frac{\langle \psi(t) | A | \psi(t) \rangle}{\langle \psi(t) | \psi(t) \rangle}$$

The Shrodinger Equation :-

The,

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$$

{operator eqⁿ for vectors in Hilbert space

where H is an operator.

(postulate :-) *statement* :-

The time rate of change of state vector $|\psi(t)\rangle$ is given by action of Hamiltonian operator on the state vector.

Note :-

$|\psi(t)\rangle$ can be uniquely determined if we have initial state vector $|\psi(0)\rangle$.

Since ' H ' is time independent, it appears that Shrodinger equation is autonomous in nature.

$|\psi(t)\rangle$ can be realized by a coloumn vector where as H can be realized by a squarre matrix in Hilbert space.

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$$

$$\frac{d}{dt} |\psi(t)\rangle = -\frac{iH}{\hbar} |\psi(t)\rangle$$

$$|\psi(t)\rangle = \psi(0) e^{-\frac{iHt}{\hbar}} |\psi(0)\rangle$$

$$|\psi(t)\rangle = e^{-\frac{iHt}{\hbar}} |\psi(0)\rangle$$

$e^{-\frac{iHt}{\hbar}}$ is operator acts on $|\psi(0)\rangle$

Suppose (t_0) is starting / initial time.

solution becomes

$$\int_{t_0}^t \frac{d(|\psi(t)\rangle)}{|\psi(t)\rangle} = \int_{t_0}^t \frac{-iH}{\hbar} dt$$

$$|\psi(t)\rangle = e^{-\frac{iH(t-t_0)}{\hbar}} |\psi(t_0)\rangle$$

$$\langle \psi(t) | = \langle \psi(t_0) | e^{\frac{iH^\dagger(t-t_0)}{\hbar}}$$

($\rightarrow$ operator)

as eigen values of H to be real, H is Hermitian i.e. $\underline{H = H^\dagger}$

$$\langle \psi(t) | = \langle \psi(t_0) | e^{\frac{iH(t-t_0)}{\hbar}}$$

so,

$$\begin{aligned} \langle \psi(t) | \psi(t) \rangle &= \langle \psi(t_0) | e^{\frac{iH(t-t_0)}{\hbar}} \cdot e^{-\frac{iH(t-t_0)}{\hbar}} | \psi(t_0) \rangle \\ &= ? \\ &\begin{cases} e^A \cdot e^B \neq e^{A+B} & \text{for } [A, B] \neq 0 \\ e^A \cdot e^{-A} = e^0 = 1 & \text{operator} \end{cases} \end{aligned}$$

$\hookrightarrow$ true always. as $\{A, -A\} = 0$

$$= \langle \psi(t_0) | \psi(t_0) \rangle$$

$$\boxed{\langle \psi(t) | \psi(t) \rangle = \langle \psi(t_0) | \psi(t_0) \rangle}$$

conservation of norm of state vector.

statement :

The analogue of conservation of phase space volume element is reflected in conservation of norm of state vector $|\psi(t)\rangle$. It is because of fact that the Hermitian conjugate of operator $e^{\frac{iH(t-t_0)}{\hbar}}$ is its inverse.

lets call

$$e^{-\frac{iH(t-t_0)}{\hbar}} = U(t, t_0) = \text{evolution operator}$$

or the time development operator. (unitary). or propogator.

Such operator $U(t, t_0)$ has interesting properties.

$$U^\dagger = U^{-1}(t, t_0)$$

(or)

$$UU^\dagger = U^\dagger U = I \quad \text{--- unitary matrix}$$

If system has t_1 time as initial time between t_0 and t_2

$$\begin{array}{c} | \quad | \quad | \\ \hline t_0 \quad t_1 \quad t_2 \end{array}$$

$$U(t_2, t_0) = U(t_1, t_0) U(t_2, t_1)$$

semi - group property.

Q) If find $f(t)$.

$$\begin{aligned} \frac{d}{dt} f(t) &= H(t) f(t) \\ \Rightarrow \frac{df(t)}{f(t)} &= H(t) dt \end{aligned}$$

$$\Rightarrow \boxed{f(t) = e^{\int_0^t H(t') dt'} f(0)}$$

→ Not the formal solⁿ as

$$e^{H(t_1)+H(t_2)} \neq e^{H(t_1)} \cdot e^{H(t_2)}$$

$$(H(t_1), H(t_2)) \neq 0$$

overview — ii prev — next ii

7 Lecture 7 :-

operators in Q.M are represented as $\hat{A}$.

* A particle moving in one dimension with no potential.

1) Analog of H in Q.M

$$\hat{H} = \frac{\hat{p}^2}{2m}$$

Shrodinger eqⁿ becomes.

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle = \frac{\hat{p}^2}{2m} |\Psi(t)\rangle$$

$$|\Psi(t)\rangle = e^{-\frac{i\hat{H}t}{\hbar}} |\Psi(0)\rangle$$

$$[\hat{x}, \hat{p}] = i\hbar$$

Just as we take coordinate system / axis for (x, p) , In Q.M. we must choose some basis. (in terms of which every other vector is expressed).

* Most basic physical measurable quantity is $\hat{x}, \hat{p}$ everything else is derived.

For convenience we should choose the basis set formed from eigen vector of operator (which is associated with each some physical measurable of system).

lets use some basis called as position basis.

$\{|x\rangle\}$ such that

$$\hat{x}|x_0\rangle = x_0|x_0\rangle$$

$\hat{x}$ = operator

$|x_0\rangle$ = eigen state ($|x_0\rangle$ is ket corresponding to a particle with precise location x_0).

x_0 = eigen value

x_0 can take any no. from $-\infty$ to $+\infty$

$$x_0 \in (-\infty, \infty).$$

So it is a continuous basis.

lets choose basis orthonormal to each other.

$$\langle x|x'\rangle = \delta(x - x') = \begin{cases} 1 & x = x' \\ 0 & x \neq x' \end{cases}$$

The completeness relation looks like

$$\int dx |x\rangle\langle x| = \mathbb{I} \quad \text{completeness.}$$

Any state vector can be written as

$$|\Psi(t)\rangle = \sum_{n=0}^{\infty} c_n |x_n\rangle \quad c_n = \langle x_n | \Psi(t) \rangle$$

but basis are continuous, so we should replace it with integral. (as $|x\rangle$ is continuous basis).

$$|\Psi(t)\rangle = \int dx \langle x_0 | \Psi(t) \rangle |x_0\rangle$$

[we could hv written as, $|\Psi(t)\rangle = \mathbb{I} |\Psi(t)\rangle = \int dx |x\rangle\langle x| \Psi(t)\rangle$

$$|\Psi(t)\rangle = \int dx \langle x | \Psi(t) \rangle |x\rangle]$$

coefficients are now labeled by continuous variable 'x'.

$$\langle x_n | \Psi(t) \rangle = c_n(t) = \Psi(x, t)$$

= probability amplitude that position is 'x' at time t.

$|\psi(x, t)|^2$ = probability density that position lies b/w x, x + dx at time t.

lets drop subscript 'n' as x is continuous.

$$\langle x | \Psi(t) \rangle = \Psi(x, t) = \text{wave function}$$

since it is in position basis, $\Psi(x, t)$ is called as position wave function.

lets choose momentum basis / momentum space set $\{|p_0\rangle\}$

$$\hat{p}|p_0\rangle = p_0|p_0\rangle$$

($\hat{p} \rightarrow$ operator, $p_0 \rightarrow$ eigen value, $|p_0\rangle =$ eigen vector.)

$$\langle p | p' \rangle = \delta(p - p') \quad \text{orthogonal } \{|p_0\rangle\} \text{ basis set}$$

$$\sum |p\rangle\langle p| \Rightarrow \int dp |p\rangle\langle p| = \mathbb{I} \quad \text{completeness.}$$

so state vector can be written as.

$$|\Psi(t)\rangle = \int dp \langle p | \Psi(t) \rangle |p\rangle$$

$$\langle p | \Psi(t) \rangle = \tilde{\Psi}(p, t)$$

($\rightarrow$ to distinguish from $\Psi(x, t)$)

$\hookrightarrow$ wave function in momentum basis. or momentum space wave function

we can write

$$\tilde{\Psi}(p, t) = \langle p | \Psi(t) \rangle$$

$$= \langle p | \left(\int dx |x\rangle \langle x| \right) | \Psi(t) \rangle \quad (\rightarrow \text{unit operator})$$

$$\tilde{\Psi}(p, t) = \int dx \langle p|x \rangle \langle x|\Psi(t) \rangle$$

$$\tilde{\Psi}(p, t) = \int dx \langle p|x \rangle \Psi(x, t)$$

where $\langle p|x \rangle$ is the probability amplitude that when the position of particle is 'x' its momentum is 'p'.

where, $\langle p|x \rangle$ will turn out to be $e^{\frac{ipx}{\hbar}}$

$$\psi(x, t) = \int dp \langle x|p \rangle \tilde{\Psi}(p, t)$$

Q) which Basis is more convenient out of $|x\rangle$ & $|p\rangle$?

Ans It is easier to work in position basis. because of following reason.

we know that,

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

if we use position basis

$$\begin{aligned} \hat{H}|x_0\rangle &= \frac{\hat{p}^2|x_0\rangle}{2m} + V(\hat{x})|x_0\rangle \\ &= \frac{\hat{p}^2|x_0\rangle}{2m} + V(x_0)|x_0\rangle \end{aligned}$$

($V(\hat{x}) \rightarrow$ operator ; $V(x_0) \rightarrow$ eigen value)

once we know how $\hat{p}$ acts on $\{|x_0\rangle\}$ we can easily compute $\hat{H}|x_0\rangle$, but in case of momentum basis $\{|p\rangle\}$.

$$\begin{aligned} \hat{H}|p_0\rangle &= \frac{\hat{p}^2}{2m}|p_0\rangle + V(\hat{x})|p_0\rangle \\ &= \frac{p_0^2}{2m} + V(\hat{x})|p_0\rangle \end{aligned}$$

$V(\hat{x})$ can be complicated function of x , thus will complicate the solution.

for $\hat{H} = \frac{p^2}{2m} + \frac{1}{2}k\hat{x}^2$ simple - Harmonic oscillator.

we can use $\{|x\rangle\}$ and $\{|p\rangle\}$ with same priority.

Q) what is $\hat{p}|x_0\rangle$?

Simultaneous Eigen states :-

Suppose we have two operators A, B such that

$$\begin{aligned} A^\dagger &= A \\ B^\dagger &= B \end{aligned} \quad \text{i.e. Hermitian operators.}$$

Q.) can I find simultaneous eigen states of A and B . can I find eigen states / eigen functions which are common to both of these operators.

Ans) In general there is no reason we can find such eigen states. But we can expect some eigen states can be common.

If $[A, A^\dagger] = 0$ i.e. if A commutes with its hermitian

such a matrix can always be diagonalised by similarity transformation.

for $A^\dagger = A$ (In all physical operator $A^\dagger = A$)

$$\Rightarrow [A, A^\dagger] = 0$$

we can simultaneously diagonalise A, B if

$$\begin{aligned} [A, B] &= 0 ; \quad AB - BA = 0 \\ \left\{ \begin{array}{l} \text{Such that } SAS^\dagger = \text{diagonal } M \\ SBS^\dagger = \text{Diagonal } N \end{array} \right. \end{aligned}$$

Here "simultaneously" means we can diagonalise with the same similarity transformation (S).

→ If $[A, B] \neq 0$

we can not find simultaneous eigen states.

i.e. we can't find complete common set of eigen states. But they may share one or more common eigen-states.

Note :- This does not mean that every eigen-state of A is also the eigen-state of B . if $[A, B] = 0$.

ex. take fun. on a line.

$$\mathbb{K} f(x) = f(x)$$

Parity operator

$$\begin{aligned} \mathbb{P} f(x) &= f(-x) \\ \mathbb{P}^2 f(x) &= P(P(f(x))) = P(f(-x)) = f(x) \\ P^2 &= \mathbb{K} \end{aligned}$$

we can see that

$$[P, \mathbb{K}] = P \cdot 1 - 1 \cdot P = 0$$

also

$$[P^2, P] = [\mathbb{K}, P] = 0$$

we know that every function is an eigen-state of unit operator ' $\mathbb{K}$ ', but every fun. is not eigen-state for parity operator.

$$\begin{aligned} \mathbb{P} \phi(x) &= \begin{cases} \phi(-x) \\ = \alpha \phi(x) \end{cases} & P^2 \phi(x) &= \alpha^2 \phi(x) = \phi(x) \\ \alpha &= \pm 1 \end{aligned}$$

$\alpha = 1$ even function

odd fun, even functions are eigen states of parity operator

$$Pf_e(x) = f(-x) = f(x) \quad \alpha = 1$$

$$Pf_o(x) = f(-x) = -f(x) \quad \alpha = -1 \quad \text{odd function.}$$

So set of eigen states of P^2 is much bigger than set of eigen functions of parity operator. / eigen state set of $\mathbb{P}$ is subset of spectrum of P^2 or $\mathbb{K}$.

since $[x, p] \neq 0$ they do not share common complete eigen value set. (But they can share few eigen values. even if $[A, B] \neq 0$).

The uncertainty in an observable :-

we know that

$$\begin{aligned} \langle \hat{A} \rangle &= \langle \Psi | \hat{A} | \Psi \rangle \quad \text{assuming } \langle \Psi | \Psi \rangle = 1 \\ \langle \hat{A}^2 \rangle &= \langle \Psi | \hat{A}^2 | \Psi \rangle \\ (\Delta \hat{A})^2 &= \left\langle \left(\hat{A} - \langle \hat{A} \rangle \right)^2 \right\rangle \\ (\Delta \hat{A})^2 &= \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2 \end{aligned}$$

for position operator $\hat{x}$

$$\begin{aligned} \langle \hat{x} \rangle &= \langle \Psi | \hat{x} | \Psi \rangle \\ |\Psi\rangle &= \int dx \psi(x) |x\rangle \\ \langle \Psi| &= \int dx' \langle x' | \psi^*(x') \\ \langle \hat{x} \rangle &= \int dx \int dx' \psi^*(x') \psi(x) \langle x' | \hat{x} | x \rangle \end{aligned}$$

($\rightarrow$ element of A in position basis.)

$$= \int dx \int dx' \psi^*(x') \psi(x) A(x', x)$$

Note :- we know that

$$\psi(x, t) = \int dp \langle x | p \rangle \tilde{\Psi}(p, t)$$

find $\langle x | p \rangle$?

we know that,

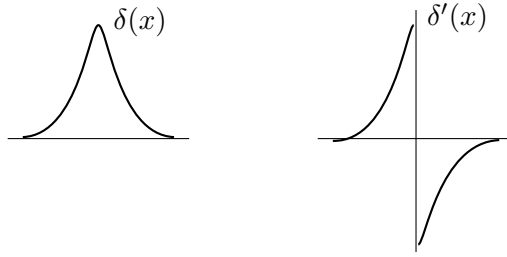
$$\hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar \mathbb{K}$$

let us find matrix element of $[\hat{x}, \hat{p}]$ in position basis.

$$\begin{aligned} \langle x | \hat{x}\hat{p} - \hat{p}\hat{x} | x' \rangle &= i\hbar \langle x | x' \rangle = i\hbar \delta(x - x') \\ x^* \langle x | \hat{p} | x' \rangle - x' \langle x | \hat{p} | x' \rangle &= i\hbar \delta(x - x') \end{aligned}$$

(x is real, so $x^* = x$; $\downarrow$ Position has to be real)

$$\begin{aligned} \langle x | \hat{p} | x' \rangle (x - x') &= i\hbar \delta(x - x') \\ \Rightarrow \langle x | \hat{p} | x' \rangle &= i\hbar \frac{\delta(x - x')}{(x - x')} \end{aligned}$$



defined :-

$$\left(\delta'(x) = \frac{-\delta(x)}{x} \right) \quad (\text{How?})$$

so

$$\delta'(x - x') = \frac{-\delta(x - x')}{(x - x')}$$

$$\Rightarrow \langle x | \hat{p} | x' \rangle = -i\hbar \frac{\partial}{\partial x} (\delta(x - x')) = -i\hbar \frac{\partial}{\partial x} (\langle x | x' \rangle)$$

for any arbitrary state $\Psi(t)$

$$\langle x | \hat{p} | \Psi(t) \rangle = -i\hbar \frac{\partial}{\partial x} \langle x | \Psi(t) \rangle$$

So it $\Rightarrow$

$$\left\{ \left(\hat{p} \longrightarrow -i\hbar \frac{\partial}{\partial x} \right) \text{ IN THE POSITION BASIS} \right.$$

$$\hat{x} | x \rangle = x | x \rangle$$

but $\hat{p} | x \rangle$ is totally different $\longrightarrow -i\hbar \frac{\partial}{\partial x} | x \rangle$

So,

$$\langle x | \hat{p} | x' \rangle = -i\hbar \frac{\partial}{\partial x} \langle x | x' \rangle$$

likewise,

$$\langle x | \hat{x} | x' \rangle = x \langle x | x' \rangle = x' \langle x | x' \rangle$$

$$\left\{ \begin{array}{ll} \langle x' | x | x \rangle = 1 & \text{if } x = x' \\ = 0 & \text{if } x \neq x' \end{array} \right. \quad \langle x | x' \rangle = \delta(x - x')$$

IN Momentum basis :-

$$\hat{p} | p_0 \rangle = p_0 | p_0 \rangle$$

what is $\hat{x} | p_0 \rangle = ?$

$$\left\{ \hat{x} = +i\hbar \frac{\partial}{\partial p} \text{ IN momentum Basis} \right.$$

In continuous basis rather than kronecker delta we use $\delta(x - x')$.

$$\int f(x') \delta(x - x') dx' = f(x)$$

$$I = \int \delta(x - x') dx' \quad ?$$

8 Lecture 8

To Find the inner product $\langle x|p\rangle$?

($\langle x| \rightarrow$ position eigen state ; $|p\rangle \rightarrow$ momentum eigen state)

given $[\hat{x}, \hat{p}] = i\hbar$

what is overlap b/w position eigen states & momentum eigen states. since $[x, p] \neq 0$ we can't find complete common eigen - states. values.

we know that

$$\langle \hat{x}|\hat{p}|\Psi(t)\rangle = -i\hbar \frac{\partial}{\partial x} \langle x|\Psi(t)\rangle$$

($\hat{x} \rightarrow$ not a operator)

instead of $\Psi(t)$, we replace it with any momentum eigen -state.

$$\langle \hat{x}|\hat{p}|p\rangle = p \langle \hat{x}|p\rangle$$

($\rightarrow$ eigen value) as $\hat{p}|p\rangle = p|p\rangle \hookrightarrow$ abstract eigen state (regardless of any basis)

also

$$\langle x|\hat{p}|p\rangle = -i\hbar \frac{\partial}{\partial x} \langle x|p\rangle = p \langle x|p\rangle$$

$$\langle x|p\rangle = \text{some fun. of } x \text{ labelled by } 'p' = f_p(x)$$

$$-i\hbar \frac{\partial}{\partial x} f_p(x) = p f_p(x)$$

$$\int \frac{df(x)}{f(x)} = \int \frac{ip}{\hbar} dx$$

$$f_p(x) \propto e^{\frac{ipx}{\hbar}}$$

($p \rightarrow$ eigen value corresponding to $\langle x|$)

$$\boxed{\langle x|p\rangle \propto e^{\frac{ipx}{\hbar}}}$$

($\rightarrow$ eigen value corresponding to $|p\rangle$)

x, p are both no. (eigen values).

$$[\langle x|p\rangle \propto e^{\frac{ip \cdot x}{\hbar}}, \langle p|x\rangle \propto e^{-\frac{ip}{\hbar}x}]$$

here p & x are eigen values of $|p\rangle$ & $|x\rangle$ respectively.]

$$\Rightarrow \langle p|x\rangle \sim e^{-\frac{ipx}{\hbar}} \quad (\text{complex conjugate of } \langle x|p\rangle)$$

we know that

$$\langle x|\Psi(t)\rangle = \Psi(x, t) = \int dp \langle x|p\rangle \langle p|\Psi(t)\rangle = \int dp e^{\frac{ipx}{\hbar}} \tilde{\Psi}(p, t)$$

$$\langle p|\Psi(t)\rangle = \tilde{\Psi}(p, t) = \int dx \langle p|x\rangle \langle x|\Psi(t)\rangle = \int dx e^{-\frac{ipx}{\hbar}} \Psi(x, t)$$

”So we can see that position space wave fun. & momentum space wave functions are fourier transforms of each other.”

In 3-D,

$$\begin{aligned}\langle \vec{r} | \vec{p} \rangle &\propto e^{\frac{i\vec{p} \cdot \vec{r}}{\hbar}} \\ \langle \vec{p} | \vec{r} \rangle &\propto e^{-\frac{i\vec{p} \cdot \vec{r}}{\hbar}}\end{aligned}$$

also

$$\langle \vec{r} | \hat{p} | \Psi(t) \rangle = -i\hbar \nabla_{\vec{r}} \langle \vec{r} | \Psi(t) \rangle$$

($\vec{r}, \hat{p}$ means 3-D)

The shrödinger equation For a particle :-

$$\begin{aligned}\hat{H}(\hat{r}, \hat{p}) &= \hat{H} = \frac{\hat{p}^2}{2m} + \hat{V}(\hat{r}) \\ i\hbar \frac{d}{dt} |\Psi(t)\rangle &= \hat{H} |\Psi(t)\rangle\end{aligned}$$

so, we want to know how above eqⁿ looks like in position basis.

taking inner product with $\langle \vec{r} | \dots \rangle$ both sides.

$$\left\langle \vec{r} \left| i\hbar \frac{d}{dt} \right| \Psi(t) \right\rangle = \langle \vec{r} | \hat{H} | \Psi(t) \rangle$$

{ Here $|\vec{r}\rangle$ or $\langle \vec{r}|$ is any particular eigen state of position basis.

above eqⁿ can be written as,

$$i\hbar \frac{d}{dt} (\langle \vec{r} | \Psi(t) \rangle) = \langle \vec{r} | \hat{H} | \Psi(t) \rangle$$

$$i\hbar \frac{\partial}{\partial t} (\Psi(r, t)) = \left\langle \vec{r} \left| \frac{\hat{p}^2}{2m} \right| \Psi(t) \right\rangle + \langle \vec{r} | V(\hat{r}) | \Psi(t) \rangle$$

{ $\frac{d}{dt}$ has to be replaced with $\frac{\partial}{\partial t}$ as derivative is only w.r.t 't' variable.

$\langle \vec{r} | \Psi(t) \rangle$ = position space wave function.

$$\Rightarrow i\hbar \frac{\partial}{\partial t} \Psi(r, t) = \frac{(-i\hbar)^2 \nabla^2}{2m} \Psi(r, t) + V(\vec{r}) \psi(r, t)$$

$$\begin{cases} \Psi(r, t) = \langle \vec{r} | \Psi(t) \rangle \\ V(r) = \text{eigen value of } V(\hat{r}) \end{cases}$$

since we deal with hermitian operators in Q.M. (to get real eigen value)

$V(\hat{r})$ should be hermitian too.

So, $V^*(\vec{r})$ is simply $V(\vec{r})$

[complex $V(r)$ is used in case where there is absorption where probabilities are not conserved.]

Find eqⁿ becomes,

$$\left\{ i\hbar \frac{\partial}{\partial t} \Psi(\vec{r}, t) = \frac{-\hbar^2}{2m} \nabla^2 \Psi(\vec{r}, t) + V(\vec{r}) \Psi(\vec{r}, t) \right\}$$

Position space shrödinger equation

Boundary condⁿ

1) Normalized $\Psi(r, t)$

$$\langle \Psi(t) | \Psi(t) \rangle = 1 = \int d^3r |\Psi(\vec{r}, t)|^2$$

($\Psi(r, t)$ is member of l_2 space)

Stationary states (eigen states) :-

stationary state of the Q.Mechanical system is by defⁿ is an eigen state of total Hamiltonian of the system.

”Any stationary state is eigen state of $\hat{H}$.

$\hat{H}$ operator has in general infinite no of eigen states”.

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = H |\Psi(t)\rangle$$

lets use $\Phi(t)$ for eigen state (of Hamiltonian) instead of $\Psi(t)$

$$i\hbar \frac{d}{dt} |\Phi(t)\rangle = H |\Phi(t)\rangle = H_0 |\Phi(t)\rangle$$

(H_0 is any eigen value of H .)

we know that Hamiltonian tells total energy so lets use E as eigen value of Hamiltonian

E is real eigen value. (as H is made up of physical measurables)

$$i\hbar \frac{d}{dt} |\Phi(t)\rangle = E |\Phi(t)\rangle$$

E = eigen value ; $\Phi(t)$ = eigen state. (any) of hamiltonian

{ there can be more eigen state for one eigen value. for that eigen value should be repeated.

solution to above eqⁿ is

$$|\Phi(t)\rangle = e^{-\frac{iEt}{\hbar}} |\Phi(0)\rangle$$

For a general state $|\Psi(t)\rangle$

we can write $|\Psi(t)\rangle$ as linear combination of basis set formed from eigen state of Hamiltonian operator.

$$|\Psi(t)\rangle = \sum_n c_n |\Phi_n(t)\rangle$$

or we can write

$$|\Psi(t)\rangle = \sum_n c_n e^{-\frac{iE_n t}{\hbar}} |\phi_n(0)\rangle$$

or

$$|\Psi(t)\rangle = \sum_n c_n(t) |\phi_n(0)\rangle$$

$$c_n(t) = c_n e^{-\frac{iE_n t}{\hbar}}$$

{ Generally we use,

$$|\Psi(t)\rangle = \sum_n c_n e^{-\frac{iE_n t}{\hbar}} |\phi_n(0)\rangle$$

} The abstract state of any system.

where, $(|\phi_n(0)\rangle)$ is a choosen time independent basis set)

$$c_n = \langle \phi_n | \Psi(t) \rangle$$

where, $\{n = \text{quantum number}\}$

In General there may be as many quantum no. as there are d.o.f of the system.

for any particle confined in 3-D potential there will be at least 3 quantum no. There may be more if there are internal quantum no. like spin, hypercharge, strangeness for elementary particles.

$$|\Psi(t)\rangle = \sum_n c_n e^{-\frac{iE_n t}{\hbar}} |\Phi_n(0)\rangle$$

is also written as

$$\int_n c_n e^{-\frac{iE_n t}{\hbar}} |\phi_n(0)\rangle$$

which changes to integral sign when n is contineous.

All quantum interference phenomena happen because of the superposition of different energy states. (Beats, Interference groups, mforms. due to different frequency superposition).

$$|\Psi(t)\rangle = \sum_n c_n e^{-\frac{iE_n t}{\hbar}} |\Phi_n(0)\rangle$$

$\hookrightarrow$ not a fourier series. $\{c_n|\phi_n(0)\rangle$ is aperiodic.

Here we have assumed that all eigen states of Hamiltonian are made orthogonal by Gramh-Schmidt orthonormalization.

lets look for stationary states in position basis $\{|\vec{r}\rangle\}$.

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$$

taking inner product.

$$\begin{aligned} \left\langle \vec{r} \left| i\hbar \frac{d}{dt} \right| \Psi(t) \right\rangle &= \langle \vec{r} | \hat{H} | \Psi(t) \rangle \\ &= \langle \vec{r} | E | \Psi(t) \rangle \end{aligned}$$

$\{|\vec{r}\rangle$ is any position eigen state $\}$.

$$i\hbar \frac{\partial}{\partial t} \Psi(\vec{r}, t) = E \Psi(\vec{r}, t)$$

$$\Rightarrow \boxed{\Psi(r, t) = e^{-\frac{iEt}{\hbar}} |\psi(r, 0)\rangle}$$

For a stationary state $|\underline{\psi(r, t)}\rangle$.

For any general state

$$\Psi(r, t) = \sum_n e^{-\frac{iE_n t}{\hbar}} |\psi_n(r, 0)\rangle$$

lets use our old notation $|\phi_n\rangle$ for stationary states.

$$|\phi_n(\vec{r}, t)\rangle = e^{-\frac{iE_n t}{\hbar}} |\phi_n(\vec{r}, 0)\rangle$$

So if work in position basis. The $|\Psi(t)\rangle$ can be written as.

$$\begin{aligned} \langle \vec{r} | \Psi(t) \rangle &= \left\langle \vec{r} \left| \sum_n c_n e^{-\frac{iE_n t}{\hbar}} \right| \phi_n(\vec{r}, 0) \right\rangle \\ &= \sum_n c_n e^{-iE_n t/\hbar} \langle \vec{r} | \phi_n(\vec{r}, 0) \rangle \end{aligned}$$

where $\langle \vec{r} | \Psi(t) \rangle$ is general state at time 't' in position basis.

So The shrödinger equation for stationary states $|\phi_n(\vec{r}, t)\rangle$ looks like

$$\begin{cases} i\hbar \frac{\partial}{\partial t} |\phi_n(\vec{r}, t)\rangle = \frac{(-i\hbar)^2}{2m} \nabla^2 \phi_n(\vec{r}, t) + V(\vec{r}) \phi_n(r, t) \\ \quad \quad \quad = E_n \phi_n(\vec{r}, t) \end{cases}$$

In fact we can remove 't' dependence by only considering

$$\frac{-\hbar^2}{2m} \nabla^2 \phi_n(\vec{r}, t) + V(\vec{r}) \phi_n(\vec{r}, t) = E_n \phi_n(\vec{r}, t)$$

where

$$\phi_n(\vec{r}, t) = \phi_n(\vec{r}, 0) e^{-\frac{iE_n t}{\hbar}}$$

using above we get,

$$\boxed{\frac{-\hbar^2}{2m} \nabla^2 \phi_n(\vec{r}, 0) + V(\vec{r}) \phi_n(\vec{r}, 0) = E_n \phi_n(\vec{r}, 0)}$$

} Time Independent Shrödinger equation for stationary state.

the same equation can be written as :-

$$\left\{ \left(\frac{-\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right) |\phi_n(\vec{r})\rangle = E_n |\phi_n(\vec{r})\rangle \right\}$$

eigen value equation. & Homogeneous too.

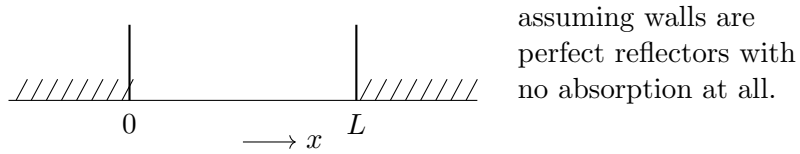
Any confinement on matter "quantizes the energy levels".

A free particle can have any energy but a confined particle in potential / box ensures that only certain packets of energy are allowed.

Q) Where is confinement in H-atom, where e^- is placed in coulomb potential?

Ans.) Hilbert noted that condⁿ that $\int |\psi|^2 d^3r = 1$ is enough to produce confinement / quantization

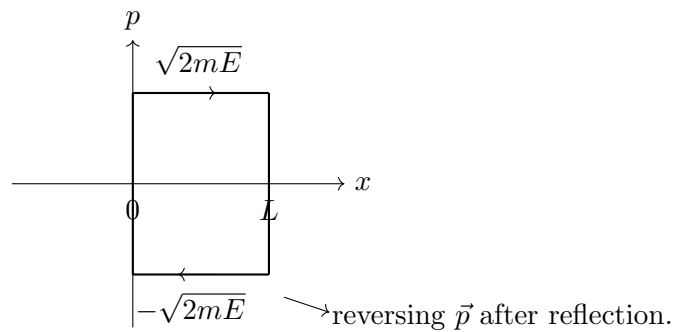
Particle in 1-D BOX :-



classically :-

$$H(x, p) = \frac{p^2}{2m} + V(x)$$

$$V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & \text{outside} \end{cases}$$



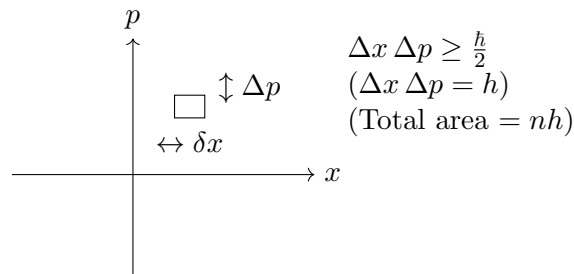
old rule of Bohr's quantization (for periodic orbits) (postulate)

Area under phase space = $\oint p dx = nh$

$$\Rightarrow 2\sqrt{2mE_n} \cdot L = nh$$

$$E_n = \frac{n^2 h^2}{8mL^2} \quad (\text{semi-classical result}) \quad \left(\oint P dx = nh \right)$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$



using time independent schrödinger eqⁿ :-

$$\left[\frac{-\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right] |\phi_n(x)\rangle = E_n |\phi_n(x)\rangle$$

for $(0 < x < L)$ inside the box. $(V(x) = 0)$

$$\begin{aligned} \frac{-\hbar^2}{2m} \frac{d^2}{dx^2} |\phi_n(x)\rangle &= E_n |\phi_n(x)\rangle \\ \left(\frac{d^2}{dx^2} + k_n^2 \right) |\phi_n(x)\rangle &= 0 \quad \left\{ k_n^2 = \frac{2mE_n}{\hbar^2} \right\} \end{aligned}$$

Solutions is trivial

$$|\phi_n(x)\rangle = A \cos k_n x + B \sin k_n x$$

Imposing boundary condⁿ :-

$$\begin{aligned} \phi_n(0) &= 0 = \phi_n(L) \\ \Rightarrow A &= 0, \quad k_n L = n\pi \end{aligned}$$

as $B \neq 0$

Because of

$$\begin{aligned} \langle \phi_n(x) | \phi_n(x) \rangle &= 1 \quad \text{for all space} \\ \int_0^L B^2 \sin^2 k_n x \, dx &= 1 \\ \Rightarrow B^2 \int_0^L \left(\frac{1 - \cos 2k_n x}{2} \right) dx &= 1 \\ \Rightarrow B^2 \left(\frac{L}{2} - \left(\frac{\sin 2k_n x}{4k_n} \right)_0^L \right) &= 1 \\ B^2 \left(\frac{L}{2} - 0 \right) &= 1 \quad \Rightarrow \quad \boxed{B = \sqrt{\frac{2}{L}}} \\ |\phi_n(x)\rangle &= \sqrt{\frac{2}{L}} \sin \left(\frac{n\pi}{L} x \right) \end{aligned}$$

and

$$E_n = \frac{\hbar^2 n^2 \pi^2}{2mL^2} \quad (n = 1, 2, 3 \dots \infty)$$

where Bohr rule also predicts same as exact eigen values.

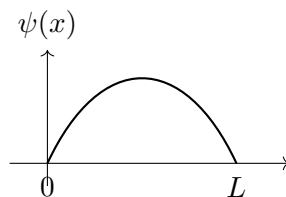
note :- ($n \neq 0$ as $|\phi_n(x)\rangle$ vanishes & normalization does not work)

what is

$$|\phi_n(x, t)\rangle = e^{-\frac{iE_n t}{\hbar}} |\phi_n(x)\rangle \quad \& \quad e^{-\frac{iE_n t}{\hbar}} |\phi_n(0)\rangle$$

Because of lack of symmetry, we can also see that for any Energy eigen value we have a unique eigen state, so there is no degeneracy in this problem.

$$\phi_n(x) = \begin{cases} \sqrt{\frac{2}{L}} \sin \left(\frac{n\pi}{L} x \right) & 0 \leq x \leq L \\ 0 & x < 0, x > L \end{cases}$$



$[\psi'(x)]$ is disc has finite discontinuity at $x = 0, L$

So, $\psi''(x)$ should have infinite discontinuity at $x = 0, L$, and that discontinuity must be compensated by $V(x)$

$$\left(\frac{-\hbar^2}{2m}\right) \frac{d^2\phi_n}{dx^2} + V(x) \phi(x) = E_n \phi(x)$$

]

$$\int_{-\infty}^{\infty} dx |\phi_n(x)|^2 = 1$$

$$[\phi_n(x)] = \frac{1}{\sqrt{\text{length}}} \quad \text{dimensions of } |\phi_n(x)\rangle$$

we do not have following const. in $\phi_n(x)$ due to some reasons.

const.	Reason for absence
c	no relativity
G	no gravity
k_B	assuming $T = 0$ K (absolute)
$\hbar$	

$[\hbar \text{ does appear in wavefunctions of H-atom } R(r) \propto e^{-(r/a_0)}, a_0 = f(\hbar)]$

$\sin(\) \rightarrow$ must depend on x . & must be dimensionless.

$\left(\sqrt{\frac{2}{L}}\right) \sin(\) \rightarrow$ To normalize.

So ' $\hbar$ ' don't appear in this problem because of dimensional reasons.

lets check if $|\phi_n(x)\rangle$ is momentum eigen state or not.

$$\hat{p}|p\rangle = p|p\rangle \quad (\rightarrow \text{eigen value})$$

$$-i\hbar \frac{\partial}{\partial x} \phi_n(x) \neq (\) \phi_n(x)$$

$$\Rightarrow |\phi_n(x)\rangle \text{ is not a momentum eigen state.}$$

But $[H, \hat{P}] = 0$ as $H = \frac{\hat{p}^2}{2m}$

So we must have some common eigen states.

But we just found out that momentum eigen states are not momentum eigen states.

[$\rightarrow$ Energy / Hamiltonian]

{ So Only solution is "Particle is not free".

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

$$\text{so } \begin{cases} [H, p^2] \neq 0 \\ [H, P] \neq 0 \end{cases}$$

what would $|p\rangle$ (momentum eigen states) looks like?

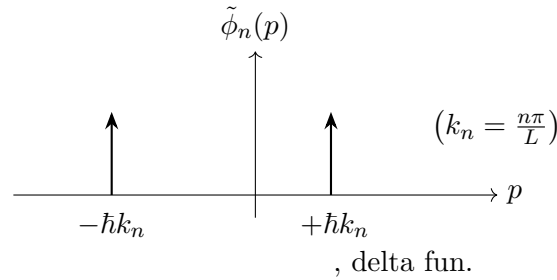
$$\hat{p} \chi(x) = p \chi(x)$$

$$\begin{aligned}\Rightarrow -i\hbar \frac{d\chi}{dx} &= p\chi \\ \Rightarrow \chi(x) &\propto e^{\frac{ipx}{\hbar}}\end{aligned}$$

so in position basis, momentum eigen states must look like this.

$$\begin{aligned}\phi_n(x) &= \sqrt{\frac{2}{L}} \left(\frac{1}{2i} \right) \left[e^{ik_n x} + e^{-ik_n x} \right] \\ \left(k_n &= \frac{n\pi}{L} \right) \\ \frac{ipx}{\hbar} &= \frac{i(n\pi/L)x}{1} \\ \boxed{p = \frac{n\pi\hbar}{L}} &\quad \& \quad p = -\frac{n\pi\hbar}{L}\end{aligned}$$

Solving problem in momentum space wave function



[Q] we know that if $\phi(x) \in l_2$, $\tilde{\phi}(p)$ must also be a member of l_2 . But here dirac functions are not square integrable so whats the solution?

$$\left(\delta(k - k_0) = \int_{-\infty}^{\infty} e^{i(k-k_0)x} dx \right)$$

iff $(-\infty < x < \infty)$ there will be $\tilde{\phi}(p) =$ delta funct at $\pm \hbar k_n$

But $0 < x < L$ so $\tilde{\phi}(p)$ is not delta fun of p

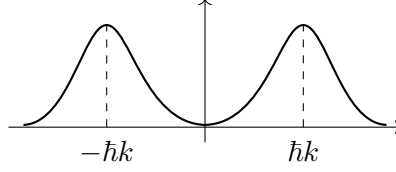
lets find $\tilde{\phi}(p)$

$\tilde{\phi}(p)$ is fourier transform of $\phi(x)$.

$$\begin{aligned}\tilde{\phi}(p) &= \int_{-\infty}^{\infty} dx \phi_n(x) e^{-\frac{ipx}{\hbar}} \\ &= \sqrt{\frac{2}{L}} \int dx \frac{e^{-\frac{ipx}{\hbar}}}{2i} \left(e^{ik_n x} - e^{-ik_n x} \right) \\ \tilde{\phi}(p) &= \sqrt{\frac{2}{L}} \int_0^L \frac{e^{-\frac{ipx}{\hbar}}}{2i} \left(e^{ik_n x} - e^{-ik_n x} \right) dx\end{aligned}$$

So,

$$\underline{\tilde{\phi}(p) = \text{smooth function of } p, \in l_2}$$



"So particle in box can have any momentum value"

$$\langle A \rangle = \frac{\langle \Psi | A | \Psi \rangle}{\langle \Psi | \Psi \rangle}$$

since Ψ is normalized $\Rightarrow \langle \Psi | \Psi \rangle = 1$

$$\langle A \rangle = \langle \Psi | A | \Psi \rangle$$

$\langle A \rangle$ is time independent. That's why we call it a stationary state.

$$\begin{aligned} \langle A \rangle &= \langle \Psi | A | \Psi \rangle \\ &= \int dx' \int dx \langle \Psi | x \rangle \langle x | A | x' \rangle \langle x' | \Psi \rangle \\ \langle \hat{x} \rangle_n &= \int dx' \int dx \phi_n^*(x) \langle x | \hat{x} | x' \rangle \phi_n(x) \end{aligned}$$

($\rightarrow$ acts upon)

$$\begin{aligned} &\begin{cases} \hat{x} | x' \rangle = x' | x' \rangle \\ \hat{x} | x \rangle = x | x \rangle \end{cases} \\ &= \int dx' \int dx \phi_n^*(x) x \langle x | x' \rangle \phi_n(x) \\ &= \int dx' \int dx \phi_n^*(x) x \delta(x - x') \phi_n(x) \end{aligned}$$

doing x' integral ; ($\delta(x - x') = 1$ for $x = x'$, $\phi_n(x') = \phi_n(x)$ at $x = x'$)

$$\boxed{\langle x \rangle_n = \int dx \phi_n^*(x) x \phi_n(x)}$$

$\hookrightarrow$ calculated in eigen state $\phi_n(x)$.

(where; x = eigen value of $\hat{x}$; $\langle x | \hat{x} = x \langle x |$; $n \rightarrow n^{\text{th}}$ eigen state of Hamiltonian operator)

so,

$$\begin{aligned} \langle x \rangle_n &= \int_0^L dx \frac{2}{L} \sin^2 \left(\frac{n\pi}{L} x \right) \cdot x \\ \langle x^2 \rangle_n &= \int_0^L dx \frac{2}{L} \sin^2 \left(\frac{n\pi}{L} x \right) \cdot x^2 \end{aligned}$$

compute $(\Delta x)_n$ uncertainty / std. deviation in position.

$$(\delta x)_n = \sqrt{\langle x^2 \rangle_n - \langle x \rangle_n^2}$$

similarly,

$$\begin{aligned}\langle p \rangle_n &= \int_{-\infty}^{\infty} dx \phi_n^*(x) (-i\hbar) \frac{d}{dx} \phi_n(x) \\ \langle p \rangle_n &= (-i\hbar) \int_0^L dx \frac{2}{L} \sin\left(\frac{n\pi}{L}x\right) \cos\left(\frac{n\pi}{L}x\right) \left(\frac{n\pi}{L}\right) = 0 \\ \langle p^2 \rangle_n &= \int_0^L dx \frac{2}{L} \sin\left(\frac{n\pi}{L}x\right) \left(\frac{n\pi}{L}\right)^2 (-i\hbar)^2 \sin\left(\frac{n\pi}{L}x\right) (-1) \\ \langle p^2 \rangle_n &= \frac{2}{L} \cdot \frac{n^2 \pi^2 \hbar^2}{L^2} \int_0^L \sin^2\left(\frac{n\pi}{L}x\right) dx \\ &= \frac{n^2 \pi^2 \hbar^2}{L^2}\end{aligned}$$

also

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} = \boxed{\frac{\langle p^2 \rangle_n}{2m}}$$

(only iff energy is only K.E.)

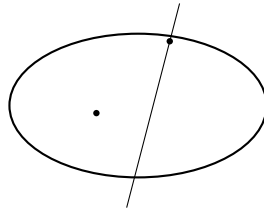
$$(\Delta p)_n = \sqrt{\langle p^2 \rangle_n - \langle p \rangle_n^2} = \frac{n\pi\hbar}{L}$$

$$(\Delta x)_n = \sqrt{\langle x^2 \rangle_n - \langle x \rangle_n^2}$$

$$\boxed{(\Delta x)_n (\Delta p)_n \geq \hbar}$$

Energy levels are not degenerate in this case.

Particle in a ring :- (circle).



$$\hat{H} = \frac{\hat{p}^2}{2m}$$

in this problem $V(x) = 0$ everywhere, so

$$[H, P] = 0 \quad \text{as} \quad [\hat{p}, p] = 0$$

[1-D simply connected space has no degeneracy, but this space is not simply connected so it may have degeneracy in 1-D. & In fact It has degeneracy of 2. (Its doubly degenerate).]

So every eigen state of H will be a momentum eigen state.

since, $\hat{p}$ and $\frac{\hat{p}^2}{2m}$ has ∞ no of eigen states so They share complete set as, common eigen state.

Periodic Boundary condition :-

$$\phi_n(0) = \phi_n(L)$$

$$\phi_n(x+L) = \phi_n(x)$$

($\phi_n(x)$ is any eigen state of $\hat{H}$.)

If function value are same at x , $x+L$ so their derivative should also be equal.

$$\phi'_n(x+L) = \phi'_n(x)$$

$$\phi''_n(x+L) = \phi''_n(x) \quad \dots \text{all derivative must match.}$$

"Are all $|p\rangle$ also eigenstates of $\hat{H}$? (yes) if so, then $|\Psi\rangle = \sum_n c_n |p\rangle$

$$\langle x|\hat{p}|p\rangle = p \langle x|p\rangle$$

$$-i\hbar \frac{\partial}{\partial x} \langle x|p\rangle = p \langle x|p\rangle \Rightarrow \langle x|p\rangle \propto e^{\frac{ipx}{\hbar}}$$

$$\frac{-\hbar^2}{2m} \frac{\partial^2 \phi_n(x)}{\partial x^2} + 0 = E_n \phi_n(x)$$

$$\frac{d^2 \phi_n(x)}{dx^2} + \frac{2mE_n}{\hbar^2} \phi_n(x) = 0$$

$$(D^2 + k^2) \phi_n(x) = 0$$

$$D = \pm ik$$

$$\phi_n(x) = Ae^{ikx} + Be^{-ikx}$$

1)

$$\phi_n(0) = \phi_n(0+L)$$

$$A + B = Ae^{ikL} + Be^{-ikL}$$

2)

$$\phi_n(x) = \phi_n(x+L)$$

$$Ae^{ikx} + Be^{-ikx} = Ae^{ik(x+L)} + Be^{-ik(x+L)}$$

$$Ae^{ikx} (1 - e^{ikL}) + Be^{-ikx} (1 - e^{-ikL}) = 0$$

$$\frac{Ae^{ikx}}{Be^{-ikx}} = \frac{e^{-ikL} - 1}{1 - e^{ikL}} \quad \dots \textcircled{1}$$

3)

$$\phi'_n(x) = \phi'_n(x+L)$$

$$[\phi_n(x) = Ae^{ikx} + Be^{-ikx}]$$

$$\phi_n(x) = a \cos k_n x + b \sin k_n x$$

$$\phi_n(0) = \phi_n(L)$$

$$a = a \cos k_n L + b \sin k_n L$$

$$k_n L = 2n\pi$$

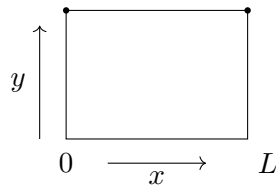
$$k_n = \frac{2n\pi}{L} = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\frac{4n^2\pi^2}{L^2} = \frac{2mE_n}{\hbar^2}$$

$$E_n = \frac{2n^2\pi^2\hbar^2}{mL^2}$$

]

Particle in 2-D box :-



$$\phi_{n_1 n_2}(x, y) = \frac{2}{L} \sin\left(\frac{n_1 \pi}{L} \cdot x\right) \sin\left(\frac{n_2 \pi}{L} \cdot y\right)$$

$$E(n_1, n_2) = \frac{\hbar^2 \pi^2}{2mL^2} (n_1^2 + n_2^2)$$

These energy levels are degenerate except ground state.

{where, $n_1, n_2 = 1, 2, 3 \dots$ }

a) Ground state $\Rightarrow n_1 = 1, n_2 = 1$ (non-degenerate).

$$E(1, 1) = \frac{\hbar^2 \pi^2}{2mL^2} (2)$$

b) First excited state (1, 2) or (2, 1) degenerate (doubly degenerate)

$$E(1, 2) = E(2, 1) = \frac{\hbar^2 \pi^2}{2mL^2} (5)$$

c) 2nd excited state (2, 2) (non-degenerate)

$$E(2, 2) = \frac{\hbar^2 \pi^2}{2mL^2} (8)$$

d) 3rd excited state (1, 3) or (3, 1). (doubly degenerate)

if potential were like a rectangle

$$\phi_{n_1 n_2}(x, y) = \frac{2}{\sqrt{L_1 L_2}} \sin\left(\frac{n_1 \pi}{L_1} x\right) \sin\left(\frac{n_2 \pi}{L_2} y\right)$$

$$E(n_1, n_2) = \frac{\hbar^2 \pi^2}{2m} \left(\frac{n_1^2}{L_1^2} + \frac{n_2^2}{L_2^2} \right)$$

Now degeneracy depends on L_1 & L_2 in energy levels.

overview — || prev — next <>

[The notebook does not carry a separate "Lec 9" heading – the material for it runs on inside the pages transcribed on the Lec 7-8 page. This page begins at the "lec 10" heading in the scans.]

9 Lecture 10

$\rightarrow$ "symmetry implies degeneracy".

→ If a problem is integrable in classical mechanics does not imply that same problem will be integrable in quantum mechanically.

ex.

$$H(x, p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 + \lambda x^4$$

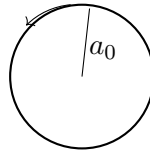
classically Integrable (1 C.O.M required for 1.D.of = H)

But quantum mechanically we can't find exact eigen values of H . we can find it with arbitrary accuracy using some perturbation method.

Nature of symmetry in Q.M has very – very deep profound implications as opposed to classical mechanics.

ex In H- atom, we assumed that e^- orbit around nucleus.

$$L \neq 0$$



according to bohr theory

$$\boxed{L \neq 0} \in L = mvr = n\hbar \quad (n = 1, 2, 3)$$

n = angular quantum no.

$$\frac{mv^2}{r} = \frac{Ze^2}{r^2}$$

$$E_n \propto \frac{-1}{n^2} \quad (n = 1, 2, 3 \dots)$$

Ground state of H. (1S) (spherically symmetric).

$$n = 1, \quad l = 0, \quad m = 0$$

eigen fun. can be written as

$$\left\{ \begin{aligned} \phi_{100}(r, \theta, \phi) &\propto e^{-r/a_0} \\ l = 0 &\Rightarrow L = 0 \end{aligned} \right.$$

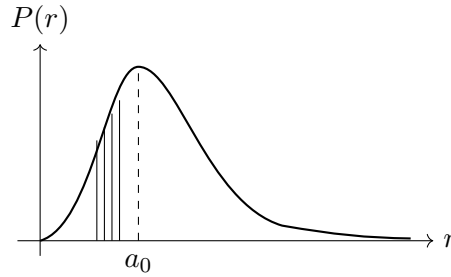
So, How is that quantum theory suggests that angular momentum of e^- in ground state is zero but classical mechanics / intuition says its not zero.

The e^- does not orbit with radius a_0 . we can find e^- anywhere b/w 0 to ∞ ($0 < r < \infty$), and probability of finding it b/w $r, r + dr$ is

$$P(r) dr = |\phi_{100}(r)|^2 dV$$

$$= e^{-\frac{2r}{a_0}} (4\pi r^2) dr$$

$$\{\phi_{100}(r, \theta, \phi) = \phi(r) \text{ (as no dependence on } \theta, \phi)$$



$\{P(r) \text{ is maximum at } r = a_0\}$

still It does not answer How $L = 0$.

$\{\text{classically, This could only happen if } e^- \text{ passes through center}\}$

How it is that the e^- predominantly at $r = a_0$ has zero angular momentum.

Classically different solutions which differ in direction of $\vec{L}$ are related by rotation transformations. we can go from one solution (rotation is in xy plane) to another solution (rotation is in yz plane) by rotation of coordinate axis.

So the group of transformations (rotation group) under which the Hamiltonian is invariant. Each element of that group of rotations takes you from one possible solution to another possible solution.

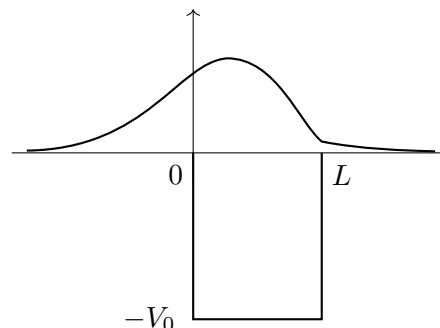
And once you fix coordinate system and specify the initial conditions, the orbit is fixed.

Now Q.M says superposition is valid. So any solution is superposition of all solutions. It says if two solutions are related by a symmetry transformation the general solution is superposition of two solutions.

Now its easy to see that if take the orbit and put it in all possible planes and add all $\vec{L}$. we get $\vec{L} = 0$.

Thats the reason in Q.M we can still sustain a zero $\vec{L}$ solution; even though the e^- has overwhelmingly probability to be at non-zero distance from origin. This is very profound & deep aspect of Q.M.

Particle in Finite well (potential well) :



what are possible values of energy states?

$$\phi_n'' + k_n^2 \phi_n - V(x) \phi_n = 0$$

$\hookrightarrow$ finite discountinuity. compensated by finite discountinuity of $\phi_n''(x)$.

There is a finite probability that a particle tunnel.

Now,

lets make the problem simpler by assuming that width of the well goes to zero & simultaneously the $V(x) \rightarrow \infty$ such that product is finite. we would like to see bound state (i.e. potential can retain the particle).

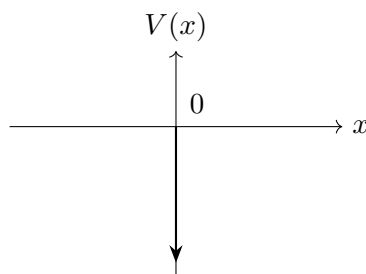
lets assume $V(x)$ is a negative delta function.

Attractive delta – fun. potential :-

$$\frac{-\hbar^2}{2m} \phi''(x) + V(x) \phi(x) = E \phi(x)$$

$\{\phi(x) = \text{eigen state of } H. ; E = \text{eigen values of } H$

lets put δ -fun at $x = 0$, so $V(x) = -\lambda \delta(x)$



$$[\lambda] = ML^3T^{-2}$$

because, $[\delta(x)] = L^{-1}$

E must depend on $\lambda, \hbar, m$

$$E \propto \lambda^\alpha \hbar^\beta m^\gamma$$

$$[ML^2T^{-2}] \propto [ML^3T^{-2}]^\alpha [ML^2T^{-1}]^\beta m^\gamma$$

$$[M^1L^2T^{-2}] = k M^{\alpha+\beta+\gamma} L^{3\alpha+2\beta} T^{-\beta-2\alpha}$$

$$\alpha + \beta + \gamma = 1 \qquad \beta = -2$$

$$2\beta + 3\alpha = 2 \qquad \Rightarrow \qquad \gamma = +1$$

$$-2\alpha - \beta = -2 \qquad \alpha = 2$$

$$\therefore E \propto \frac{m\lambda^2}{\hbar^2}$$

For $x > 0) \Rightarrow V(x) = 0$

$$\left(\frac{-\hbar^2}{2m} \phi''(x) = E \phi(x) \right) \Rightarrow \phi''(x) = -\frac{2mE}{\hbar^2} \phi(x)$$

$\{E < 0$ for bound states so better to write

$$\phi''(x) = +\frac{2m|E|}{\hbar^2} \phi(x)$$

$$\phi''(x) = \frac{2m|E|}{\hbar^2} \phi(x)$$

$$\phi''(x) - k^2 \phi(x) = 0 \quad \left(k = \sqrt{\frac{2m|E|}{\hbar^2}} \right)$$

$$(D^2 - k^2)\phi(x) = 0 \quad (D = \pm k) \quad (D = d/dx)$$

$$\phi(x) = Ae^{kx} + Be^{-kx}$$

$$\lim_{x \rightarrow \infty} \|\phi(x)\| \rightarrow \text{finite} \quad \text{so} \quad A = 0$$

$$\phi(x) = Be^{-kx}$$

for $x < 0$)

$$\phi(x) = Ce^{kx} + De^{-kx}$$

normalizability $\Rightarrow D = 0$ [$\lim_{x \rightarrow -\infty} \|\phi(x)\| \rightarrow \text{finite}$]

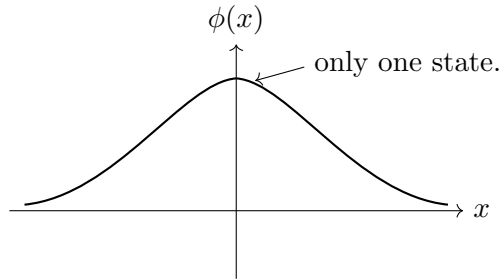
$$\phi(x) = Ce^{kx}$$

Since $\phi(x)$ is continuous.

$$\begin{aligned} \phi(x) \Big|_{x \rightarrow 0^-} &= \phi(x) \Big|_{x=0} = \phi(x) \Big|_{x \rightarrow 0^+} \\ \Rightarrow \quad &\boxed{C = B} \end{aligned}$$

So,

$$\phi(x) = Be^{kx}$$



$$\phi'(x(0^-)) \neq \phi'(x(0^+))$$

$$\Rightarrow \phi''(x(0)) \text{ has infinite discontinuity.}$$

for $x = 0$)

$$\phi''(x) + \frac{2m\lambda}{\hbar^2} \delta(x) \phi(x) = k^2 \phi(x) \quad \left\{ k^2 = \frac{-2mE}{\hbar^2} \right\}$$

Integrating from $x = -\epsilon$ to $x = +\epsilon$ & let $\epsilon \rightarrow 0$

$$\int_{-\epsilon}^{\epsilon} dx \left\{ \phi''(x) + \frac{2m\lambda}{\hbar^2} \delta(x) \phi(x) \right\} = \int_{-\epsilon}^{\epsilon} k^2 \phi(x) dx$$

$$\left(\frac{d\phi(x)}{dx} \right)_{0^-}^{0^+} + \frac{2m\lambda}{\hbar^2} \phi(0) = k^2 \left(\int_0^{\epsilon} Be^{-kx} dx + \int_{-\epsilon}^0 Be^{kx} dx \right) = 0$$

$$\begin{aligned}
& \left(\frac{d\phi}{dx} \Big|_{x \rightarrow 0^+} - \frac{d\phi}{dx} \Big|_{x \rightarrow 0^-} \right) + \frac{2m\lambda}{\hbar^2} B = 0 \quad (\phi(0) = B) \\
& \begin{cases} x > 0, \phi(x) = Be^{-kx}, & \phi'(x) = -kBe^{-kx} \\ x < 0, \phi(x) = Be^{kx}, & \phi'(x) = +kBe^{+kx} \end{cases} \\
& -2kB + \frac{2m\lambda}{\hbar^2} B = 0 \\
& B \left(2k - \frac{2m\lambda}{\hbar^2} \right) = 0 \\
& B \neq 0, \quad k = \frac{m\lambda}{\hbar^2}; \quad k^2 = \frac{2m|E|}{\hbar^2} = \frac{m^2\lambda^2}{\hbar^4} \\
& \Rightarrow |E| = \frac{m\lambda^2}{2\hbar^2}
\end{aligned}$$

So, for bound state $E < 0$

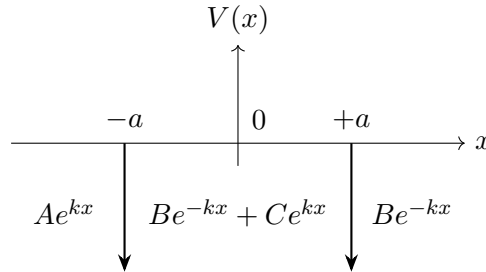
Note :-

$$E = -\frac{m\lambda^2}{2\hbar^2}$$

(only one bound state) as no quantum no exists.

So a single point of ∞ -discontinuity in potential supports one B. state

WHAT IF we have two such delta – potential wells



$$V(x) = -\lambda [\delta(x - a) + \delta(x + a)]$$

{5 eqⁿ unknown (A, B, C, D, k)}

$\phi(-a)$ is continuous & differentiable $\Rightarrow$ 2 equations

$\phi(+a) \Rightarrow$ 2 equations ; (normalization)

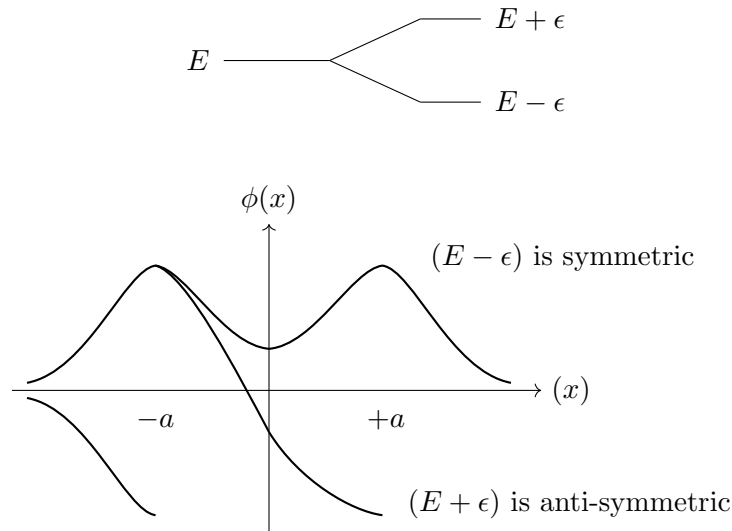
For two states which are independent

$$H \sim \begin{pmatrix} E & 0 \\ 0 & E \end{pmatrix} \quad \text{in some vector space.}$$

now, if we switch on the coupling b/w the two

$$H \sim \begin{pmatrix} E & \varepsilon \\ \varepsilon & E \end{pmatrix}$$

eigen values $\Rightarrow E_{1,2} = E \pm \varepsilon$



overview — || prev — next ||

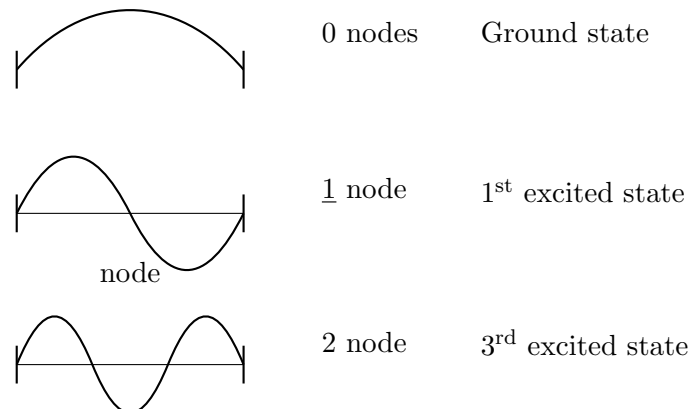
10 Lecture 11

In 1-D problems there is a theorem which says,

Ground state has no node, 1st excited state has one node 2nd excited state has 2 nodes.

so Question is, why does the no. of nodes increase as Energy increases.

To have wave fun. normalizable. It should be clamped at both ends.

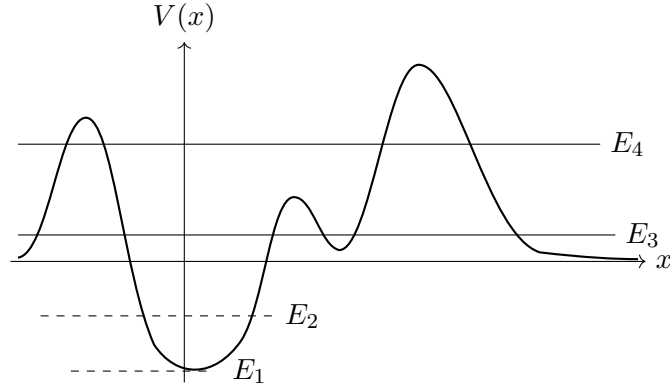


It is because of curvature keeps changing as energy increases. $\phi''(x)$ contributes more and more as curvature changes.

$$\frac{-\hbar^2}{2m} \nabla^2 \phi(x) = E \phi(x)$$

(curvature (1-D))

So we can see that how ∇^2 contributes to increase of Energy eigen value.



let E_i be total energy of system. $\{E_i\} = \{E_1, E_2, E_3, \dots\}$

($E_1 < 0$) and $E_1 < V(\min)$:-

Such case does not exist as we will see that $\langle \text{K.E} \rangle$ is always a real positive no.

$$\left[\frac{\hat{p}^2}{2m} + V(\hat{x}) \right] |\phi(x)\rangle = E|\phi(x)\rangle$$

Since x, p does not commute, $[x, p] \neq 0$ we can not have definite value of K.E and P.E simultaneously in the same state. as.

$$\left[\frac{p^2}{2m}, V(x) \right] \neq 0.$$

(or)

In general the eigen states of $\hat{H}$ are not the eigen states of K.E or the P.E separately.

(or)

In a stationary state, the particle can not have definite value either of its K.E or its P.E. But only sum of the two has definite value of measurement.

So we can't say given the total energy so much is K.E and so much is P.E.

In arbitrary state $|\Psi\rangle$.

$\langle \text{K.E} \rangle$ or $\langle \hat{p}^2 \rangle$ is given by

$$\langle \Psi | \hat{p}^2 | \Psi \rangle = \langle \Psi | \hat{p}^\dagger \hat{p} | \Psi \rangle$$

(as $\hat{p}$ is hermitian ($\hat{p} = \hat{p}^\dagger$), ($\hat{p}^2 = \hat{p}^\dagger \hat{p}$))

$$= \|\hat{p}|\Psi\rangle\|^2 \geq 0$$

(length of vector)

so $\langle \text{K.E} \rangle$ is always a positive real no. i.e. $\langle \text{K.E} \rangle \geq 0$.

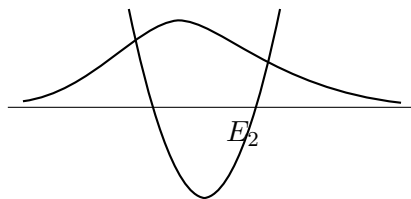
so we can't have total energy lesser than P.E because that would imply $\langle \text{K.E} \rangle < 0$.

"A particle can not have definite total energy when it is at particular place. because $[\hat{H}, \hat{x}] \neq 0$. also,

$[\hat{H}, \hat{p}] \neq 0 \Rightarrow$ A particle can't have definite Energy for a given momentum eigen value.

$(E_2) (E_2 > V(x)) :$

The wave function looks like this :-

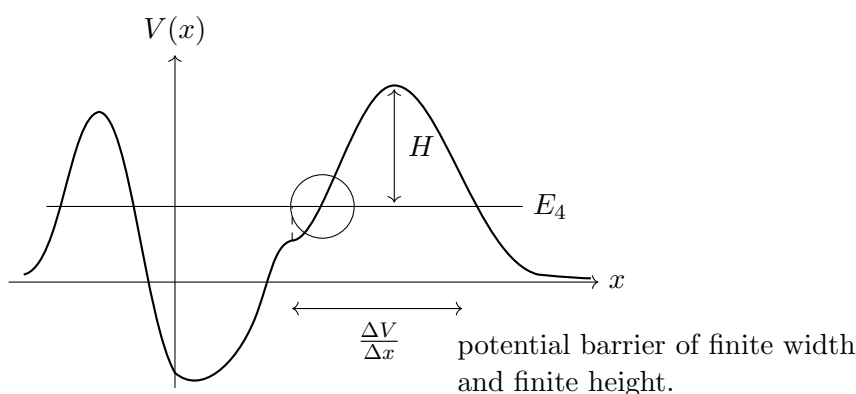


Most of the time we will find particle inside the well, but there is a finite non-zero probability that it can exist outside the well (tunneling).

$(E_3) \& (E_4) :-$

can we have such energy value / level?

Not strictly, These can not be strict stationary states or bound states. Because of the finite width of potential barrier (ΔV), there exists finite probability of the particle to exist outside the barrier / (tunnel through barrier).



so $\exists \int |\Psi(x)|^2 > 0$ for $x >$ barrier width.

so there can't be fixed / bound state.

E_4) If we place particle in such potential. initially. As time goes along this state would evolve under such $\hat{H}$, such that we end up having finite probability to find it outside the well.

Note :-

If we place particle near origin well. and iff we have very large width and height; then we have tunneling probability to be exceedingly low. And we might find the particle inside this well for very – very long time but eventually it is *bound to escape*. Such a state is called metastable state.

Note :- (probability of tunneling)

”Probability of tunneling becomes zero iff the barrier has finite width and infinite height”

$$\lim_{H \rightarrow \infty} (\text{Tunneling Probability}) \rightarrow 0$$

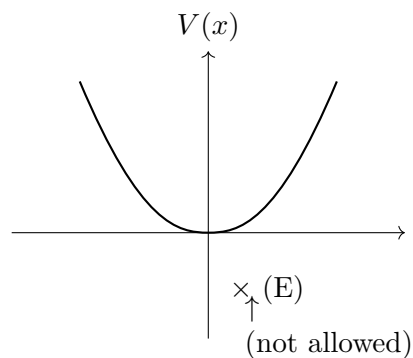
Suppose we have delta fun ($\delta(x)$) barrier, could particle tunnel through it?

Yes because width is infinitesimally small and height is infinite such that their product is finite.

If we have width of barrier infinite from either side, then we can't have E_2 energy level. But still it should have finite probability to tunnel to nearby finite barrier wells. (?) (?) (?)

Question :- What happens if potential has certain symmetry ?

How it is reflected in the eigen states / eigen functions of $\hat{H}$.



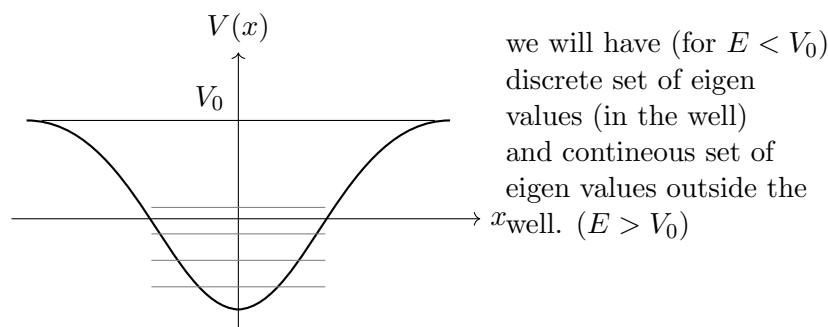
$V(x) \neq V(-x)$ say,

So what kind of eigen states or energy eigen value you would expect?

($E < 0$ not allowed)

$E > 0$ allowed. for (stationary / eigen states of $\hat{H}$) and discrete set of energy eigen values.

For we such potential



For $E > V_0$ eigen functions are non – normalizable.

Each eigen function must obey :-

$$-\frac{\hbar^2}{2m} \frac{d^2 \phi}{dx^2} + V(x) |\phi(x)\rangle = E |\phi(x)\rangle$$

let $x' = -x$, or ($x = -x'$)

eqⁿ becomes.

$$-\frac{\hbar^2}{2m} \frac{d^2 (\phi(x'))}{dx'^2} + V(-x') \phi(-x') = E \phi(-x')$$

that would be proved $V(-x') = V(x')$ (or odd)

$$\frac{-\hbar^2}{2m} \frac{d^2}{dx'^2} |\phi(-x')\rangle + V(x') |\phi(-x')\rangle = E |\phi(-x')\rangle$$

$\Rightarrow$ if $\phi(x)$ is solution with corresponding eigen value E so is $\phi(-x)$.

"since there is no degeneracy in 1-D, so every eigen value should have unique eigen fun / state.

$$\Rightarrow \phi(-x) = c \phi(x)$$

$\{|\Psi\rangle \text{ \& } \alpha|\Psi\rangle$ are same thing in Q.M

or

$$\phi(x) = c \phi(-x)$$

$$\Rightarrow \phi(x) = c(c(\phi(x)))$$

$$\phi(x) [1 - c^2] = 0 \Rightarrow \boxed{c = \pm 1}$$

so,

$$\phi(-x) = \pm \phi(x)$$

$$\phi(-x) = \begin{cases} \phi(x), & \text{a even function} \\ -\phi(x), & \phi(x) \text{ is odd function} \end{cases}$$

If potential has symmetry (reflection / mirror symmetry) / Parity invariance then solutions have definite parity. (either odd / even parity).

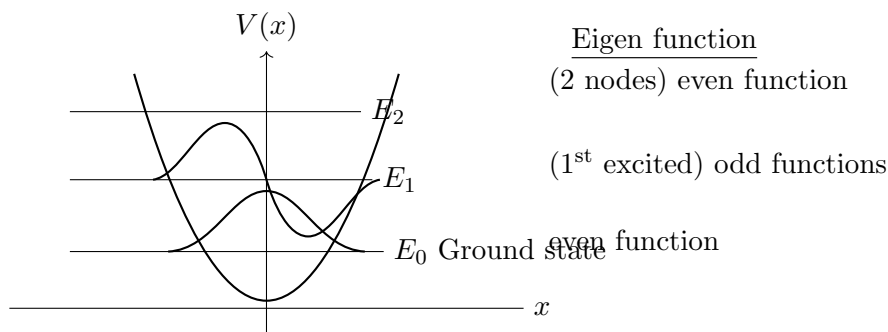
If potential is even function, wave function must be even or odd, but it can't be a mixed function.

$$[\hat{H}, \hat{P}] = 0 \quad \hat{P} = \text{Parity operator}$$

What would the ground state be? (even or odd)?

Even because, if it were odd, wave fun must be continuous and because it is an odd fun it must have zero value at $x = 0$, so it means it has node. Ground state does not have node.

so ground state will be even function.



$$[\hat{H}, \hat{P}] = 0$$

$\Rightarrow$ All eigen functions of $\hat{H}$ are eigen functions of parity operator.

But converse is not true (All odd/even functions are not eigen fun. of this particular $\hat{H}$).
(simultaneously).

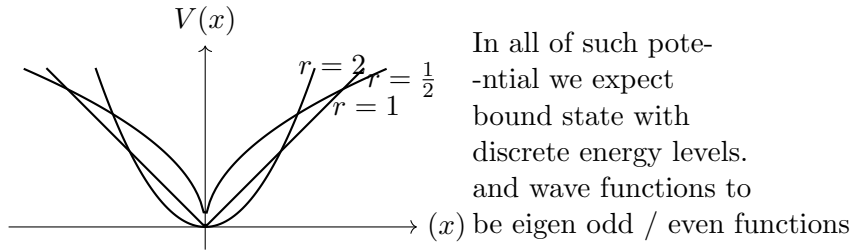
eigen fun. of $\hat{H}$ are subset of eigen fun. of Parity operator.

Question :- what kind of eigen functions / eigen values we expect?

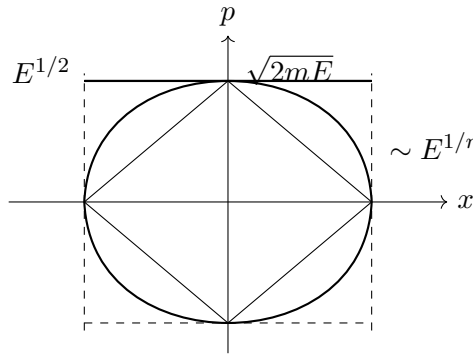
let the general

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x) = \frac{\hat{p}^2}{2m} + \lambda|x|^r$$

$\hookrightarrow$ for it to be ($V(x)$ even fun.)



classical (phase trajectories looks like)



lets find out semiclassically.

lets find,

$$\oint P dx = nh$$

so, area goes like

$$E^{\frac{1}{2} + \frac{1}{r}} \in \left(E^{\frac{1}{2}} \cdot E^{\frac{1}{r}} \right)$$

$$\oint P dx \sim E^{\frac{r+2}{2r}} \sim nh$$

$$E^{\frac{r+2}{2r}} \sim n$$

$$E_n \sim n^{\frac{2r}{r+2}}$$

for large quantum no (n) we expect results of quantum theory to be matched with semiclassical result.

for $r = 2$

$$E_n \sim n\hbar\omega$$

so we expect that for $n \gg 1$ energy levels are proportional to 'n' itself
for $r = 1$

$$E_n = n^{2/3}$$

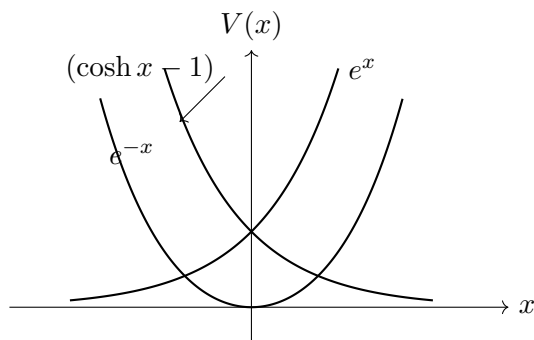
and

as, $r \rightarrow \infty$

$$\underline{E_n \propto n^2}$$

(This is what happens in 1-D well). with ∞ potential barriers

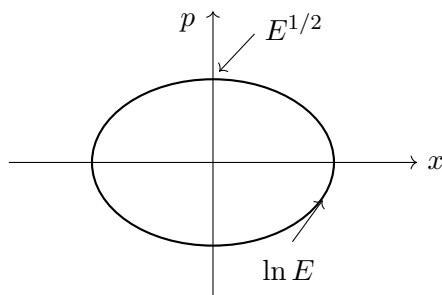
what would happen if $V(x) = \lambda(\cosh x - 1)$



$$[\cosh x = \frac{e^x + e^{-x}}{2}]$$

So how does energy values should go like.

E_n must increase slower than n^2 as $n \uparrow$ and faster than $(n^{2-\epsilon})$ for arbitrary small ϵ .



$$\frac{p^2}{2m} + \lambda(\cosh x - 1) = E$$

$$e^{x_0} \gg \frac{p^2}{2m}$$

$$E \sim e^{x_0}$$

$$(x_0 \sim \ln E)$$

[(struck out) [?]]

$$\oint P dx = nh$$

$$E^{\frac{1}{2}} \ln E \sim n \quad \text{---(1) (transcendental eqⁿ)}$$

$$\frac{1}{2} \ln E + \ln(\ln E) = \ln n$$

but $\ln(\ln E)$ is much smaller

so

$$\frac{1}{2} \ln E \simeq \ln n$$

$$\Rightarrow E \simeq n^2$$

$$E^{1/2} \ln n^2 \sim n$$

using first root in eq (1)

$$E^{\frac{1}{2}} \ln(n^2) \sim n$$

$$E^{1/2} \ln n \sim n$$

(Ignoring factor of 2)

$$E^{\frac{1}{2}} \sim \frac{n}{\ln n}$$

$$E_n = \frac{n^2}{(\ln n)^2}$$

$$\Rightarrow \boxed{E \sim \frac{n^2}{(\ln n)^2}}$$

Finally lets solve

Linear Harmonic Oscillator

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

$$[\hat{X}, \hat{P}] = i\hbar$$

In position basis $\hat{p}$ is differential operator $(-i\hbar \frac{d}{dx})$

Note :-

"The energy eigen values are independent of what basis we choose.

Eigen values & corresponding eigen states are independent of the basis we choose to solve the problem."

Note :- It is not possible to write $\hat{x}, \hat{p}$ in terms of finite-dimensional matrices. as we can see, $(xp - px = i\hbar I)$

$$\text{Tr}(xp - px) = i\hbar \text{Tr}(I)$$

$$0 \neq i\hbar N$$

[N = matrix dimension]

"we need *infinite dimensions* to represent operators if we choose to use matrix representation."

For use of differential operator then there is no questions of matrices & traces.

For eigen value of energy E , & eigen state $|\phi(x)\rangle$

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} |\phi(x)\rangle + \frac{1}{2} m \omega^2 x^2 |\phi(x)\rangle = E |\phi(x)\rangle \quad \text{---(1)}$$

we already know that $E \geq 0$

using boundary condⁿ

$$\phi(x) \longrightarrow 0 \quad \text{as} \quad x \rightarrow \pm\infty$$

after that

$$\int_{-\infty}^{\infty} |\phi(x)|^2 dx < \infty$$

redefine $|\phi(x)\rangle$ such that

$$\int_{-\infty}^{\infty} |\phi(x)|^2 dx = 1$$

since eqⁿ (1) is non-linear in ' x ', principle of superposition does not imply here, so we have to solve it using different method.

Eq. (1) is a kind of Hermite equation.

Dirac's Approach

Paul dirac broke the non-linear $\hat{H}$ into two linear operators (which commute with each other).

[1st order factors]

$$\hat{H} = \left[\frac{-\hbar^2 \nabla^2}{2m} + \frac{1}{2} m \omega^2 x^2 \right] = A \cdot B = \text{two first order factors.}$$

such that $\hat{H} \phi(x) = E \phi(x)$

(How to factorize $x^2 + p^2 \Rightarrow (x + ip)(x - ip)$ thats what exactly dirac did.).

[$x + ip$ is not allowed because of diamencional reasons. so we have to make it dimensionless.]

(1)

1st step is to get rid of ' $i\hbar$ ' in below eqⁿ.

$$[\hat{x}, \hat{p}] = i\hbar \mathbb{K}$$

parameters in problem are $m, \omega, \hbar$

so quantity of length can be written as

$$[L] = \sqrt{\frac{\hbar}{m\omega}} = m^{-1/2} \omega^{-1/2} \hbar^{1/2}$$

$$[P] = \hbar^{1/2} m^{1/2} \omega^{1/2} \quad \left(P \propto \sqrt{\hbar m \omega} \right)$$

Note :- we have to make factors diamentionless, as ' $x + ip$ ' for ' $x - ip$ ' is not allowed

[↗ diamentionless]

lets define operator

$$(a) = \left(\frac{x}{\sqrt{2\hbar/m\omega}} + \frac{i p}{\sqrt{2m\omega\hbar}} \right)$$

and

$$a^\dagger = \frac{x}{\sqrt{2\hbar/m\omega}} - \frac{i p}{\sqrt{2m\omega\hbar}}$$

such that / now

$$\begin{aligned} a \cdot a^\dagger &= \frac{x^2}{(2\hbar/m\omega)} - \frac{i^2 p^2}{2m\omega\hbar} - \frac{i(i\hbar)\mathbb{K}}{2\hbar} \\ &= \frac{1}{2} \frac{m\omega x^2}{\hbar} + \frac{p^2}{2m(\omega\hbar)} + \frac{1}{2}\mathbb{K} \\ &= \left(\frac{1}{\omega\hbar} \right) \left[\frac{1}{2} m\omega^2 x^2 + \frac{p^2}{2m} \right] + \frac{1}{2} I \end{aligned}$$

since $a \neq a^\dagger$, a is not hermitian operator

so its not a physical observable.

similarly

$$a^\dagger a = \frac{x^2}{2\hbar/m\omega} + \frac{i(xp - px)}{2\hbar} + \frac{p^2}{2m\omega\hbar} = (\hat{H}) \frac{1}{\omega\hbar} - \frac{1}{2}(\mathbb{K})$$

what is

$$\begin{aligned} [a, a^\dagger] &= aa^\dagger - a^\dagger a = \mathbb{K} \\ H &= \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2 = \left(aa^\dagger - \frac{1}{2}\mathbb{K} \right) \omega\hbar \end{aligned}$$

[$\rightarrow$ unit operator]

$$\begin{aligned} &= \left(a^\dagger a + \frac{\mathbb{K}}{2} \right) \omega\hbar \\ H &= \left(a^\dagger a + \frac{1}{2}\mathbb{K} \right) \omega\hbar \end{aligned}$$

[Heisenberg algebra: $[a, a] = 0 = [a^\dagger, a^\dagger]$; $[a^\dagger, a] = I$]

" $a^\dagger a$ is not finite diamentional." as

$$\text{Trace} \left(aa^\dagger - a^\dagger a \right) \neq \text{Trace}(I).$$

Dirac's solution

Define $a^\dagger a = N$ (number operator has eigen values will turn out to be $0, 1, 2 \dots$ Natural numbers)

$$[N, a] = [a^\dagger a, a]$$

$$\left\{ \begin{aligned} [A, BC] &= [A, B]C + B[A, C] \\ [AB, C] &= [A, C]B + A[B, C] \end{aligned} \right\} \text{commutator rules}$$

so, $[N, a]$ becomes

$$= [a^\dagger a, a]$$

$$\begin{aligned}
&= [a^\dagger, a]a + a^\dagger \underbrace{[a, a]}_0 \\
&= -a
\end{aligned}$$

similarly

$$[N, a^\dagger] = [a^\dagger a, a^\dagger] = \underbrace{[a^\dagger, a^\dagger]}_0 a + a^\dagger [a, a^\dagger] = +a^\dagger$$

Now our job is to find eigen-states of

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger a + \frac{1}{2} \cdot I \right)$$

since eigen states of $\hat{H}$ is trivial (any fun. is eigen-state)

we have to look for eigen states of ' $a^\dagger a$ '.

"All eigen-states of $a^\dagger a$ will be eigen states of $\hat{H}$ "

Let, $|\lambda\rangle$ be eigen state of N ($a^\dagger a$) with eigen value λ .

$$N|\lambda\rangle = \lambda|\lambda\rangle$$

$\therefore$ ' λ ' has to be real as $a^\dagger a$ is hermitian.

we know, $[N, a] = -a$

so, lets apply to same eigen state $|\lambda\rangle$

$$\begin{aligned}
[N, a]|\lambda\rangle &= -a|\lambda\rangle \\
\Rightarrow Na|\lambda\rangle - aN|\lambda\rangle &= -a|\lambda\rangle \\
\Rightarrow N(a|\lambda\rangle) - a\lambda|\lambda\rangle &= -a|\lambda\rangle
\end{aligned}$$

$[\hat{a}$ is linear operator $a\lambda|\lambda\rangle = \lambda(a|\lambda\rangle)]$

$$\Rightarrow N(\hat{a}|\lambda\rangle) = (\lambda - 1)(\hat{a}|\lambda\rangle)$$

This implies that if $|\lambda\rangle$ is eigen state of $\hat{N}$. so is the $\hat{a}|\lambda\rangle$ an eigen state of $\hat{N}$. but with eigen value reduced by unity.

exactly same way

$$\begin{aligned}
[N, a^\dagger]|\lambda\rangle &= a^\dagger|\lambda\rangle \\
\Rightarrow N(a^\dagger|\lambda\rangle) &= (\lambda + 1)(a^\dagger|\lambda\rangle)
\end{aligned}$$

if $|\lambda\rangle$ is eigen state of $\hat{N}$, so is the $a^\dagger|\lambda\rangle$ an eigen state of $\hat{N}$, with eigen value increased by 1.

so, $a^\dagger$ is called raising operator,

and a is called as lowering operator.

suppose we have state (eigen) $|\lambda\rangle$ of operator $\hat{N}$. lets see about $a|\lambda\rangle$ state, then $a|\lambda\rangle$ state is also an eigen state of $\hat{N}$ but with reduced eigen value by 1.

eigen state of $\hat{N}^$	eigen value of state	Energy
$ \lambda\rangle$	λ	$\hbar\omega(\lambda)$
$a \lambda\rangle$	$\lambda - 1$	$\hbar\omega(\lambda - 1)$
$a^2 \lambda\rangle$	$\lambda - 2$	$\hbar\omega(\lambda - 2)$
$\vdots$	$\vdots \geq 0$	$\vdots \geq 0$

eventually $(a^n|\lambda\rangle) = \lambda - n$ can hit negative.

But $\hat{N}$ can not have negative eigen values.

$$\begin{aligned}\langle N \rangle &= \langle \Psi | N | \Psi \rangle = \langle \Psi | a^\dagger a | \Psi \rangle \\ &= \|(a|\Psi)\rangle\| \geq 0\end{aligned}$$

after all, $\langle N \rangle =$ arithmetic avg of all eigen values.

so no eigen values should be negative.

so, λ has to be positive integers.

$$\lambda \in I^+$$

(or)

$$\begin{aligned}(\lambda = n) \\ (n = 0, 1, 2 \cdots \infty)\end{aligned}$$

$\exists$ a state $|0\rangle$ such that,

$$a|0\rangle = 0$$

so there must be a state with eigen values zero such that $a|0\rangle = 0$. and we can't lower the eigen states & values further.

' $|0\rangle$ ' is called ground state. in Q.M

———— vacuum in Q.F.T.

and anihilation operator a anihilates the vaccum.

$$(a|0\rangle = 0.)$$

$|0\rangle$ is not a null vector here. it is a eigen state $|\phi_0(x)\rangle$ whose eigen value is zero.

$$a|\phi_0(x)\rangle = 0.$$

To Find eigen states $|n\rangle$.

[let stationary states of $\hat{H}$ is represented as $|n\rangle$.]

$$a^\dagger a |n\rangle = n |n\rangle$$

$$\hat{H}|n\rangle = \hbar\omega \left(N + \frac{1}{2}I \right) |n\rangle = \boxed{\hbar\omega \left(n + \frac{1}{2} \right)} |n\rangle$$

so $\hbar\omega \left(n + \frac{1}{2} \right)$ are energy levels.

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right) \quad n = 0, 1, 2 \cdots \infty$$

Ground state

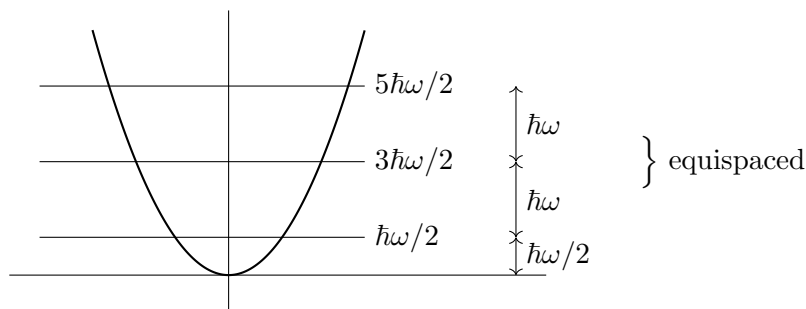
$$E_0 = \frac{\hbar\omega}{2}$$

1st excited state,

$$E_1 = \frac{3}{2}\hbar\omega$$

2nd excited state,

$$E_2 = \frac{5}{2}\hbar\omega.$$



lets say we have find ground state $|0\rangle$, which is normalized. ($\langle 0|0\rangle = 1$)

lets find 1st excited state.

$$|1\rangle = a^\dagger |0\rangle \quad (\text{raising operator})$$

lets put normalization const. c .

$$|1\rangle = c a^\dagger |0\rangle$$

In general.

$$|n+1\rangle = c_n a^\dagger |n\rangle \quad \langle n+1| = c_n^* \langle n| a$$

$$\langle n+1|n+1\rangle = 1 = |c_n|^2 \langle n| a a^\dagger |n\rangle$$

$$(a a^\dagger - a^\dagger a = \mathbb{I})$$

$$a a^\dagger - N = I \Rightarrow (a a^\dagger = N + I)$$

$$\langle n+1|n+1\rangle = 1 = |c_n|^2 \langle n|(N+I)|n\rangle$$

[assuming $\langle n|n\rangle = 1$]

$$= |c_n|^2 (n+1) \langle n|n\rangle$$

$$\Rightarrow c_n = \frac{1}{\sqrt{n+1}}$$

so

$$|n+1\rangle = c_n a^\dagger |n\rangle = \frac{1}{\sqrt{n+1}} a^\dagger |n\rangle$$

$$\Rightarrow a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

In exactly same way,

$$|n-1\rangle = d_n a |n\rangle$$

$$\begin{aligned}
1 &= \langle n-1 | n-1 \rangle = |d_n|^2 \langle n | a^\dagger a | n \rangle \\
&= |d_n|^2 \cdot n \langle n | n \rangle \\
1 &= |d_n|^2 n \\
d_n &= \frac{1}{\sqrt{n}} \\
|n-1\rangle &= \frac{1}{\sqrt{n}} a |n\rangle \\
a |n\rangle &= \sqrt{n} |n-1\rangle
\end{aligned}$$

for $n=0$, $a|0\rangle=0$ and there is no state $|-1\rangle$

lets see how the wave function looks like.

$$\begin{aligned}
\hat{a}|0\rangle &= 0 \\
\hat{a} &= \left(\frac{x}{\sqrt{2\hbar/m\omega}} + \frac{i p}{\sqrt{2m\omega\hbar}} \right) |0\rangle = 0 \\
\Rightarrow (m\omega x + ip)|0\rangle &= 0. \\
\phi_0(x) &= \langle x|0\rangle
\end{aligned}$$

representation of ground state in position basis.

so

$$\begin{aligned}
\langle x | m\omega x + ip | 0 \rangle &= 0 \\
\Rightarrow m\omega \langle x | x + i \left(-i\hbar \frac{d}{dx} \right) | 0 \rangle &= 0 \quad \Rightarrow m\omega x \langle x | 0 \rangle + \hbar \frac{d}{dx} \langle x | 0 \rangle = 0 \\
\Rightarrow \hbar \frac{d}{dx} \phi_0(x) + m\omega x \phi_0(x) &= 0 \\
\phi_0(x) &= A_0 e^{-\frac{m\omega x^2}{2\hbar}} \\
\boxed{|0\rangle \text{ or } \phi_0(x) = A_0 e^{-\frac{m\omega x^2}{2\hbar}}}
\end{aligned}$$

} no nodes, gaussian

Now, ($A_0=?$)

$$\begin{aligned}
\int_{-\infty}^{\infty} |\phi_0(x)|^2 dx &= A_0^2 \sqrt{\frac{\pi (2\hbar^2)}{m\omega}} = 1 \quad (\text{normalization}) \\
A &= \left(\frac{m\omega}{2\pi\hbar^2} \right)^{1/4} \\
\boxed{\phi_0(x) = \left(\frac{m\omega}{2\pi\hbar^2} \right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}}}
\end{aligned}$$

Now lets find $|1\rangle$.

$$\begin{aligned}
|1\rangle &= \frac{1}{\sqrt{1}} a^\dagger |0\rangle \\
\langle x | 1 \rangle = \phi_1(x) &= \frac{1}{\sqrt{1}} \langle x | a^\dagger | 0 \rangle
\end{aligned}$$

$$= \langle x \left| \frac{x}{\sqrt{2\hbar/m\omega}} - \frac{ip}{\sqrt{2m\omega\hbar}} \right| 0 \rangle$$

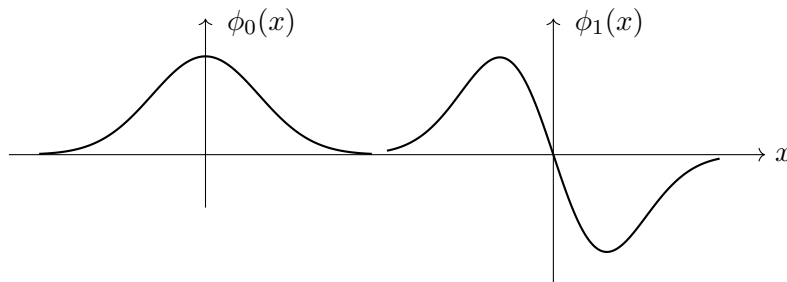
$$\phi_1(x) = \sqrt{\frac{2\hbar}{m\omega}} x \phi_0(x) - \frac{\hbar}{\sqrt{2m\omega\hbar}} \frac{d}{dx} \phi_0(x)$$

$$[|n\rangle = \frac{1}{\sqrt{n}} a^\dagger |n-1\rangle]$$

(We don't have to normalize it again, it is previously normalized)

since we have $\phi_0(x)$, we insert it in above eqⁿ and get *normalized* $\phi_1(x)$.

we can see that $\phi_0(x)$ is even function, and $\phi_1(x)$ is odd function.



$$\boxed{\phi_n(x) = A_n H_n \left(x \sqrt{\frac{m\omega}{2\hbar}} \right) e^{-\frac{m\omega x^2}{2\hbar}}}$$

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right) \quad (n = 0, 1, 2 \dots \infty).$$

11 Lecture 12

Wavefunctions of Harmonic oscillator (Hermite polynomials)

Wavefunctions of Harmonic oscillator are expressed as

$$\phi_n(x) = A_n H_n \left(x \sqrt{\frac{m\omega}{2\hbar}} \right) e^{-\frac{m\omega x^2}{2\hbar}}$$

$$[\text{Let } x_0 = \sqrt{\frac{2\hbar}{m\omega}}]$$

$$= A_n e^{-\frac{x^2}{x_0^2}} H_n \left(\frac{x}{x_0} \right)$$

where,

$$\begin{cases} H_n(x) = \text{Hermite polynomial of order 'n'}. \\ H_0(x) = 1 \\ H_1(x) = 2x \\ \vdots \\ H_n(x) = \begin{cases} \text{odd powers of 'x'} & n = \text{odd} \\ \text{even powers of 'x'}, & n = \text{even} \end{cases} \end{cases}$$

It is family of orthogonal functions. (Mutually orthogonal)

{condⁿ for orthogonality

$$\int_{-\infty}^{\infty} dx e^{-x^2} H_n(x) H_m(x) = (\text{---}) \delta_{nm}$$

$\hookrightarrow$ weight factor.

{Rodrigue's formula} where,

$$H_n(x) = e^{+x^2} \cdot \left(\frac{-d}{dx} \right)^n e^{-x^2}$$

$$H_0(x) = e^{+x^2} e^{-x^2} = 1$$

$$H_1(x) = e^{+x^2} \left(\frac{-d}{dx} \right) e^{-x^2} = 2x e^0 = 2x$$

$$\begin{aligned} H_2(x) &= e^{+x^2} \left(\frac{-d}{dx} \right) \left(\frac{-d}{dx} \right) e^{-x^2} = e^{+x^2} \left[\frac{-d}{dx} (e^{-x^2} \cdot 2x) \right] \\ &= e^{+x^2} e^{-x^2} \cdot 2x (2x + x) \\ &= 4x(x+1) \quad 4x^2 + 2 \\ &\quad 4x^2 - 2 \end{aligned}$$

[the expressions $4x(x+1)$ and $4x^2+2$ are struck out in the notebook; the answer written below them is $4x^2-2$]

Generating function :

$$e^{2tx-t^2} = \sum_{n=0}^{\infty} \frac{H_n(x) t^n}{n!}$$

Generating functions generates power of 't' whose coefficients are precisely the $\left(\frac{\text{Hermite polynomials}}{n!} \right)$.

One more thing $\rightarrow$

Hermite polynomials are regular solutions of the below differential equations.

Recurrence formula }

$$\frac{d^2 H_n}{dx^2} - 2x \frac{dH_n}{dx} + 2n H_n = 0$$

we can solve it using power series solution.

Note :-

The Hamiltonian of Harmonic oscillator

$$H = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2$$

This equation is symmetric in both $\hat{x}$ & $\hat{p}$.

In position basis :- $(\hat{p} = -i\hbar \frac{\partial}{\partial x})$, $\hat{x} = \hat{x}$

$$\hat{H} = \left(\frac{-\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 \hat{x}^2 \right)$$

In momentum basis :- $\hat{x} = i\hbar \frac{\partial}{\partial p}$, $\hat{p} = \hat{p}$

$$\hat{H} = \left(\frac{\hat{p}^2}{2m} - \frac{1}{2}m\omega^2\hbar^2 \frac{d^2}{dp^2} \right)$$

So the solution of wave fun./eigen states in momentum basis are also going to be $(H_n(p) \cdot e^{-p^2/2m})$.

$$\tilde{\phi}_n(p) = B_n e^{-\frac{p^2}{2m\omega\hbar}} \cdot H_n\left(\frac{p}{\sqrt{2m\omega\hbar}}\right)$$

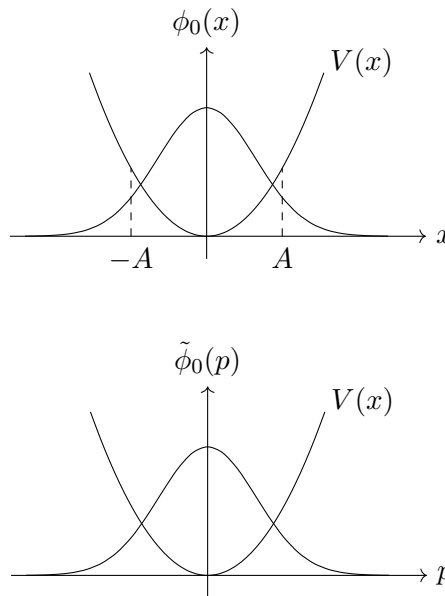
Ground state of Harmonic oscillator, ladder operators & uncertainty

The ground state of Harmonic oscillator is gaussian, both in $\{|x\rangle\}$ basis and $\{|p\rangle\}$ basis.

$$\tilde{\phi}_0(p) = B_0 e^{-\frac{p^2}{2m\omega\hbar}}$$

$$\phi_0(x) = A_0 e^{-\frac{x^2 m\omega}{2\hbar}}$$

also we can see both are fourier transform of each other.



Note :- classically particle does to and fro motion b/w $(A, -A)$.

But q.mechanics says $\exists$ non zero finite probability to find it anywhere on x -axis.

let us find $(\Delta x)_n = \langle (x - \langle x \rangle)^2 \rangle_n^{1/2}$

first, let me find out, $\langle n|a|n \rangle$

$\{|n\rangle = n^{\text{th}}$ eigen state of H .

where

$$a = x\sqrt{\frac{m\omega}{2\hbar}} + ip\left(\frac{1}{\sqrt{2m\omega\hbar}}\right)$$

$$a^+ = x\sqrt{\frac{m\omega}{2\hbar}} - i\frac{p}{\sqrt{2m\omega\hbar}}$$

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right), \quad H = \hbar\omega \left(a^+a + \frac{I}{2}\right)$$

also we know $a^+a|n\rangle = n|n\rangle$; $a^+a = N$ (Number operator)

we also know that,

$$a|n\rangle = \sqrt{n}|n-1\rangle$$

$$a^+|n\rangle = \sqrt{n+1}|n+1\rangle$$

{and all $|n\rangle$ are orthonormal.

Given that, what is $\langle n|a|n\rangle$.

$$\langle n|\sqrt{n}|n-1\rangle = \sqrt{n}\langle n|n-1\rangle = 0$$

similarly, $\langle n|a^+|n\rangle = 0$

we can represent $a, a^+, \hat{x}, \hat{p}$ as ∞ -dimensional matrices.

{So what looks like $-i\hbar\frac{d}{dx}$ operator in one basis, same operator looks simply p in another basis, & same operator in energy basis / no. operator basis looks like ∞ -dimensional matrices. So These are abstract operator which wear many clothes.

since matrix element of a in $\{|n\rangle\}$ basis can be written as $a_{ij} = \langle i|a|j\rangle$ or $\langle n|a|m\rangle = a_{nm}$

since

$$\left. \begin{array}{l} \langle n|a|n\rangle = 0 \\ \langle n|a^+|n\rangle = 0 \end{array} \right\} \Rightarrow \text{diagonal elements of } a, a^+ \text{ are zero so is the true for } \hat{x}, \hat{p} \text{ too}$$

as

$$a = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} + \frac{i\hat{p}}{\sqrt{2m\omega\hbar}}$$

$$a^+ = \text{---} - \frac{i\hat{p}}{\sqrt{2m\omega\hbar}}$$

$$\hat{x} = \frac{a + a^+}{2\sqrt{\frac{m\omega}{2\hbar}}}, \quad \hat{p} = \left(\frac{a - a^+}{i2}\right) \sqrt{2m\omega\hbar}$$

also don't have diagonal elements.

lets choose units such that $(m, \omega, \hbar = 1)$

$$a = \frac{x + ip}{\sqrt{2}}$$

$$a^+ = \frac{x - ip}{\sqrt{2}}$$

$$\Rightarrow x = \frac{a + a^+}{\sqrt{2}}$$

$$p = \frac{a - a^+}{i\sqrt{2}}$$

since a, a^+ are not hermitian.

but, x, p are hermitian operators, $\left. \begin{array}{l} x^+ = x, \\ p^+ = p \end{array} \right\}$

↓

(physical operators)

lets find

$$\left. \begin{array}{l} \langle n|x|n\rangle = 0 = \langle x\rangle \\ \langle n|p|n\rangle = 0 = \langle p\rangle \end{array} \right\} \text{ diagonal elements are zero.}$$

$$\begin{aligned} \langle x^2 \rangle &\Rightarrow \langle n|x^2|n\rangle = \left\langle n \left| \frac{(a + a^+)^2}{2} \right| n \right\rangle \\ &= \left\langle n \left| \frac{a^2 + aa^+ + a^+a + a^{+2}}{2} \right| n \right\rangle \\ &= \left\langle n \left| \frac{a^2}{2} \right| n \right\rangle + \left\langle n \left| \frac{aa^+}{2} \right| n \right\rangle + \left\langle n \left| \frac{a^+a}{2} \right| n \right\rangle + \left\langle n \left| \frac{a^{+2}}{2} \right| n \right\rangle \\ &= \frac{n}{2} \langle n|n\rangle + \left\langle n \left| \frac{aa^+}{2} \right| n \right\rangle \\ &= \frac{n}{2} + \left\langle n \left| \frac{I + a^+a}{2} \right| n \right\rangle \\ &= \frac{n}{2} + \frac{n}{2} + \frac{1}{2} = n + \frac{1}{2} \end{aligned}$$

Now putting $m, \omega, \hbar$,

$$\langle x^2 \rangle = \left(n + \frac{1}{2}\right) \frac{\hbar}{m\omega}$$

$$\begin{aligned} \langle p^2 \rangle &= \langle n|p^2|n\rangle \\ &= \left\langle n \left| \frac{(a - a^+)^2}{2(i)^2} \right| n \right\rangle \\ &= \left\langle n \left| \frac{a^2 - aa^+ - a^+a + a^{+2}}{(-2)} \right| n \right\rangle \\ &= \left(\frac{-1}{2}\right) \langle n|a^2|n\rangle + \frac{1}{2} \langle n|aa^+|n\rangle + \frac{1}{2} \langle n|a^+a|n\rangle + \frac{1}{2} \langle n|a^{+2}|n\rangle \\ &= \left(n + \frac{1}{2}\right) \end{aligned}$$

$$\left\{ \begin{array}{l} \langle x \rangle = \langle n|\hat{x}|n\rangle = 0 \\ \langle p \rangle = \langle n|\hat{p}|n\rangle = 0 \\ \hat{x} = \frac{a + a^+}{\sqrt{2}} \\ \hat{p} = \frac{a - a^+}{\sqrt{2}i} \end{array} \right.$$

putting back $(m, \omega, \hbar)$

$$\langle p^2 \rangle = \left(n + \frac{1}{2}\right) \cdot m\omega\hbar$$

$$\begin{aligned}\Delta x &= \langle (x - \langle x \rangle)^2 \rangle^{1/2} = \sqrt{\langle x^2 \rangle - 2x\langle x \rangle + \langle x \rangle^2} \\ &= \sqrt{\langle x^2 \rangle - 2\langle x \rangle^2 + \langle x \rangle^2} \\ &= \sqrt{\langle x^2 \rangle - \langle x \rangle^2}\end{aligned}$$

$$(\Delta x)_n (\Delta p)_n = \sqrt{\left(n + \frac{1}{2}\right) \frac{\hbar}{m\omega}} \cdot \sqrt{\left(n + \frac{1}{2}\right) m\omega\hbar} = \left(n + \frac{1}{2}\right) \hbar$$

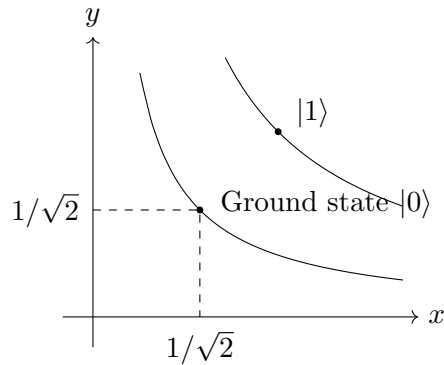
In ground state, $n = 0$;

$$(\Delta x)_0 (\Delta p)_0 = \left(0 + \frac{1}{2}\right) \hbar = \frac{\hbar}{2}$$

also, uncertainty principle says. $\Delta x \Delta p \geq \frac{\hbar}{2}$

”which implies ground state of Harmonic oscillator is minimum uncertainty state.”

If we plot $\left(\frac{\Delta x}{\sqrt{\hbar/m\omega}} \rightarrow x\right) \left(\frac{\Delta p}{\sqrt{m\omega\hbar}} \rightarrow y\right)$



Number operator states $|n\rangle$ are called – folk space states.

”Important thing in Harmonic oscillator is that energy levels are equally spaced. Therefore when we quantize fields (including E-m field), it turns out every time we add some energy, it is added in quantum / const amt of energy.

Eigen states of $a, a^\dagger$:- coherent states

Question

What are eigen-states of $a, a^\dagger$. (?)

As these operators are not hermitian, we do not expect real eigen values. (may be complex).

$$\begin{aligned}a^+ &= \frac{x - ip}{\sqrt{2}} \xrightarrow{\text{in } \{|x\rangle \text{ basis}\}} \frac{1}{\sqrt{2}} \left(x + i^2 \frac{d}{dx} \right) \\ &\quad \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right)\end{aligned}$$

lets say $\chi(x)$ be eigen state of a^+ .

$$a^+ |\chi(x)\rangle = \lambda |\chi(x)\rangle$$

$$\frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right) |\chi(x)\rangle = \lambda |\chi(x)\rangle$$

Exercise {Find $\chi(x)$ }

eigen states of a^+ :-

$$a^+|\chi\rangle = \lambda|\chi\rangle$$

we know that, $|\chi\rangle = \sum_{n=0}^{\infty} c_n |n\rangle$ uniquely in $\{|n\rangle\}$ basis

$$\begin{aligned} |\chi\rangle &= c_0|0\rangle + c_1|1\rangle + c_2|2\rangle + \dots \\ a^+|\chi\rangle &= c_0a^+|0\rangle + c_1a^+|1\rangle + c_2a^+|2\rangle + \dots \\ a^+|\chi\rangle &= c_0\sqrt{1}|1\rangle + c_1\sqrt{2}|2\rangle + c_2\sqrt{3}|3\rangle + \dots \quad - (1) \end{aligned}$$

But

$$a^+|\chi\rangle = \lambda|\chi\rangle = \lambda(c_0|0\rangle + c_1|1\rangle + c_2|2\rangle + \dots) \quad (2)$$

Comparing coefficients of $|0\rangle, |1\rangle, \dots$

$$\left. \begin{aligned} c_0 &= 0 \\ \lambda c_1 &= c_0 = 0 \\ \lambda c_2 &= c_1\sqrt{2} = 0 \\ \lambda c_3 &= c_2\sqrt{3} = 0 \end{aligned} \right\} \text{ so solution is there are no eigen functions.}$$

But there are solutions to the differential eqⁿ

$$\begin{aligned} \frac{1}{\sqrt{2}} \left(x - \frac{d}{dx} \right) \chi(x) &= \lambda \chi(x) \\ \frac{d\chi}{dx} &= \frac{(x - \lambda\sqrt{2}) \chi(x)}{\sqrt{2}} \\ \Rightarrow \chi(x) &= A e^{\frac{1}{\sqrt{2}} \left(\frac{x^2}{2} - \lambda x \sqrt{2} \right)} \end{aligned}$$

So what is resolution to this ?

” $\chi(x)$ from above diff. eqⁿ is not normalizable ”.

$$\text{or, } \left(\int_{-\infty}^{\infty} A^2 e^{\sqrt{2} \left(\frac{x^2}{2} - \lambda x \sqrt{2} \right)} dx \neq \text{finite} \right)$$

eigen -states of a :-

$$\begin{aligned} a &= \frac{x + ip}{\sqrt{2}} \\ a|\alpha\rangle &= \alpha|\alpha\rangle \quad \Rightarrow \quad \frac{1}{\sqrt{2}} \left(x + \frac{d}{dx} \right) \alpha(x) = \alpha \cdot \alpha(x) \end{aligned}$$

$\uparrow \langle x|\alpha\rangle$

$$\frac{1}{\sqrt{2}} \frac{d(\alpha(x))}{dx} = \left(\lambda - \frac{x}{\sqrt{2}} \right) \alpha(x)$$

$$\int \frac{d\alpha(x)}{\alpha(x)} = \int (\sqrt{2}\lambda - x) dx$$

$$\alpha(x) = A e^{\sqrt{2}\lambda x - \frac{x^2}{2}}$$

$$|\alpha\rangle = \sum_{n=0}^{\infty} c_n |n\rangle$$

$$= c_0|0\rangle + c_1|1\rangle + c_2|2\rangle + \dots$$

$$a|\alpha\rangle = c_0 a|0\rangle + c_1 a|1\rangle + c_2 a|2\rangle + \dots$$

$$= 0 + c_1$$

$$= 0 + c_1 \sqrt{1} |0\rangle + c_2 \sqrt{2} |1\rangle + c_3 \sqrt{3} |2\rangle + \dots \quad (1)$$

$$a|\alpha\rangle = \alpha|\alpha\rangle = \alpha c_0|0\rangle + \alpha c_1|1\rangle + \alpha c_2|2\rangle + \dots \quad (2)$$

comparing coefficients from eqⁿ (1) & (2).

$$\alpha c_0 = c_1 \quad \Rightarrow \quad c_1 = \frac{\alpha c_0}{\sqrt{1}}$$

$$\alpha c_1 = c_2 \sqrt{2} \quad \Rightarrow \quad c_2 = \frac{\alpha^2}{\sqrt{1 \cdot 2}} \cdot c_0$$

$$\alpha c_2 = c_3 \sqrt{3} \quad \Rightarrow \quad c_3 = \frac{\alpha^3}{\sqrt{1 \cdot 2 \cdot 3}} c_0$$

$$c_n = \frac{\alpha^n}{\sqrt{n!}} \cdot c_0$$

So

$$|\alpha\rangle = c_0 \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

(α = any complex no.)

all no. in complex planes are eigen value of annihilation operator ($\hat{a}$). & They have coorespondng normalized states.

$$\langle \alpha | \alpha \rangle = 1 = |c_0|^2 \sum \frac{|\alpha|^{2n}}{n!} \langle n | n \rangle$$

provided $\langle n | m \rangle = \delta_{mn}$

$$1 = |c_0|^2 e^{|\alpha|^2}$$

$$c_0 = e^{-\frac{|\alpha|^2}{2}}$$

therefore, we can say

$\hat{a}$ has normalizable eigen states $\{|\alpha\rangle\}$.

$$a|\alpha\rangle = \alpha|\alpha\rangle \quad \alpha \in \mathbb{C}$$

where,

$$|\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

↓

$$|n\rangle = \frac{(a^+)^n}{\sqrt{n!}} |0\rangle \quad (? \text{ How}).$$

$$\left\{ \begin{array}{l} |1\rangle = \frac{1}{\sqrt{1}} a^+ |0\rangle \\ |2\rangle = \frac{1}{\sqrt{2}} a^+ |1\rangle = \frac{(a^+)^2 |0\rangle}{\sqrt{2 \cdot 1}} \\ \vdots \\ |n\rangle = \frac{(a^+)^n}{\sqrt{n!}} |0\rangle \end{array} \right.$$

$$|\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2} \sum_{n=0}^{\infty} \frac{(\alpha a^+)^n}{n!} |0\rangle$$

$$|\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2} \cdot e^{\alpha a^+} \cdot |0\rangle$$

$$\langle \alpha | \alpha \rangle = 1$$

These eigen states $\{|\alpha\rangle\}$ are called coherent states. (Fundamental role in quantum optics).

Every normalizable $|\Psi\rangle$ can be written as.

$$|\Psi\rangle = \sum_{n=0}^{\infty} c_n |n\rangle \quad \{c_0, c_1, c_2 \dots\}$$

(Q) Are those coherent states orthogonal to each other?

are eigen states of a . i.e. $|\alpha\rangle$ orthonormal to each other. ?

$$\langle \alpha | \beta \rangle = ?$$

$$\langle \beta | \alpha \rangle = \left(\sum_n \sum_m \frac{(\beta^*)^m}{\sqrt{m!}} \frac{\alpha^n}{\sqrt{n!}} \underbrace{\langle m | n \rangle}_{\delta_{nm}} \right) \cdot e^{-\frac{1}{2}(|\alpha|^2 + |\beta|^2)}$$

($n = m$)

$$= e^{-\frac{1}{2}(|\alpha|^2 + |\beta|^2)} \cdot \sum_n \frac{\alpha \beta^*}{n!}$$

$$= e^{-\frac{1}{2}(|\alpha|^2 + |\beta|^2)} e^{\alpha \beta^*}$$

$$\langle \beta | \alpha \rangle = e^{-\frac{1}{2}(|\alpha|^2 + |\beta|^2) + \alpha \beta^*} \neq 0.$$

(check for ($\alpha = \beta$) , ($\langle \alpha | \alpha \rangle = 1$))

So, states of a , **are not orthonormal**.

(over – completeness = $\int \frac{d^2\alpha}{\pi} |\alpha\rangle \langle \alpha| = I$)

set $\{|\alpha\rangle\}$ is an overcomplete set of states

Q.) How does $|\alpha\rangle$ look like in $|x\rangle$ basis.

$$\begin{aligned}\alpha(x) &= \langle x|\alpha\rangle = \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \langle x|n\rangle \\ &= \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \phi_n(x) \\ &= \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \left(A_n e^{-\frac{x^2}{2}} \cdot H_n(x) \right)\end{aligned}$$

12 Lecture 13

Factorization method / method of inter-twining operators

Using Dirac approach we factorized the $\hat{H}$ (which was 2nd order & non-linear) into two linear first order operators (a, a^+).

$$\begin{aligned}a &= \frac{x + ip}{\sqrt{2}} \\ a^+ &= \frac{x - ip}{\sqrt{2}}\end{aligned}$$

} in suitable units, $p = -i\hbar \frac{\partial}{\partial x}$

$$\hat{H} = \hbar\omega \left(a^+ a + \frac{I}{2} \right)$$

[which was earlier $\frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2$]

[Factorization]

This can't be done with any potential, it can be done with only few potentials. This method is called the **factorization method / method of inter-twining operators**.

matrix representation of a, a^+, x, p (in SHO) in Fock basis.

we know that,

$$\begin{aligned}a|n\rangle &= \sqrt{n}|n-1\rangle \\ a^+|n\rangle &= \sqrt{n+1}|n+1\rangle\end{aligned}$$

matrix elements :-

$$\begin{aligned}\langle \ell|a|n\rangle &= a_{\ell n} = \sqrt{n} \langle \ell|n-1\rangle = \sqrt{n} \delta_{\ell, n-1} \\ \hat{a} &= \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 & 0 & \cdots \\ 0 & 0 & \sqrt{2} & 0 & 0 & \cdots \\ 0 & 0 & 0 & \sqrt{3} & 0 & \cdots \\ 0 & 0 & 0 & 0 & \sqrt{4} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}\end{aligned}$$

$(\infty \times \infty)$ dimensional. implies.

All diagonal elements has to be zero.

$(a_{\ell n} = \sqrt{n}$ for $\ell = n - 1)$

a^+ = simply hermitian conjugate of $\hat{a}$.

$$a = x \sqrt{\frac{m\omega}{2\hbar}} + \frac{i p}{\sqrt{2m\omega\hbar}}$$

$$\hat{a}^+ = x \sqrt{\frac{m\omega}{2\hbar}} - \frac{i p}{\sqrt{2m\omega\hbar}}$$

$$x = \frac{a + a^+}{2} \left(\sqrt{\frac{2\hbar}{m\omega}} \right) = \frac{(a + a^+)}{\sqrt{2}} \sqrt{\frac{\hbar}{m\omega}}$$

$$p = \frac{(a - a^+)}{i\sqrt{2}} \sqrt{m\omega\hbar}$$

(x, p, a^+) can be represented in matrix form in same way

since

$$[a, a^+] = I = aa^+ - a^+a$$

$$\text{Trace}(I) = \text{Trace}(aa^+ - a^+a)$$

$\{\text{Trace}(AB) = \text{Trace}(BA)$ even if $[A, B] \neq 0$.

(nonzero $\neq$ zero) so we can't represent them in terms of finite dimensional matrices.

x, p are not bounded operators, their eigen values goes upto ∞ .

General uncertainty principle :-

{for two operators $\hat{A}$ & $\hat{B}$ }

A and B are compatible if –

$$[A, B] = 0$$

(or)

common basis set of eigen vectors such that simultaneous states exist.

(or)

(Implication) $\downarrow [A, B] = 0$

$$\begin{array}{ccc} A & B & A \\ \downarrow & \downarrow & \downarrow \\ \textcircled{a_n} & b_\ell & \textcircled{a_n} \end{array} \quad |\Psi\rangle$$

getting same state / state does not change by measurment of different measurable in midde.

a_n – eigen value of $\hat{A}$

b_ℓ – eigen value of $\hat{B}$

what if $[A, B] \neq 0$, what is $(\Delta A) \cdot (\Delta B)$?

$$\Delta A \cdot \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$$

uncertainty principle.

$$\Delta A = \langle A^2 \rangle - \langle A \rangle^2 = \langle (A - \langle A \rangle)^2 \rangle$$

lets call

$$\begin{aligned} A - \langle A \rangle &= A' \\ B - \langle B \rangle &= B' \end{aligned}$$

then, what is $\langle [A, B] \rangle$?

$$\langle [A, B] \rangle = \langle [A', B'] \rangle$$

as $\langle A \rangle$ is a no which is subtructed from A . ($A - \langle A \rangle = A'$)

$$\begin{aligned} \Rightarrow \quad \langle \Psi | [A' B'] | \Psi \rangle &= \langle [A, B] \rangle \\ \Rightarrow \quad \langle \Psi | A' B' | \Psi \rangle - \langle \Psi | B' A' | \Psi \rangle &= \langle [A, B] \rangle \end{aligned}$$

since A', B', A, B are hermitian

$$\left(\begin{array}{l} A, B \text{ are hermitian} \\ \text{so } \Rightarrow A', B' \text{ are too.} \end{array} \right)$$

$$\begin{aligned} [A, B] &= AB - BA \\ &= (A' + \langle A \rangle) (B' + \langle B \rangle) - (B' + \langle B \rangle) (A' + \langle A \rangle) \\ &= A' B' - B' A' + A' \langle B \rangle + B' \langle A \rangle + \langle A \rangle \langle B \rangle \\ &= A' B' - B' A' = [A', B'] \end{aligned}$$

[the cross terms $A' \langle B \rangle$, $B' \langle A \rangle$ and $\langle A \rangle \langle B \rangle$ are struck out]

$$\begin{aligned} \Rightarrow \quad \langle A' \Psi | B' \Psi \rangle - \langle B' \Psi | A' \Psi \rangle &= \langle [A, B] \rangle \\ \Rightarrow \quad \langle A' \Psi | B' \Psi \rangle - \langle A' \Psi | B' \Psi \rangle^* &= \langle [A, B] \rangle \end{aligned}$$

$$\left\{ \begin{array}{l} x = a + ib \\ x^* = a - ib \\ x - x^* = 2ib \end{array} \right.$$

$$\begin{aligned} 2i \operatorname{Im} (\langle A' \Psi | B' \Psi \rangle) &= \langle [A, B] \rangle \\ |\operatorname{Im} (\langle A' \Psi | B' \Psi \rangle)| &= \frac{1}{2} |\langle [A, B] \rangle| \end{aligned}$$

also.

$$\begin{aligned} |\operatorname{Im} (\langle A' \Psi | B' \Psi \rangle)| &\leq |\langle A' \Psi | B' \Psi \rangle| \\ \frac{1}{2} |\langle [A, B] \rangle| &\leq |\langle A' \Psi | B' \Psi \rangle| \end{aligned}$$

$\{\langle v|w\rangle \leq |v| |w| \text{ cauchy inequality} \}$

$$\begin{aligned}\frac{1}{2} |\langle [A, B] \rangle| &\leq \langle A' \Psi | A' \Psi \rangle^{1/2} \langle B' \Psi | B' \Psi \rangle^{1/2} \\ \frac{1}{2} |\langle [A, B] \rangle| &\leq \langle \Psi | A' A' | \Psi \rangle^{1/2} \langle \Psi | B' B' | \Psi \rangle^{1/2} \\ &\leq \Delta A \cdot \Delta B\end{aligned}$$

$$\{\langle \Psi | A'^2 | \Psi \rangle = \langle (A - \langle A \rangle)^2 \rangle = \Delta A^2$$

so

$$\boxed{\Delta A \cdot \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|}$$

Question ?

even if $[A, B] \neq 0$ we can have $\langle [A, B] \rangle = 0$ (? How)

if,

$$\begin{aligned}A &= x \\ B &= p\end{aligned} \quad [A, B] = [x, p] = i\hbar I$$

$$\{\langle I \rangle = 1$$

$$\Delta A \Delta B \geq \frac{\hbar}{2}$$

$$\boxed{\Delta x \Delta p \geq \frac{\hbar}{2}}$$

The shrodinger and Heisenberg pictures :-

[the notebook writes the heading as "The shrödinger and Heisenberg pictures :-"]

what we did so far is shrödinger's picture. In which state of any system can be represented by state vector $|\Psi(t)\rangle$ in some hilbert space.

$|\Psi(t)\rangle$ is explicitly time dependent, and the time dependence is specified by the shrödinger's equation.

$$\begin{aligned}i\hbar \frac{d}{dt} |\Psi(t)\rangle &= \hat{H} |\Psi(t)\rangle \\ |\Psi(t)\rangle &= e^{\frac{-iH(t-t_0)}{\hbar}} \cdot |\Psi(t_0)\rangle\end{aligned}$$

where $|\Psi(t)\rangle$ is obtained from action of unitary operator

$$U(t, t_0) = e^{-\frac{iH(t-t_0)}{\hbar}}$$

which takes $|\Psi(t_0)\rangle$ to $|\Psi(t)\rangle$

It is unitary operator because it preserves norm of vector just like unitary operator does.

$$\langle \Psi(t) | \Psi(t) \rangle = \langle \Psi(t_0) | \Psi(t_0) \rangle$$

$$UU^+ = U^+U = I$$

Note :-

$$U(t, t_0) = e^{-\frac{iH(t-t_0)}{\hbar}}$$

only in case of autonomous Hamiltonian

Since Norm of state vector is preserved, it looks like state vector is rotating in Hilbert space.

remember $|\Psi\rangle$ and $e^{i\alpha}|\Psi\rangle$ have same norm.

It is as if $|\Psi\rangle$ is rotating on surface of sphere, except it is happening in ∞ -dimensional hilbert space. *It is strictly not a rotation but a unitary transformation.*

unitary transformation is counter part of canonical transformation which preserves phase-space volume.

$$\{q_0, p_0\} = 1$$

$$\{q_t, p_t\} = 1$$

In same way just as classical Hamiltonian dynamics is gradual unfolding of sequence of infinitesimal canonical transformation, in exactly same way the Quantum evolution is gradual unfolding of sequence of unitary transformation on state vector.

$$\langle A \rangle(t) = \langle \Psi(t) | A | \Psi(t) \rangle$$

In shrodinger picture $\hat{A}$ are time independent but $|\Psi(t)\rangle$ is time dependant, lets use subscript 's' for shrödinger picture.

$$\langle A_s \rangle(t) = \langle \Psi_s(t) | A_s | \Psi_s(t) \rangle$$

$$|\Psi_s(t)\rangle = e^{-\frac{iHt}{\hbar}} |\Psi_s(0)\rangle$$

$$\langle \Psi_s(t) | = \langle \Psi_s(0) | e^{\frac{+iHt}{\hbar}}$$

$$\langle A_s \rangle(t) = \left\langle \Psi_s(0) \left| e^{\frac{iHt}{\hbar}} A e^{-\frac{iHt}{\hbar}} \right| \Psi_s(0) \right\rangle$$

"Now Heisenberg comes and says, operator itself is time dependent, where as wave function / state vector does not have dependence."

$$\langle A \rangle(t) = \langle \Phi | A(t) | \Phi \rangle$$

$$A(t) = e^{\frac{+iHt}{\hbar}} A e^{-\frac{iHt}{\hbar}}$$

$$\boxed{\langle A \rangle(t) = \langle \Psi_H | A_H(t) | \Psi_H \rangle}$$

It must be true for every observable $\hat{A}$ and every state vector $|\Psi_H\rangle$.

These two are completely consistent with each other, provided,

$$\left. \begin{aligned} |\Psi_H\rangle &= |\Psi_s(0)\rangle \\ A_H(t) &= e^{\frac{iH_s t}{\hbar}} A_s e^{-\frac{iH_s t}{\hbar}} \end{aligned} \right\}$$

therefore,

$$A_H(0) = A_s \quad (\text{by def}^n).$$

so, our $A_H(t)$ becomes

$$A_H(t) = e^{\frac{iH_s t}{\hbar}} A_H(0) e^{-\frac{iH_s t}{\hbar}}$$

lets $A = H$, find $H_H(t) = ?$

$$H(t) = e^{\frac{iH_s t}{\hbar}} H_s e^{\frac{-iH_s t}{\hbar}}$$

since $\left[e^{\frac{iH_s t}{\hbar}}, H_s \right] = 0$

$$\boxed{H(t) = H_s} = H, \text{ say}$$

So, Hamiltonian operator is same in both pictures so let us call it H .

$$A_H(t) = e^{\frac{iHt}{\hbar}} A_H(0) e^{\frac{-iHt}{\hbar}}$$

Hamiltonian operator is a special operator which does not change with time.

$$H(t) = e^{\frac{iHt}{\hbar}} H_H(0) e^{\frac{-iHt}{\hbar}} = H(t=0).$$

let us go backward and find,

$$\begin{aligned} \frac{d}{dt} A_H(t) &= ? & A_H(t) &= e^{\frac{iHt}{\hbar}} A_H(0) e^{\frac{-iHt}{\hbar}} \\ \frac{dA_H(t)}{dt} &= \frac{iH}{\hbar} A_H(t) + e^{\frac{iHt}{\hbar}} A_H(0) e^{\frac{-iHt}{\hbar}} \left(\frac{-iH}{\hbar} \right) \\ &= \frac{iH}{\hbar} A_H(t) + A_H(t) \left(\frac{-iH}{\hbar} \right) \\ &= \frac{i}{\hbar} (H A_H(t) - A_H(t) H) \\ \frac{dA_H(t)}{dt} &= \frac{i}{\hbar} [H, A_H(t)] \\ i\hbar \frac{dA_H(t)}{dt} &= -[H, A_H(t)] \\ \boxed{i\hbar \frac{d}{dt} A_H(t) = [A_H(t), H]} &\quad \text{or} \quad \frac{dA_H(t)}{dt} = \frac{[A_H(t), H]}{i\hbar} \end{aligned}$$

[← Heisenberg equation of motion.]

solution to this equation is simply

$$A_H(t) = e^{\frac{iHt}{\hbar}} A_H(0) e^{\frac{-iHt}{\hbar}}$$

classically if $H = H(q, p)$

$$\frac{dA(q, p)}{dt} = \{A, H\}$$

So indeed the poisson brackett has been replaced by $\frac{[A, H]}{i\hbar}$

	Shrodinger picture	Heisenberg picture
$\hat{A}$	Time-independent (A_s)	Time dependent $A_H(t)$
$ \Psi\rangle$	time dependent $ \Psi(t)\rangle$	Time independent $ \Psi_H\rangle$
E.O.M.	$i\hbar \frac{d}{dt} \Psi(t)\rangle = \hat{H} \Psi(t)\rangle$	$i\hbar \frac{dA_H(t)}{dt} = [A, H]$

$$\langle A \rangle(t) = \langle \Psi_s(t) | A_s | \Psi_s(t) \rangle = \langle \Psi_H | A_H(t) | \Psi_H \rangle$$

Note :- That is like saying when average value changes either probability distribution changes and variable itself is not time dependent, or I say no, distribution is fixed for all, where as variables has undergone dynamical evolution.

Heisenberg and shrodinger pictures are connected by unitary transformation.

Also, prove that

$$\begin{aligned} \frac{d}{dt} \langle A_s \rangle &\equiv \left\langle \frac{dA_H}{dt} \right\rangle \\ \langle A_s \rangle &= \langle \Psi_s(t) | A_s | \Psi_s(t) \rangle \bigg/ \underbrace{\langle \Psi(t) | \Psi(t) \rangle}_{1 \text{ (normalized assumptn)}} \\ \frac{d}{dt} \langle A_s \rangle &= \frac{d}{dt} \langle \Psi_s(t) | A | \Psi(t) \rangle + \left\langle \Psi(t) | A | \frac{d}{dt} | \Psi(t) \right\rangle \end{aligned}$$

using,

$$\begin{aligned} i\hbar \frac{d}{dt} | \Psi(t) \rangle &= H | \Psi(t) \rangle \\ \frac{d}{dt} \langle \Psi(t) | &= \frac{\langle \Psi(t) | H}{-i\hbar} \\ \frac{d}{dt} \langle A_s \rangle &= \left[-1 \left\langle \Psi(t) \left| \frac{HA}{i\hbar} \right| \Psi(t) \right\rangle + \left\langle \Psi(t) \left| \frac{AH}{i\hbar} \right| \Psi(t) \right\rangle \right] \\ &= \left\langle \Psi(t) \left| \frac{AH - HA}{i\hbar} \right| \Psi(t) \right\rangle \\ &= \left\langle \Psi(t) \left| \frac{[A, H]}{i\hbar} \right| \Psi(t) \right\rangle \\ \frac{d}{dt} \langle A_s \rangle &= \left\langle \Psi(t) \left| \frac{dA_H(t)}{dt} \right| \Psi(t) \right\rangle = \left\langle \frac{dA_H}{dt} \right\rangle \end{aligned}$$

In general if shrödinger equation has operators which are explicit dependent on time.

$$A = x^2 t + p t^3$$

then

$$\boxed{\frac{dA_H(t)}{dt} = \frac{[A, H]}{i\hbar} + \frac{\partial A}{\partial t}}$$

($\frac{\partial A}{\partial t} = x^2 + 3pt^2$ in above case)

exactly like $\frac{dA}{dt} = \{A, H\} + \frac{\partial A}{\partial t}$ from classical mechanics.

overview — ii prev — next ⚡

[The scan page that opens this set begins with the tail of the previous lecture – ”if $H = H(t)$ non-autonomous quantum system, Heisenberg eqⁿ takes the form $\frac{dA_H(t)}{dt} = \frac{[A_H(t), H_H(t)]}{i\hbar} + \frac{\partial A_H(t)}{\partial t}$, Most general Heisenberg equation of motion.” – that material belongs to the Lec 12-13 page. This page begins at the ”lec 14” heading in the scans.]

13 Lecture 14

let us go back a little bit to Schrödinger eqⁿ of a particle in 3-D. and show associated with probability density $|\psi|^2$, there is a probability current.

from classical,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0$$

ρ = density

J = current density

Schrödinger eqⁿ of motion in position basis.

$$\psi^* \left[i\hbar \frac{\partial \psi(r, t)}{\partial t} \right] = H|\psi(r, t)\rangle = \frac{-\hbar^2}{2m} \nabla^2 \psi(r, t) + V(r) \cdot |\psi(r, t)\rangle \quad \text{--- ①}$$

complex conjugate eqⁿ.

$$\psi \left(-i\hbar \frac{\partial \psi^*(r, t)}{\partial t} = \frac{-\hbar^2}{2m} \nabla^2 \psi^*(r, t) + V(r) \psi^*(r, t) \right) \quad \text{--- ②}$$

$$\text{①} * \psi^* - \text{②} * (\psi) \Rightarrow$$

$$i\hbar \frac{\partial}{\partial t} (\psi^* \psi) = \frac{-\hbar^2}{2m} [\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*] + \psi^* V(r) \psi - \psi V(r) \psi^*$$

$(\psi^* V(r) \psi = \psi V(r) \psi^*)$ How (?)

$$i\hbar \frac{\partial}{\partial t} |\psi|^2 = \frac{-\hbar^2}{2m} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*)$$

$$\boxed{i\hbar \frac{\partial}{\partial t} |\psi|^2 = \frac{-\hbar^2}{2m} \nabla \cdot (\psi^* \nabla \psi - \psi \nabla \psi^*)}$$

$$\{ \nabla \cdot (\psi^* \nabla \psi - \psi \nabla \psi^*)$$

$$\Rightarrow \nabla \psi^* \cdot \nabla \psi + \psi^* \nabla^2 \psi - \nabla \psi \cdot \nabla \psi^* - \psi \nabla^2 \psi^*$$

$$\frac{\partial}{\partial t} |\psi|^2 = \frac{-\hbar^2}{(i\hbar)2m} \nabla \cdot (\vec{J})$$

$$\frac{\partial}{\partial t} |\psi|^2 + \frac{\hbar}{2mi} \nabla \cdot \vec{J} = 0$$

$$\rho = |\psi|^2$$

$$J = (\psi^* \nabla \psi - \psi \nabla \psi^*) \frac{\hbar}{2mi} = \text{probability current density.}$$

spread of a wave packet :-

we have a particle (free or in a potential), its propagating in a direction, If we prepare an initial state of this particle such that it is localized around some point x_0 , (so a wave packet peaked at x_0) and let go. This particle propagate under schrödinger eqⁿ, let us ask, "what happens to shape of this wave packet ?" as time goes along.

Let us assume we have a state vector $|\Psi(t)\rangle$

What happens to mean value of position $\langle x \rangle$. ?

or mean squared value of the position $\langle x^2 \rangle$. and so on.

Now we know that if we have a gaussian wave packet. the peak position is actual value of the mean, if it spreads out or it contracts, we know that wave packet is either being dispersed or getting tighter.

let us ask what is $\langle x^2 \rangle(t)$?

$$H = \frac{p^2}{2m} + V(x)$$

$$\langle x^2 \rangle(t) = \langle \Psi(t) | x^2 | \Psi(t) \rangle$$

This can be calculated either from schrödinger picture or Heisenberg picture.

$$\frac{d}{dt} \langle x^2 \rangle = \left(\frac{d}{dt} \langle \Psi(t) | \right) x^2 | \Psi(t) \rangle + \langle \Psi(t) | x^2 \left(\frac{d}{dt} | \Psi(t) \rangle \right)$$

using schrödingers eqⁿ of motion

$$i\hbar \frac{d}{dt} | \Psi \rangle = H | \Psi \rangle$$

$$-i\hbar \frac{d}{dt} \langle \Psi | = \langle \Psi | H^\dagger = \langle \Psi | H$$

{ H is hermitian

so,

$$\begin{aligned} \frac{d}{dt} \langle x^2 \rangle &= \left\langle \Psi(t) \left| \frac{Hx^2}{-i\hbar} \right| \Psi(t) \right\rangle + \left\langle \Psi(t) \left| \frac{x^2 H}{i\hbar} \right| \Psi(t) \right\rangle \\ &= \left\langle \Psi(t) \left| \frac{x^2 H - Hx^2}{i\hbar} \right| \Psi(t) \right\rangle \\ &= \left\langle \Psi(t) \left| \frac{[x^2, H]}{i\hbar} \right| \Psi(t) \right\rangle \quad \text{--- --- --- } \textcircled{1} \end{aligned}$$

$$\left[\left\langle \frac{dx^2}{dt} \right\rangle \right]$$

Let us find

$$\begin{aligned} [x^2, H] &= \left[x^2, \frac{p^2}{2m} + V(r) \right] = \left[x^2, \frac{p^2}{2m} \right] + [x^2, V(r)] \nearrow 0 \\ &= \left[x^2, \frac{p^2}{2m} \right] \\ &= \frac{1}{2m} [x^2, p^2] \\ &= \frac{1}{2m} [x \cdot x, p^2] = \frac{1}{2m} [x[x, p^2] + [x, p^2]x] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2m} (x(p[x, p] + [x, p]p) + (p[x, p] + [x, p]p)x) \\
&= \frac{1}{2m} (xp(i\hbar I) + x(i\hbar)Ip + i\hbar px + i\hbar px) \\
&= \frac{2i\hbar}{2m} (xp + px) = \frac{i\hbar}{m} (xp + px)
\end{aligned}$$

eqⁿ ① becomes.

$$\begin{aligned}
\frac{d}{dt}\langle x^2 \rangle &= \left\langle \Psi(t) \left| \frac{i\hbar(xp + px)}{i\hbar m} \right| \Psi(t) \right\rangle \\
\frac{d}{dt}\langle x^2 \rangle &= \frac{1}{m} \langle xp + px \rangle
\end{aligned}$$

since x , x^2 are physical observables so is xp , px and $(xp + px)$.

”So we can also find $\frac{d}{dt}\langle xp + px \rangle$ and again we will have new physical measurables, and the series of generation of such new physical operators do not truncate, and end up having ∞ — of such operators.”

$$\begin{aligned}
\frac{1}{m} \langle xp + px \rangle &\equiv \frac{1}{m} \langle \Psi(t) | xp + px | \Psi(t) \rangle \quad (\text{in position basis}) \\
&= \frac{-i\hbar}{m} \\
&= \frac{1}{m} \int_{-\infty}^{\infty} dx \psi^*(x, t) \cdot \left(\frac{\partial}{\partial x} (x \psi(x, t)) \right) \left(\frac{-i\hbar}{m} \right)
\end{aligned}$$

[the two lines above are struck out in the notebook]

$$\begin{aligned}
&= \frac{-i\hbar}{m} \int_{-\infty}^{\infty} dx \psi^*(x, t) \left\{ x \frac{\partial}{\partial x} \psi(x, t) + \frac{\partial}{\partial x} (x \psi(x, t)) \right\} \\
&= \frac{\hbar}{mi} \left(\int_{-\infty}^{\infty} dx \cdot x \left(\psi^* \frac{\partial \psi}{\partial x} - \frac{\partial \psi^*}{\partial x} \psi \right) \right)
\end{aligned}$$

we have

$$\begin{aligned}
\frac{d}{dt}\langle x^2 \rangle &= \frac{1}{m} \langle xp + px \rangle \\
\frac{d^2}{dt^2}\langle x^2 \rangle &= \frac{1}{m} \left\langle \frac{[xp + px, H]}{i\hbar} \right\rangle
\end{aligned}$$

$$\left\{ \frac{dA}{dt} = \left\langle \frac{[A, H]}{i\hbar} \right\rangle \right.$$

$$\left. \left\{ \frac{dA}{dt} = \frac{(A, H)}{i\hbar} \right. \right.$$

for free particle $V(x) = 0$

$$\begin{aligned}
\frac{d^2}{dt^2}\langle x^2 \rangle &= \frac{1}{mi\hbar} \langle [xp + px, H] \rangle \\
&= \frac{1}{mi\hbar} \langle [xp, H] + [px, H] \rangle
\end{aligned}$$

since $\hat{H} = \frac{p^2}{2m}$

$$\begin{aligned}
&= \frac{1}{2m^2 i \hbar} \langle [xp, p^2] + [px, p^2] \rangle \\
&= \frac{1}{2m^2 i \hbar} \langle [x, p^2]p + x[p, p^2] \nearrow^0 + [p, p^2] \nearrow^0 x + p[x, p^2] \rangle \\
&= \frac{1}{2im^2 \hbar} \langle ([x, p]p + p[x, p])p + p(p[x, p] + [x, p]p) \rangle \\
&= \frac{1}{2im^2 \hbar} \langle 2i\hbar p^2 + 2i\hbar p^2 \rangle \\
&= \frac{2}{m^2} \langle p^2 \rangle
\end{aligned}$$

$\left\langle \frac{p^2}{2m} \right\rangle$ is just the energy

$\Rightarrow \langle p^2 \rangle$ is independent of time.

$$\frac{d}{dt} \langle x^2 \rangle = \frac{2}{m^2} \langle p^2 \rangle t + \left. \frac{d}{dt} \langle x^2 \rangle \right|_{t=0}$$

[integrating]

$$\langle x^2 \rangle(t) = ?$$

integrating once again

$$\langle x^2 \rangle(t) = \frac{2}{m^2} \langle p^2 \rangle t^2 + \left(\left. \frac{d}{dt} \langle x^2 \rangle \right|_{t=0} \right) t + \langle x^2 \rangle \Big|_{t=0}$$

lets find :-

$$\begin{aligned}
\frac{d}{dt} \langle x \rangle &= \frac{1}{i\hbar} \langle [x, H] \rangle \\
\frac{d}{dt} \langle x \rangle &= \frac{1}{i\hbar} \left\langle \left[x, \frac{p^2}{2m} + V(x) \right] \right\rangle = \frac{1}{i\hbar} \left\langle \left[x, \frac{p^2}{2m} \right] + [x, V(x)] \nearrow^0 \right\rangle \\
\frac{d}{dt} \langle x \rangle &= \frac{1}{2im\hbar} \langle [x, p^2] \rangle \\
&= \frac{1}{2mi\hbar} \langle p[x, p] + [x, p]p \rangle \\
&= \frac{1}{2im\hbar} \langle 2i\hbar p \rangle = \frac{\langle p \rangle}{m} \\
\frac{d}{dt} \langle x \rangle &= \frac{\langle p \rangle}{m}
\end{aligned}$$

integrating once. (For free particle)

[$\langle p \rangle$ is time independent and we can integrate directly].]

$$\langle x \rangle(t) = \frac{\langle p \rangle}{m} t + \langle x \rangle(t=0)$$

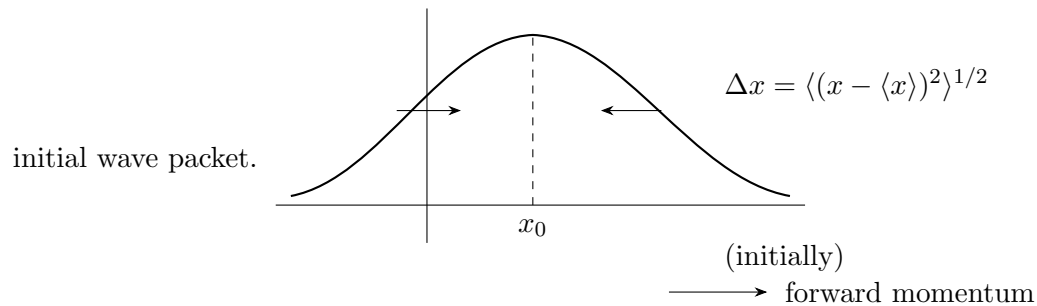
This eqⁿ resembles with classical picture of $x = ut + x_0$

So quantum mechanical equation of motion expectation values obey classical equation of motion.
 } ernfest theorem

This theorem breaks down when choose occurs in classical dynamics.

(How uncertainty varies in time) ?

At $t = 0$:-



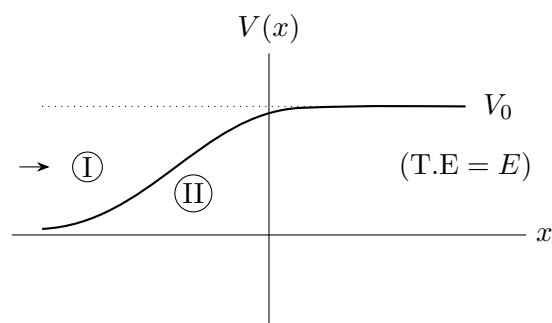
$$\begin{aligned}
 (\Delta x)_{(t)} &= \langle x^2 \rangle(t) - (\langle x \rangle(t))^2 \\
 &= \left[\frac{2}{m^2} \langle p^2 \rangle t^2 + \frac{d\langle x^2 \rangle}{dt} \Big|_{t=0} + \langle x^2 \rangle(0) \right]^2 - \left(\frac{\langle p \rangle t}{m} + \langle x \rangle(0) \right)^2 \\
 &\geq 0
 \end{aligned}$$

so, Δx will increase with time. $\Delta x(t)$ is increasing fun.

There is dispercion, the wave packet is actually increase with time.

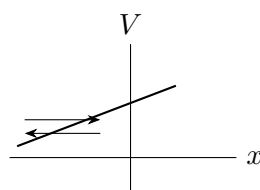
14 Lecture 15

scattering from a potential barrier (1-D) :-



$(E < V_0)$

classically any particle with energy $< V_0$, will climb up the hill and roll back.



Quantum mechanically, if we shoot a particle in x -direction, If total energy $< V_0$, then particle comes along and portion of state vector / wave function $\psi(r, t)$, actually exists in the III^{rd} region but dies down exponentially. and there is also a reflected wave fun. But there would be exactly total internal reflection, The incident wave would be reflected back, & there is no transmission as such.

$(E > V_0)$

classically if a particle is having T.E $> V_0$, the particle climbs up the hill and moves in $+x$ direction with reduced velocity. $(K = T - V)$ and continues to move upto $+\infty$.

Quantum mechanically something interesting happens, A particle with T.E $> V_0$, a portion of its wave is transmitted and a portion of its incident wave is reflected back. And the amount which is transmitted or reflected will tell us, transmission coefficient and reflection co-efficient.

(let us solve first for $E > V_0$)

Its incident wave fun. (stationary) $\sim e^{ikx}$

[approximation because of some potential]

where,

$$\frac{2mE}{\hbar^2} = k^2$$

(This state is not normalizable, its extends from $-\infty$ to $+\infty$)

Transmitted wave fun. $\sim Ae^{ik'x}$

where,

$$k'^2 = \frac{2m(E - v)}{\hbar^2}$$

Reflected wave function : $\sim Be^{-ikx}$

$$\text{as } x \rightarrow (-\infty) \quad \left(\psi(x) = e^{ikx} + Be^{-ikx} \right)$$

$$x \rightarrow (+\infty) \quad \left(\psi(x) = Ae^{ik'x} \right)$$

$$\text{Probability current} = \frac{\hbar}{2mi} \left(\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right)$$

$$\text{for } \psi = e^{ikx}, \quad \text{current} = \frac{\hbar k}{m}$$

so let us define :-

$$(T) \text{ Transmission coefficient} = \frac{\text{Transmitted Probability current}}{\text{Incident Probability current}}$$

$$= \frac{\frac{\hbar k'}{m} |A|^2}{\left(\frac{\hbar k}{m} \right)} = \frac{k'}{k} |A|^2.$$

$$(R) \text{ Reflection coefficient} = \frac{\text{Reflected Probability current}}{\text{Incident probability current}}$$

$$= |B|^2 \frac{k}{k} = |B|^2$$

Now Probability must be conserved, If initially our wave fun. is normalized, then Probability sum of reflected and transmitted should also be equal to 1.

$$\Rightarrow T + R = 1$$

$$\left(\frac{k'}{k} |A|^2 + |B|^2 = 1 \right)$$

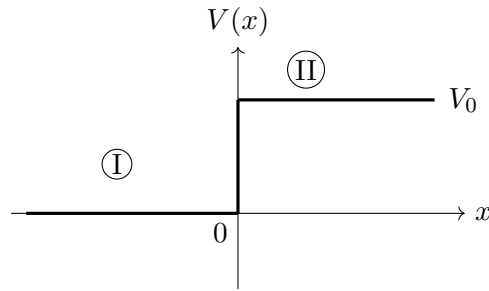
To find actual value of $|A|^2$, $|B|^2$ one should solve shrödinger equation, with that specific form of potential, then find out what happens asymptotically.

”Let us look into possibility of absorption. Absorption only happens if a reaction is causing particles to disappear.

Suppose we incident a beam of neutrons, and Nucleus absorbs some of it, then the total flux of incident neutrons decreases / disappear, then we indeed have real absorption coefficient and part of it is scattered. This is taken into account by making the potential complex (optical potential method).

where imaginary part of $V(x)$ is related with absorption coefficient”.

Let's take a simple potential like



In ①

$$\psi(x) = e^{ikx} + Be^{-ikx}$$

In ②

$$\psi(x) = Ae^{ik'x}$$

where,

$$\begin{cases} k^2 = \frac{2mE}{\hbar^2} \\ k'^2 = \frac{2m(E - V_0)}{\hbar^2} \end{cases}$$

Since $(T + R = 1)$

$$\frac{k'}{k} |A|^2 + |B|^2 = 1$$

But to find $|A|$, $|B|$ we have to solve for shrödingers eqⁿ and apply some boundary condⁿ.

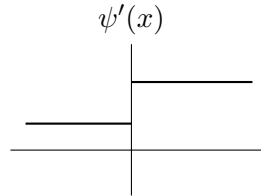
$$\psi(0^-) = \psi(0^+) \quad (\text{wave fun is contineous})$$

$$1 + B = A$$

we know that for any stationary state / eigen state of H .

$$\frac{-\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\psi(x)$$

↑ finite discontinuity ; ↑ finite discontinuity ; ↑ continuous



slopes will be same $\psi'(0^-) = \psi'(0^+)$

such that $\psi''(0^-) \neq \psi''(0^+)$.

(finite discontinuity).

So, using $\psi'(0^-) = \psi'(0^+)$

$$ik \left(e^{ikx} - Be^{-ikx} \right) \Big|_{x=0} = (ik')A e^{ik'x}$$

$$k(1 - B) = k'A$$

$$k(1 - B) = k'(1 + B)$$

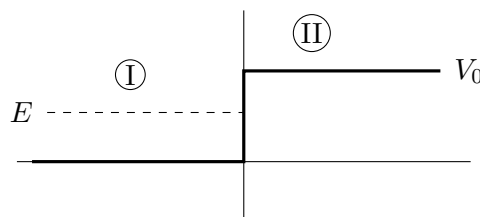
$$\frac{k}{k'} = \frac{1 + B}{1 - B} \Rightarrow B(k' + k) = (k - k')$$

$$\left(B = \frac{k - k'}{k' + k} \right) \quad A = 1 + B = \frac{2k}{k' + k}$$

checking ($T + R = 1$)

$$\begin{aligned} & \left(\frac{k'}{k} |A|^2 + |B|^2 \right) = ? \\ & = \frac{k'}{k} \frac{4k^2}{(k' + k)^2} + \frac{(k' - k)^2}{(k' + k)^2} \\ & = \frac{4kk' + k'^2 + k^2 - 2kk'}{(k' + k)^2} = \frac{(k' + k)^2}{(k' + k)^2} = 1. \end{aligned}$$

What happens if ($E < V_0$).



since

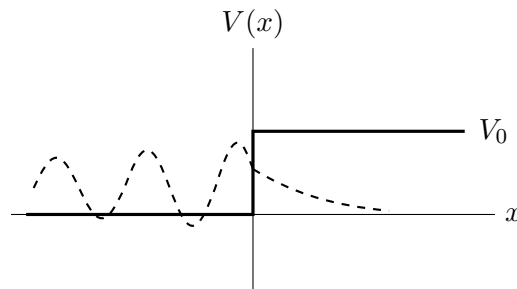
$$k'^2 = \frac{2m(E - V_0)}{\hbar^2}$$

$$k' = \sqrt{\frac{2m(E - V)}{\hbar^2}}$$

for $E < V_0$, k' becomes imaginary

$$\psi(x) \text{ in } \textcircled{\text{II}} \text{ region} = Ae^{ik'x} \nearrow Ae^{-|k'|x}$$

So wave dies down exponentially



So although Transmisson coefficient is zero (in this case) [i.e. as $x \rightarrow \infty$ there is no wave function) (no flux of particles). However there is a penetration in this barrier, and it dies down exponentially fast (with an exponent $\sqrt{\frac{2m(V_0-E)}{\hbar^2}}$.)

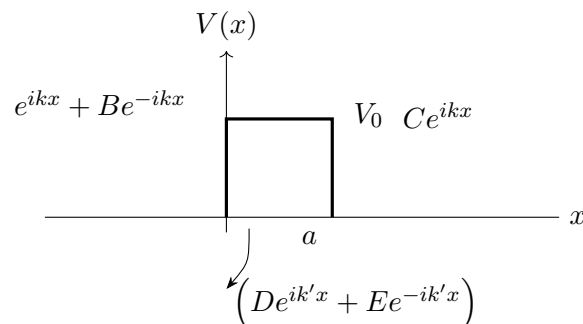
"Since $\psi_I(x)$ is not normalized for infinite domain, but we can put whole thing in a box, so that e^{ikx} is member of L_2 (normalizable). so that $T + R = 1$."

But if we put the whole thing in huge box $[-L \text{ to } +L]$, then momentum gets quantized (because it gotta satisfy boundary conditions), called box normalization, let's extend the box to infinity then the levels become essentially continuous."

one more rigerous way of doing it (because we can't find sharp momentum eigen state. i.e. plane wave is a fiction), so let's find a wave packet localized about some wave number (k_0) maybe a gaussian wave function, we shoot that and it comes back and we can do same thing for superposition of waves.

"Also in case of total internal reflection there is a wave which penetrates another medium, but it also dies down exponentially"

What happens if we cut the barrier at $x = a$.

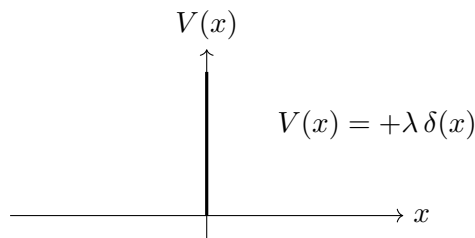


[equate wave fun. and its derivative $x = 0, a.$]

$$\begin{cases} \psi(0^-) = \psi(0^+) \\ \psi'(0^-) = \psi'(0^+) \end{cases}$$

$$\begin{cases} \psi(a^-) = \psi(a^+) \\ \psi'(a^-) = \psi'(a^+) \end{cases}$$

what happens if we decrease width and increase height to ∞ such that product remains const. (we get a delta fun.)



$$-\frac{\hbar^2}{2m} \frac{d^2 \phi_E(x)}{dx^2} + V(x) \phi_E(x) = E \phi_E(x)$$

$\uparrow$ (∞ discontinuity) ; $\uparrow$ (∞ discontinuity) ; $\uparrow$ continuous

If ϕ'' has ∞ -discontinuity, this means $\phi'(x)$ has finite jump, which implies $\phi(x)$ has cusp at origin. so only boundary condⁿ. left is.

$$\phi(0^-) = \phi(0^+)$$

$$\int_{-\epsilon}^{+\epsilon} \left(\frac{-\hbar^2}{2m} \phi''(x) + V(x) \phi(x) \right) dx = \int_{-\epsilon}^{+\epsilon} E \phi(x) dx$$

Now discontinuity in $\phi'(x)$ will be related to $\delta(x)$ function.

(It is given as an exercise) [solve it]

particle will get through the barrier in $\delta(x)$ potential.

$$\left(\psi_T \sim e^{ikx} \text{ plane wave} \right)$$

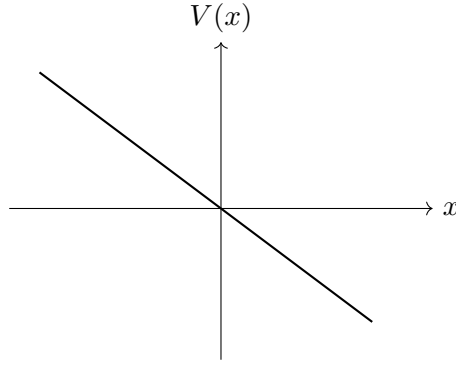
Particle in a constant Force field

$$V(x) = -Fx \quad (F = \text{const})$$

so that

$$\text{Force} = -\frac{dV}{dx} = F \text{ (constant force).}$$

(i.e. could be electric force qE if $E=\text{const}$).



Every Energy is allowed ($-\infty < E < +\infty$)

classically any particle shot in $(-x)$ will climb up the hill, loose the K.E and will roll back accelerated upto $+\infty$.

(Quantum mechanically what would we expect ?)

$$-\infty < E < \infty$$

and No bound states (no confining potential)

so, solution would not be normalizable.

so what kind of wave would it be ?

let ϕ_E be stationary state of H . satisfying

$$\frac{-\hbar^2}{2m} \frac{d^2 \phi_E(x)}{dx^2} + (-Fx) |\phi_E(x)\rangle = E |\phi_E(x)\rangle$$

$$\phi_E''(x) + \frac{2m}{\hbar^2} (E + Fx) \phi_E(x) = 0$$

Let

$$\xi = \frac{2m}{\hbar^2} (E + Fx), \quad x = \left(\frac{\hbar^2}{2m} \frac{\xi}{F} - \frac{E}{F} \right)$$

[side working in the margin:]

$$\frac{d}{dx} \left(\frac{d\phi_E'(\xi)}{d\xi} \cdot \frac{d\xi}{dx} \right) + \xi \phi_E = 0$$

$$\frac{d}{d\xi} \left(\frac{d\phi'(\xi)}{d\xi} \cdot \frac{2mF}{\hbar^2} \right) \frac{d\xi}{dx} + \xi \phi(\xi) = 0$$

$$\phi''(\xi) + \frac{\hbar^4}{(2mF)^2} \xi \phi(\xi) = 0$$

$$\phi_E''(\xi) + c \xi \phi_E(\xi) = 0$$

(Airy's equation)

$$\left(c = \frac{\hbar^4}{(2mF)^2} \right) \text{ which is independent of } E$$

$$\phi_E(\xi) = (\text{const.}) Ai(-\xi)$$

where

$$Ai(z) = \frac{1}{\sqrt{\pi}} \int_0^\infty du \cdot \cos\left(\frac{u^3}{2} + u\xi\right)$$

(since c is independent of E , every energy eigenstate has same $\phi(x)$ }

Let's work in momentum basis $\rangle$ (This problem is easier in $|p\rangle$).

$$\begin{aligned} H\hat{\phi}_E(p) &= E\tilde{\phi}_E(p) \quad - \textcircled{1} \\ H &= \frac{p^2}{2m} - F\hat{x} = \frac{p^2}{2m} - F\left(i\hbar\frac{d}{dp}\right) \end{aligned}$$

so above eqⁿ $\textcircled{1}$ becomes

$$\begin{aligned} \left(\frac{p^2}{2m} - i\hbar F\frac{d}{dp}\right)\tilde{\phi}_E(p) &= E\tilde{\phi}_E(p) \\ \frac{d\tilde{\phi}_E(p)}{dp} &= \frac{i\left(E - \frac{p^2}{2m}\right)\tilde{\phi}_E(p)}{\hbar F} \\ \int \frac{d\tilde{\phi}_E(p)}{\tilde{\phi}_E(p)} &= \frac{1}{\hbar F} \int i\left(E - \frac{p^2}{2m}\right) dp \\ \ln\left(\tilde{\phi}_E(p)\right) &= \frac{i}{\hbar F} \left(Ep - \frac{p^3}{6m}\right) + c \\ \boxed{\tilde{\phi}_E(p) &= A e^{\frac{i}{\hbar F} \left(Ep - \frac{p^3}{6m}\right)}} \end{aligned}$$

normalization condⁿ :-

$$\int_{-\infty}^{\infty} dp \tilde{\phi}_E^*(p) \tilde{\phi}_{E'}(p) = \delta(E - E')$$

[normalization condⁿ for the wavefun. which are not normalizable in position space.]

[orthonormality condⁿ]

[Energy normalization]

$$\begin{aligned} \text{as } \xi \rightarrow -\infty \quad \phi &\longrightarrow e^{-|\xi|^{3/2}} \\ \xi \rightarrow +\infty \quad \phi &\longrightarrow \xi^{-1/4} \sin([\?]) \end{aligned}$$

charge particle in constant B. (magnetic field) :-

classically :-

$$L(q, \dot{q}) = \frac{1}{2}m\vec{V}^2 + e(\vec{A} \cdot \vec{V}) - e\phi$$

$$\vec{A} = \vec{A}(r, t)$$

$$\phi = \phi(r, t)$$

$$\vec{P} = \frac{\partial L}{\partial \vec{V}} = m\vec{V} + e\vec{A}$$

$$H \text{ (free particles)} = \frac{p^2}{2m} + e\phi$$

$$H(B) = \frac{(\vec{p} - e\vec{A})^2}{2m} + e\phi$$

since we only have const B .

$$(\phi = 0)$$

$$H = \frac{(\vec{p} - e\vec{A})^2}{2m}$$

$\{[p, A(r)] \neq 0$ Because $[P, r] \neq 0$

lets first find $[p, A] = ?$

$$\begin{aligned} \vec{p} \cdot \vec{A}(r) - \vec{A}(r) \cdot \vec{p} \\ = -[A, p] = -i\hbar \nabla \cdot \vec{A}(r) \end{aligned}$$

{if $[x, P_x] = i\hbar$

then what is $[f(x), P_x]$

$= i\hbar f'(x)$ (How?)

$$\begin{aligned} f(x) \left(-i\hbar \frac{\partial}{\partial x} \phi \right) + \left(i\hbar \frac{\partial}{\partial x} \right) f(x) \cdot \phi(x) \\ = [f(x), P_x] \phi(x) \\ = -i\hbar f(x) \frac{\partial \phi}{\partial x} + \phi(x) i\hbar \frac{\partial f}{\partial x} + f(i\hbar) \frac{\partial \phi}{\partial x} \\ = (i\hbar) f'(x) \cdot \phi(x) \end{aligned}$$

$$\boxed{[f(x), p] = i\hbar f'(x)}$$

"we can always work in gauge where $\nabla \cdot \vec{A} = 0$ (coulomb gauge)"

We know that

$$\nabla \cdot \vec{B} = 0 \quad \Rightarrow \quad \vec{B} = \nabla \times \vec{A}(r)$$

But $\vec{B}$ must be const, so what is $\vec{A}$ in terms of B .

$$\vec{A}(\vec{r}) = \left(\frac{1}{2}\right) \cdot \vec{B} \times \vec{r} \quad - - - - \text{(How)}.$$

Is $\nabla \cdot \vec{A} = 0$

$$\begin{aligned} \nabla \cdot \vec{A} &= \partial_i A_i \\ &= \frac{1}{2} \partial_i \epsilon_{ijk} x_k = 0 \end{aligned}$$

so $\nabla \cdot \vec{A} = 0$ for const $\vec{B}$.

$$\Rightarrow [p, A(r)] = 0$$

$\vec{p}$ & $\vec{A}$ commute.

But A is not fixed, we can choose any A'
such that

$$A' = A + \nabla\chi(\vec{r})$$

so that

$$(\nabla \times A' = \nabla \times A) = B.$$

By doing so, $\hat{H}$ will also change, but it will be a canonical transformation.

$$H = \frac{(p - eA)^2}{2m}$$

$$H' = \frac{(p - eA')^2}{2m}$$

provided schrödinger equation does not change.

$$H\phi = E\phi \quad (\text{should not change})$$

"|lec 16> is written after |lec 17>"

[The notebook's "lec 17" heading begins here, part-way down scan page qm6-05; the material that follows it is transcribed on the Lec 16-17 page.]

[The notebook writes these two lectures out of order: the "lec 17" heading appears first (immediately below a marginal remark saying so), and the "lec 16)" heading only comes several pages later. This page follows the notebook's physical order – Lecture 17 first, Lecture 16 second.]

15 Lecture 17

["—lec 16) is written after —lec 17)"]

The general $\hat{H}$ of a charge particle in EM field induced by $A(\vec{r}, t)$ & $\phi(r, t)$ is

$$H = \frac{(\vec{P} - e\vec{A})^2}{2m} + e\phi \quad \begin{cases} A = A(r, t) \\ \phi = \phi(r, t) \end{cases}$$

[e = charge of particle]

Schrödinger eqⁿ in position representation.

$$H\psi(r, t) = i\hbar \frac{\partial \psi(r, t)}{\partial t} = \frac{(-i\hbar \nabla - eA) \cdot (-i\hbar \nabla - eA)}{2m} \psi(\vec{r}, t) + e\phi \psi(\vec{r}, t)$$

let us work in Heisenberg picture, where rate of change of position operator can be simply

$$\frac{dx_i}{dt} = \frac{[x_i, H]}{i\hbar} = \frac{\left[x_i, \frac{p^2}{2m} + e\phi \right]}{i\hbar}$$

$$\begin{cases} [x_i, A_i] = 0 \\ (x_i, \phi_i) = 0 \end{cases}$$

$$\begin{aligned}
\frac{dx_i}{dt} &= \frac{1}{2mi\hbar} ([x_i, P_\ell]P_\ell + P_\ell[x_i, P_\ell]) \\
&= \frac{[x_i, p_\ell]P_\ell - e \underbrace{[x_i, A_\ell]}_0 P_\ell + P_\ell[x_i, p_\ell] - P_\ell \underbrace{[x_i, eA_i]}_0}{2mi\hbar} \\
\frac{d\hat{x}_i}{dt} &= \frac{i\hbar P_\ell + P_\ell(i\hbar)}{2mi\hbar} = \frac{P_\ell}{m} = \frac{p_i - eA_i}{m}
\end{aligned}$$

find acceleration,

$$\begin{aligned}
\frac{d^2 x_i}{dt^2} &= \frac{\left[\frac{dx_i}{dt}, H \right]}{i\hbar} \\
&= \frac{\left[\frac{p_i - eA_i}{m}, \frac{(\vec{p} - e\vec{A})^2}{2m} + e\phi \right]}{i\hbar} \quad (P_i = p_i - eA_i)
\end{aligned}$$

$$\frac{d^2 \vec{r}}{dt^2} = \frac{e}{m} \left[\vec{E} + \frac{1}{2} \left(\frac{d\vec{r}}{dt} \times \vec{B} - \vec{B} \times \frac{d\vec{r}}{dt} \right) \right]$$

$$\left[a = \frac{e}{m} \left(E + \vec{v} \times \vec{B} \right) \right]$$

$\frac{d\vec{r}}{dt} \times \vec{B}$ is not hermitian so to take care of that it is represented as $\frac{1}{2} \left(\frac{d\vec{r}}{dt} \times \vec{B} - \vec{B} \times \frac{d\vec{r}}{dt} \right)$ which is hermitian.

L in Q.M

$$\vec{L} = \vec{r} \times \vec{p}$$

we know that $\hat{r}, \hat{p}$ do not commute with each other so how do we say $\vec{L}$ is hermitian?

reason is because non-canonical conjugate pairs do commute with each other.

$$\begin{aligned}
\vec{L} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix} = \hat{i}(yp_z - zp_y) - \hat{j}(xp_z - zp_x) + \hat{k}(xp_y - yp_x) \\
L &= L^\dagger = \hat{i}(yp_z - zp_y)^\dagger - \hat{j}(xp_z - zp_x)^\dagger + \hat{k}(xp_y - yp_x)^\dagger = L \\
&= L, \text{ (as } [r_i, p_j] = i\hbar \delta_{ij} \text{)}
\end{aligned}$$

$[x, p_z] = 0$ & so on

$(x, p_y) = 0$

Radial momentum :-

$$p_r = \vec{p} \cdot \hat{r} \Rightarrow p_r = \frac{\vec{p} \cdot \vec{r}}{|\vec{r}|} = \frac{(xp_x + yp_y + zp_z)}{\sqrt{x^2 + y^2 + z^2}}$$

Q). is the above defn compatible with Q.M? No!

radial momentum is represented as :-

$$p_r = \alpha \frac{\vec{p} \cdot \vec{r}}{|\vec{r}|} + (1 - \alpha) \frac{\vec{r}}{|\vec{r}|} \cdot \vec{p}$$

$$p_r^\dagger = p_r \quad \text{at} \quad \alpha = \frac{1}{2}; \quad p_r = \frac{1}{2} \left(\frac{\vec{p} \cdot \vec{r} + \vec{r} \cdot \vec{p}}{|\vec{r}|} \right)$$

p_r can't be represented as $\vec{p} \cdot \hat{r}$ as it is not hermitian.

$$\left(\frac{\vec{p} \cdot \vec{r}}{|\vec{r}|} \right)^\dagger = \frac{(p_x^\dagger x^\dagger + p_y^\dagger y^\dagger + p_z^\dagger z^\dagger)}{r} \neq$$

$$= \frac{p_x x + p_y y + p_z z}{r} \neq \frac{xp_x + yp_y + zp_z}{r}$$

as $[x, p_x] \neq 0, [y, p_y] \neq 0, [z, p_z] \neq 0$

where as

$$p_r = \frac{(\vec{p} \cdot \vec{r} + \vec{r} \cdot \vec{p})}{2|\vec{r}|} \quad \text{is hermitian.} \quad \left(\alpha = \frac{1}{2} \right).$$

(L, p_r) are physical measurables, so upon measurement we should get real eigen values, which is possible easily if operator is hermitian).

similarly, in laplace – runge vector (A)

we should write it as

$$A = \frac{1}{2} \left(\vec{p} \times \vec{L} - \vec{L} \times \vec{p} \right) - \frac{mk\vec{r}}{|\vec{r}|}$$

comments on gauge transformation :-

→ When we make a gauge transformation, No physical variables change. of E , & B .

→ In the quantum version it implies that the physical quantities like probabilities ($|\psi|^2$ & so on) should not change when we go to a different gauge. (in A, ϕ).

(Q) what is the change that occurs, when we go from one gauge to another. let's ask, what happens to probability current when we switch on the E-m field.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0$$

where, $\begin{cases} \rho = |\psi|^2 & \text{probability density} \\ J = \text{probability current} \end{cases}$

$$\begin{aligned}
J &= \frac{\hbar}{2mi} [\psi^* \nabla \psi - \psi \nabla \psi^*] \\
&= \frac{\hbar}{m} \text{Im}(\psi^* \nabla \psi)
\end{aligned}$$

Yet, there is another way of doing it, which is important in other applications of Q.M. & that's the following

let's write $\psi(r, t)$ as.

as $(|\psi|^2 = \rho)$

$$\psi(r, t) = \sqrt{\rho} e^{\frac{iS(r, t)}{\hbar}}$$

$S(r, t) \longleftrightarrow$ Action or phase

so what is $\vec{J}$ in this form?

$$\begin{aligned}
\nabla \psi &= \nabla(\sqrt{\rho}) e^{iS/\hbar} + \sqrt{\rho} \frac{i}{\hbar} \nabla(S) e^{iS/\hbar} \\
\nabla \psi^* &= \nabla(\sqrt{\rho}) e^{-iS/\hbar} - \sqrt{\rho} \frac{i}{\hbar} \nabla(S) e^{-iS/\hbar} \\
J &= \frac{\hbar}{2mi} [\psi^* \nabla \psi - \psi \nabla \psi^*] \\
&= \frac{\hbar}{2mi} \left(\sqrt{\rho} \nabla(\sqrt{\rho}) + \rho \frac{i}{\hbar} \nabla(S) - \sqrt{\rho} \nabla(\sqrt{\rho}) + \frac{i}{\hbar} \rho \nabla(S) \right)
\end{aligned}$$

[the two $\nabla(\sqrt{\rho})$ terms are struck out]

$$\begin{aligned}
&= \frac{2i}{\hbar} \frac{\rho \nabla(S) \hbar}{2mi} \\
&\boxed{J = \frac{\rho}{m} \nabla(S)}
\end{aligned}$$

Q) what is $\nabla(S) = ?$ in presence of E-m field :-

$$\begin{aligned}
J &= \frac{\hbar}{2mi} [\psi^* \nabla \psi - \psi \nabla \psi^*] \\
\vec{p} &\longrightarrow \vec{p} - e\vec{A} \\
-i\hbar \nabla &\longrightarrow -i\hbar \nabla - e\vec{A} \\
\nabla &\longrightarrow \nabla - \frac{ie\vec{A}}{\hbar} \\
J' &= \frac{\hbar}{2mi} \left[\psi^* \left(\nabla - \frac{ie\vec{A}}{\hbar} \right) \psi - \psi \left(\nabla - \frac{ie\vec{A}}{\hbar} \right) \psi^* \right] \\
J' &= J - \frac{\hbar}{2mi} \frac{2ieA |\psi|^2}{\hbar} \\
J' &= J - \frac{eA}{m} |\psi|^2 = \frac{\rho}{m} (\nabla(S) - e\vec{A}) = \frac{\rho}{m} \nabla(S)'
\end{aligned}$$

$$\nabla S' \longrightarrow \nabla S - e\vec{A}$$

[we know that the $|\psi|^2$ should not change upon varying the (A, ϕ) But Action/phase(s), J , changes upon transforming $(A, \phi) \rightarrow (A', \phi')$.]

[If ρ plays role of charge density, $\nabla(s)$ plays role of velocity, then J becomes current density]

so this is all happens when we have EM field, This phase fun. s . its $\nabla(s)$ is shifted by $(e\vec{A})$.

$$s \longrightarrow s - e \int \vec{A} \cdot d\vec{l}$$

so it suggest, phase of EM field is path dependent. (line integral)

let's go back to angular momentum (algebra), find its (eigen values/fun. study its some properties, then we will come back and apply it on charge particle sitting in EM field.

Digression :- Angular momentum :-

Q.Mechanical angular momentum turns out to be somewhat different from classical angular momentum.

For a single particle angular momentum was defined as

$$\vec{L} = \vec{r} \times \vec{p} \quad \text{--- (classical).}$$

It turns out the same definition is true for Q.M except r, p, L are operators.

classically we know,

$$\begin{aligned} \{x_i, p_j\} &= \delta_{ij} \\ \{L_i, L_j\} &= \epsilon_{ijk} L_k \end{aligned}$$

$$\begin{cases} \epsilon_{ijk} \text{ is 1 if } i, j, k \text{ are consecutive} \\ \text{else is } -1 \text{ --- odd sequenced} \\ = 0 \text{ if } i = j = k. \end{cases}$$

[a small cyclic (circular) arrow is drawn beside "consecutive"]

Quantum mechanically angular momentum is defined by these commutation relations.

$\vec{J} = (J_1, J_2, J_3)$ are angular momentum operators satisfying

$$[J_i, J_j] = i\hbar \epsilon_{ijk} J_k$$

So any quantum mechanical angular momentum of any system whatsoever is represented by three **hermitian** operators (J_1, J_2, J_3) which satisfy this commutation relation.

$$[J_i, J_j] = i\hbar \epsilon_{ijk} J_k$$

You can see dimensions are matching.

The above statement is definition and all the properties of J . will come from it.

This angular momentum algebra is also called SO(3) algebra, because it has to do with rotation in 3-D. It is a lie – algebra.

$$[J_1, [J_2, J_3]] + \text{cyclic permutations} = 0 \quad (\text{Jacobi identity})$$

Given this information what can we say about angular momentum states of any system.

(remarkably this question can be answered from this algebra and nothing more).

we need few observations:

$$J^2 = J_1^2 + J_2^2 + J_3^2 \quad (\text{by def}^n).$$

lets ask, what is $[J, J^2] = ?$

$$\begin{aligned} [J, J^2] &= [J_1, J_1^2 + J_2^2 + J_3^2] = \underbrace{[J_1, J_1^2]}_0 + [J_1, J_2^2] + [J_1, J_3^2] \\ &\quad + [J_1, J_3]J_3 + J_3[J_1, J_3] \\ [J_1, J^2] &= i\hbar(J_3J_2 + J_2J_3) + -i\hbar(J_2J_3 + J_3J_2) \\ &= i\hbar(J_3J_2 - J_3J_2) + i\hbar(J_2J_3 - J_2J_3) = 0 \end{aligned}$$

similarly

$$\begin{aligned} [J_2, J^2] &= [J_3, J^2] = 0 \\ \Rightarrow [J_i, J^2] &= 0 \quad (i = 1, 2, 3) \end{aligned}$$

It also implies any linear combination of J_1, J_2, J_3 will commute with J^2 .

$$\boxed{[\vec{J} \cdot \hat{n}, J^2] = 0} \quad \text{for any } \hat{n}$$

So, component of J in any direction commutes with J^2 . Therefore we can simultaneously find eigen-states of J_i and J^2 . Generally we use J_3 as axis of quantisation rather than $\vec{J} \cdot \hat{n}$. But remember any direction $(\hat{n})\vec{J}$ can be used as axis of quantisation. No two components commute with each other.

$$[\vec{J} \cdot \hat{n}, \vec{J} \cdot \hat{n}'] \neq 0 \quad (\text{if } \hat{n} \neq \hat{n}')$$

Note :- $[\vec{J} \cdot \hat{n}, J^2] = 0$ forms maximal set of commuting operator so we can write angular momentum states depending upon only $(j \text{ \& } j_n)$.

”we would like to find out what is spectrum of J, J_1, J_2, J_3, J^2 or what are possible values of eigen value of J^2, J, J_1, J_2, J_3 based on these commutation relations”?

we must do exactly what we did for harmonic oscillator. we have to do this purely algebraically. ”(using trick of bosonisation”

In case of Harmonic oscillator we found $a^\dagger, a$ (ladder operators), can we find similar operators here? (yes we can).

$$J_+ = J_1 + iJ_2$$

$$J_- = J_1 - iJ_2 = (J_+)^{\dagger} = (J_1 + iJ_2)^{\dagger} = J_1^{\dagger} - iJ_2^{\dagger}$$

what is,

$$\begin{aligned} [J_+, J_-] &= [J_1 + iJ_2, J_1 - iJ_2] \\ &= \underbrace{[J_1, J_1]}_0 + (-i)[J_1, J_2] + i[J_2, J_1] + [J_2, J_2] \\ &= (-iJ_3 + i(-J_3))(i\hbar) \\ &= 2\hbar J_3 \\ &\boxed{[J_+, J_-] = 2\hbar J_3} \\ [J_+, J_3] &= [J_1 + iJ_2, J_3] \\ &= [J_1, J_3] + i[J_2, J_3] \\ &= (-i\hbar J_2 + i(i\hbar)J_1) \\ &= -\hbar[J_1 + iJ_2] = -\hbar J_+ \\ &\boxed{[J_+, J_3] = -\hbar J_+} \\ [J_-, J_3] &= [J_1 - iJ_2, J_3] = [J_1, J_3] - i[J_2, J_3] \\ &= -i\hbar J_2 - i(i\hbar)J_1 \\ &= \hbar[J_1 - iJ_2] = \hbar J_- \\ &\boxed{[J_-, J_3] = \hbar J_-} \end{aligned}$$

[we can Rewrite the angular momentum algebra

$$\begin{aligned} (J_+ J_-) &= 2\hbar J_3 \\ [J_+, J_3] &= -\hbar J_+ \\ [J_-, J_3] &= \hbar J_- \end{aligned}$$

advantage is that we do not have complex no "i".]

J. Schwinger method.

consider two different harmonic oscillator which do not talk with each other.

$[(A, B)]$

$$\left. \begin{aligned} A &\rightarrow a, a^{\dagger} \\ B &\rightarrow b, b^{\dagger} \end{aligned} \right\} \text{raising \& lowering operators}$$

such that

$$[a, a^{\dagger}] = [b, b^{\dagger}] = I$$

$$[a, b] = 0 = [a, b^\dagger] = [a^\dagger, b^\dagger] = 0$$

$[a, a^\dagger] / [b, b^\dagger]$ are chosen to be dimensionless.]

If,

$$\begin{aligned}
J_+ &= (a^\dagger b) \hbar \\
J_- &= (J_+)^{\dagger} = (a^\dagger b)^{\dagger} = (b^\dagger a) \hbar = (ab^\dagger) \hbar \\
[J_+, J_-] &=? \\
&= \hbar^2 [a^\dagger b, ab^\dagger] \\
&= \left(a^\dagger [b, ab^\dagger] + [a^\dagger, ab^\dagger] b \right) \hbar^2 \\
&= \hbar^2 \left(a^\dagger \left(\underbrace{[b, a]}_0 b^\dagger + a [b, b^\dagger] \right) + \left([a^\dagger, a] b^\dagger + a \underbrace{[a^\dagger, b^\dagger]}_0 \right) b \right) \\
&= \hbar^2 \left(a^\dagger a [b, b^\dagger] + [a^\dagger, a] b^\dagger b \right) \\
&= \hbar^2 \left(a^\dagger a I + (-I) b^\dagger b \right) \\
&\boxed{[J_+, J_-] = \hbar^2 (a^\dagger a - b^\dagger b)} \\
\begin{cases} J_1 = \frac{J_+ + J_-}{2} = \hbar \left(\frac{a^\dagger b + ab^\dagger}{2} \right) = \text{hermitian} \\ J_2 = \frac{J_+ - J_-}{2i} = \hbar \left(\frac{a^\dagger b - ab^\dagger}{2i} \right) = \text{hermitian} \\ J_3 = \frac{[J_+, J_-]}{2\hbar} = \hbar \left(\frac{a^\dagger a - b^\dagger b}{2} \right) = \text{hermitian} \end{cases}
\end{aligned}$$

This automatically implies

$$\begin{aligned}
[J_i, J_j] &= i\hbar \epsilon_{ijk} J_k \\
J^2 &= J_1^2 + J_2^2 + J_3^2 \\
\frac{4}{\hbar^2} J^2 &= \left(a^\dagger b + ab^\dagger \right)^2 - \left(a^\dagger b - ab^\dagger \right)^2 + \left(a^\dagger a - b^\dagger b \right)^2 \\
&= a^\dagger b a^\dagger b + a^\dagger b a b^\dagger + a b^\dagger a^\dagger b + a b^\dagger a b^\dagger \\
&\quad - a^\dagger b a^\dagger b + a^\dagger b a b^\dagger + a b^\dagger a^\dagger b - a b^\dagger a b^\dagger \\
&\quad + a^\dagger a a^\dagger a - a^\dagger a b^\dagger b - b^\dagger b a^\dagger a + b^\dagger b b^\dagger b \\
\frac{4}{\hbar^2} J^2 &= 2 \left(a^\dagger b a b^\dagger + a b^\dagger a^\dagger b \right) + \left(a^\dagger a \right)^2 - 2 a^\dagger a b^\dagger b + \left(b^\dagger b \right)^2
\end{aligned}$$

lets call

$$\begin{aligned}
N_a &= a^\dagger a \\
N_b &= b^\dagger b
\end{aligned}$$

we know that $ab = ba$, $b^\dagger a^\dagger = a^\dagger b^\dagger$

$$\begin{aligned}
 \frac{4}{\hbar^2} J^2 &= 2 \left(a^\dagger a b b^\dagger + a a^\dagger b^\dagger b \right) + (N_a)^2 - 2N_a N_b + N_b^2 \\
 &= 2 \left(N_a(1 + N_b) + (N_a + 1)N_b \right) + (N_a - N_b)^2 \\
 &= N_a^2 + N_b^2 + 2N_a N_b + 2(N_a + N_b) \\
 &= (N_a + N_b)^2 + 2(N_a + N_b) \\
 J^2 &= \frac{\hbar^2}{4} (N_a + N_b) ((N_a + N_b) + 2) \\
 &= \frac{\hbar^2 (N_a + N_b)}{2} \left[\frac{(N_a + N_b)}{2} + 1 \right]
 \end{aligned}$$

$[N_a, N_b]$ are operators / number operators where as n_a, n_b are eigen states of these number operators]

let's call

$$\frac{N_a + N_b}{2} = j \quad \left(j = \left(\frac{n_a + n_b}{2} \right) \right) \rightarrow \text{(eigen values)}$$

$J^2 = \hbar^2 j(j+1)$

$$\text{where } \left. \begin{matrix} (n_a) & n_a = 0, 1, 2, \dots \\ (n_b) & n_b = 0, 1, 2, \dots \end{matrix} \right\} \text{ are eigen values of } N_a \text{ \& } N_b$$

whatever be the physical system, the possible value of eigen values of J^2 are given by

$$J^2 = \hbar^2 j(j+1) \quad \Bigg| \quad J^2 = \hbar^2 j(j+1)$$

where "j" can only have

$$0, \frac{1}{2}, 1, \frac{3}{2}, \dots$$

so we see half integers appears naturally (spins have half integral values). physical interpretation is still missing.

what are eigen values of J_3 .

$$J_3 = \frac{\hbar (a^\dagger a - b^\dagger b)}{2} = \frac{\hbar (N_a - N_b)}{2}$$

eigen values of $J_3 = \frac{\hbar}{2} (n_a - n_b)$

$$\left\{ \begin{array}{l} \text{given } j, \frac{(N_a + N_b)}{2}, \text{ what are eigen values of } J_3, \text{ i.e.} \\ \text{eigen values of } \frac{(N_a - N_b)}{2} \hbar ? \end{array} \right.$$

["lec 18 is written after lec 16)"]

16 Lecture 16

charged particle in a constant uniform magnetic field.

$$H = \frac{(p - eA)^2}{2m}$$

we would like to find eigen values and eigen states of this hamiltonian. (in the position basis)

Since, $\vec{A} = \vec{A}(\vec{r})$

therefore in general it does not commute with p , $[A, p] \neq 0$

However we found that $[A(r), p] = -i\hbar \nabla \cdot \vec{A}(r)$

if $\nabla \cdot \vec{A} = 0$

$$\Rightarrow [\vec{p}, \vec{A}] = 0$$

for constant uniform $\vec{B}$.

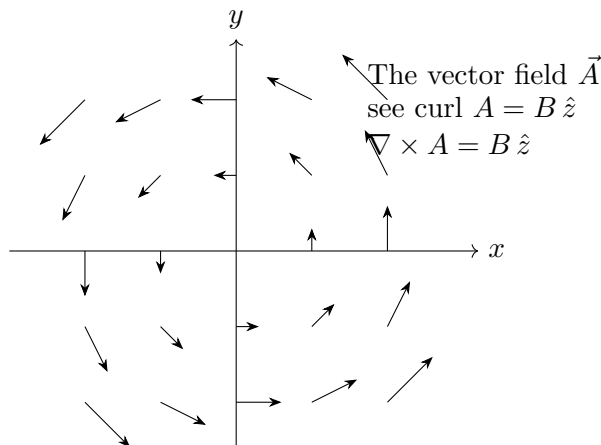
$$\vec{A} = \frac{1}{2} (\vec{B} \times \vec{r})$$

it is easily verified $\nabla \times A$ gives us back B .

if $B = B \hat{e}_z$

$$A = \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & B \\ x & y & z \end{vmatrix} = \frac{1}{2} (\hat{i}(-BY) - \hat{j}(-BX), 0\hat{k})$$

$$\vec{A} = \left(\frac{-1}{2} \cdot Y \cdot B, \frac{1}{2} \cdot X \cdot B, 0 \right)$$



so the general rule is if we want a $\vec{B}$ in particular direction you can choose the vector potential to be transverse to it, and the transversed components do not involve co-ordinate along $\vec{B}$.

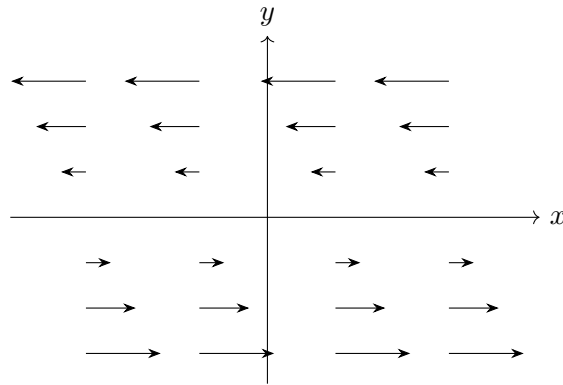
(Is $\vec{A}$ unique ?)

(No we can always add $A' = A + \nabla\chi$)

Is it possible to get rid of one of the components of $\vec{A}$? to make it just x component.

$$\begin{aligned}
A' &= A + \nabla\chi(r) \\
\chi &= -\frac{1}{2}BXY. \\
\nabla\chi &= \frac{\partial\chi}{\partial x}\hat{i} + \frac{\partial\chi}{\partial y}\hat{j} + \frac{\partial\chi}{\partial z}\hat{k} \\
&= \left(\frac{-BY}{2}\hat{i}\right) + \left(\frac{-BX}{2}\hat{j}\right) + 0 \\
A' &= \left(\frac{-BY}{2}, \frac{+XB}{2}, 0\right) + \left(\frac{-BY}{2}, \frac{-XB}{2}, 0\right) \\
A' &= (-BY, 0, 0)
\end{aligned}$$

$\begin{cases} \text{no physical quantity should change} \\ |\psi|^2, H, \text{ if we work in } A' = A + \nabla\chi. \end{cases}$
 (plot A')



let's look at what happens to H .

$$H = \frac{(p - eA)^2}{2m} \quad A = (-BY, 0, 0)$$

$$\begin{cases} \text{is } \nabla \cdot \vec{A} = 0 \\ \vec{\nabla} \cdot \vec{A} = \frac{\partial}{\partial x}A_x + \frac{\partial}{\partial y}A_y + \frac{\partial}{\partial z}A_z \\ = 0 + 0 + 0 = 0 \\ [p, A] = 0 \end{cases}$$

$$\begin{aligned}
p^2 &= p_x^2 + p_y^2 + p_z^2 \\
H &= \frac{(p - eA)(p - eA)}{2m} \\
&= \frac{p^2 - 2e\hat{A} \cdot \vec{p} + e^2 A^2}{2m} \\
&= \frac{(p_x^2 + p_y^2 + p_z^2) - 2e A_x p_x + e^2 A_x^2}{2m}
\end{aligned}$$

$$H = \frac{(p_x + eYB)^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m} \quad \leftarrow (A_x = -BY)$$

$$\begin{cases} p^2 = (\vec{p}_x + \vec{p}_y + \vec{p}_z) \cdot (\vec{p}_x + \vec{p}_y + \vec{p}_z) \\ \quad = p_x^2 + p_y^2 + p_z^2 + 2p_x \cdot p_y + 2p_y p_z + 2p_z p_x \\ p^2 = p_x^2 + p_y^2 + p_z^2 \\ \quad \because (p_i p_j = \delta_{ij}) \end{cases}$$

Now we can solve for schrödinger eqⁿ in position basis by writing $p_x = -i\hbar \frac{\partial}{\partial x}$ & so on but there is a easier way

▷ (beginning of digression)

(let's ask what happens to velocity of this particle?-)

$$p = m\vec{v} + e\vec{A} \quad \left(m\vec{v} = \vec{p} - e\vec{A} = \vec{\pi} \right)$$

$$\vec{v} = \frac{\vec{p} - e\vec{A}}{m}$$

what is

$$\begin{aligned} [v_x v_y] &= \frac{1}{m^2} [p_x - eA_x, p_y - eA_y] \\ &= \frac{1}{m^2} \left(\underbrace{[p_x p_y]}_0 - e[p_x A_y] - e[A_x p_y] + e^2 \underbrace{[A_x A_y]}_0 \right) \\ &= \frac{1}{m^2} (e[A_y p_x] - e[A_x p_y]) \end{aligned}$$

$$\{[f(x), p_x] = i\hbar f'(x)$$

$$\begin{aligned} [v_x v_y] &= \frac{e}{m^2} \left(i\hbar \frac{\partial A_y}{\partial x} - i\hbar \frac{\partial A_x}{\partial y} \right) \\ &= \frac{i\hbar e}{m^2} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \\ [v_x v_y] &= \frac{i\hbar e}{m^2} B_z \end{aligned}$$

$$\boxed{[v_i v_j] = \frac{i\hbar e}{m^2} \epsilon_{ijk} B_k} \quad [\pi_i \pi_j] = i\hbar e \epsilon_{ijk} B_k$$

$$\left[\begin{array}{ccc} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{array} \right]$$

$$[v_i v_j] \neq 0$$

→ we can't measure two cartesian component of velocity simultaneously with arbitrary precision for a charged particle in const. $\vec{B}$.

◁ (end of digression)

For any const B . and choosing $A(A_x, A_y, 0)$. ($A_z = 0$)

The Hamiltonian can be written as.

$$H = \frac{\pi_x^2 + \pi_y^2}{2m} + \frac{p_z^2}{2m}$$

{it is kind of Harmonic oscillator form of H . $\frac{1}{2}(x^2 + p^2)$ }

To find eigen values of H , we need to diagonalize it, so just like we did in SHO,

$$a = x + ip$$

$$a^\dagger = x - ip$$

[a is written as $\frac{x+ip}{\sqrt{2}}$ with the $\sqrt{2}$ struck out]

so that $a a^\dagger$ is diagonalizable.

Cyclotron frequency and Landau levels

similarly

$$a = \frac{\pi_x + i\pi_y}{\sqrt{2eB\hbar}}$$

so,

$$a^\dagger = \frac{\pi_x - i\pi_y}{\sqrt{2eB\hbar}}$$

{ p, A are hermitian so π is also $= \pi^\dagger$ }

$$[\pi_x, \pi_y] = i\hbar e B_k \quad (B)$$

$$[\pi^2] = [\hbar e B] ; \quad [\pi] = \boxed{\sqrt{\hbar e B}}$$

so,

$$\begin{aligned} a^\dagger a &= \frac{1}{(2eB\hbar)} \{ \pi_x^2 + \pi_y^2 + i(\pi_x \pi_y - \pi_y \pi_x) \} \\ &\quad \frac{1}{(2eB\hbar)} \{ \pi_x^2 + \pi_y^2 + i^2 \hbar e B \} \\ \left(a^\dagger a + \frac{1}{2} \right) &= \frac{\pi_x^2 + \pi_y^2}{(2eB\hbar)} \quad \downarrow \text{(Landau levels)} \end{aligned}$$

using value of $\pi_x^2 + \pi_y^2$ in H .

$$H = \frac{(2eB\hbar)}{2m} \left(a^\dagger a + \frac{1}{2} \right) + \frac{p_z^2}{2m}$$

$$H = \left(\frac{eB\hbar}{m} \right) \left(a^\dagger a + \frac{1}{2} \right) + \frac{p_z^2}{2m}$$

So we now know the eigen values of $a^\dagger a$ so we know eigen values of Half of H .. We have reduced the problem into Harmonic oscillator plus a free particle moving in z -direction.

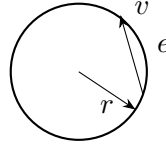
And these levels are quantized.

▷

$$\left(\frac{eB}{m} = \text{cyclotron frequency} \right) \quad (W_c)$$

$$qVB = \frac{mv^2}{r}$$

$$\left(\frac{1}{T} = \frac{v}{2\pi r} \right) = \frac{eB}{m}$$



$$L = mr^2 \cdot \omega = \frac{mr^2 \cdot 2\pi}{T}$$

$$\mu = iA = \frac{e}{T} \cdot \pi r^2 = \frac{er^2\omega}{2} = \frac{e}{2} \left(\frac{L}{m} \right) = \frac{eL}{2m}$$

$$\left(\frac{\mu}{L} = \frac{e}{2m} = \text{gyromagnetic ratio} \right)$$

◁

$$H = \hbar W_c \left(a^\dagger + \frac{I}{2} \right) + \frac{p_z^2}{2m} \quad \left(W_c = \frac{eB}{m} \right)$$

$$E(n, k_z) = \hbar W_c \left(n + \frac{1}{2} \right) + \frac{\hbar^2 k_z^2}{2m} \quad [a^\dagger a] = 1(I)$$

↓ (Landau levels)

”So if we confine the particle in XY plane then levels would look quantized like those of simple Harmonic oscillator, but if not there is free motion in z direction it drifts in $\hat{z}$ direction with any real no. k_z .

These levels look non degenerate but actually are degenerate. (more the one eigen state for a single eigen value).

These landau levels have acquired enormous importance, because now it is possible to take e^- and confine them essentially to two – dimensions. (MOSFET divices) $\{e^-$ gas in 2-D behaves very different than 3D).

overview — || prev — next ||

[Editorial note : the explicit "Lec 18" heading in the notebook stands at the top of scan qm7-04. The three pages before it (qm7-01 .. qm7-03) carry no heading and simply finish off the Landau-level / charged-particle-in-a-magnetic-field calculation that was already running on the last page of the previous notebook (qm6-24, which breaks off mid-sentence with "So, to write down wave function explicitly, we must choose the vector potential explicitly. The wave vector potential ($\vec{A}(r)$)). Those three pages are transcribed here, first, as a carry-over section so that they are not lost; the "Lec 18" material proper begins at the heading "Angular momentum states from two oscillators" below.]

17 Lecture 18

Charged particle in a magnetic field – contd.

should not present in wave functions, because final wave fun. energy eigen values should be independent of any gauge you choose.

So let's choose $\vec{A}(\vec{r}) = (-By, 0, 0)$

$$H = \frac{(\vec{p} - e\vec{A})^2}{2m}$$

$$[A = (-By, 0, 0)]$$

$$[p = (p_x, p_y, p_z)]$$

$$[p^2 = p_x^2 + p_y^2 + p_z^2]$$

$$\begin{aligned} H &= \frac{p^2 - e\vec{p} \cdot \vec{A} - e\vec{A} \cdot \vec{p} + e^2 A^2}{2m} \\ H &= \frac{p_x^2 + e^2 A^2 - 2e\vec{A} \cdot \vec{p}}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m} \\ H &= \frac{(p_x + eBy)^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m} \\ &= \frac{p_x^2 + p_y^2 + p_z^2}{2m} + \frac{eBy}{m} p_x + \frac{e^2 B^2 y^2}{2m} \end{aligned}$$

Therefore

$$\begin{aligned} H\phi(\vec{r}) &= E\phi(\vec{r}) \\ p_x &= -i\hbar \frac{\partial}{\partial x} \\ p_y &= -i\hbar \frac{\partial}{\partial y} \\ p_z &= -i\hbar \frac{\partial}{\partial z} \\ -\frac{\hbar^2}{2m} \left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \right) + \frac{eBy}{m} \left(-i\hbar \frac{\partial \phi}{\partial x} \right) + \frac{e^2 B^2 y^2}{2m} \phi(r) &= E\phi(r) \\ \left(\frac{-\hbar^2}{2m} (\nabla^2) + \frac{ie\hbar B}{m} y \frac{\partial}{\partial x} + \frac{e^2 B^2}{2m} y^2 \right) \phi(r) &= E\phi(\vec{r}) \end{aligned}$$

let the trial guess be

$$\phi(\vec{r}) = e^{ik_x x} e^{k_z z} \cdot f(y)$$

(non-trivial to guess)

$\hookrightarrow$ as we have derivatives for eigen value.

$$\frac{d(e^{ax})}{dx} = a e^{ax}$$

so we assumed it would be a fun. of exponentials.

the guess $\phi(\vec{r})$ should be a solution of above shrödinger equation.

$$\begin{aligned} \left(\frac{\hbar^2 k_z^2}{2m} + \left\{ \frac{\hbar^2}{2m} k_x^2 - \frac{\hbar^2}{2m} \frac{d^2 f}{dy^2} + \frac{eB\hbar}{m} k_x y + \frac{e^2 B^2}{2m} y^2 \right\} \right) e^{ik_x x} \left(e^{ik_z z} \cdot f \right) &= E e^{ik_x x} e^{ik_z z} \cdot f \\ \Rightarrow \left(E - \frac{\hbar^2 k_z^2}{2m} \right) f &= \left(\frac{\hbar^2 k_x^2}{2m} - \frac{\hbar^2}{2m} \frac{d^2 f}{dy^2} + \frac{eB\hbar}{m} k_x y + \frac{e^2 B^2}{2m} y^2 \right) \\ - \frac{\hbar^2}{2m} \frac{d^2 f}{dy^2} + \left(\frac{e^2 B^2}{2m} y^2 + \frac{\hbar e B}{m} y + \frac{\hbar^2 k_x^2}{2m} \right) f &= \left(E - \frac{\hbar^2 k_z^2}{2m} \right) f \end{aligned}$$

[the bracket on the right is called $\rightarrow \varepsilon$]

$$\begin{aligned} \frac{-\hbar^2}{2m} \frac{d^2 f}{dy'^2} + \frac{1}{2} m \omega_c^2 y'^2 f &= \varepsilon f(y') \\ y' &= (y + \dots k_x) \end{aligned}$$

$\downarrow$

eqⁿ of simple harmonic oscillator in coordinate 'y'.

$$\phi(\vec{r}) = A_n e^{ik_x x} e^{ik_z z} \cdot e^{-\frac{y'^2 m \omega}{2\hbar}} \cdot H_n \left(y \sqrt{\frac{m \omega}{2\hbar}} \right)$$

$\hookrightarrow$ solⁿ from SHO.

$$\begin{aligned} E(k_x, k_z) &= \hbar \omega_c \left(n + \frac{1}{2} \right) + \frac{k_z^2 \hbar^2}{2m} \\ H_n \left(y \sqrt{\frac{m \omega}{2\hbar}} \right) &= \text{hermite polynomials} \\ H_n(x) &= e^{-x^2} \left(\frac{-d}{dx} \right)^n e^{x^2} \end{aligned}$$

where is the centre of circle?

centre can appear anywhere in xy plane, so this infinite degeneracy appear through ' k'_x ' which does not appear in energy at all.

every level is infinitely degenerate, in the sense that for every different value of $k_x \exists$ a different eigen state for the same energy level.

as long as $L \gg \left(\sqrt{\frac{2\hbar}{m\omega}} = \lambda \text{ say} \right)$.

(size of box) $\hookrightarrow$ radius of cyclotron orbit

Then it is a good approximation and degeneracy $\propto \left(\frac{L}{\lambda}\right)^2$

when box $\rightarrow \infty$

degeneracy becomes infinite degeneracy.

If box is very small we have to put potential because of box. in H .

Angular momentum states from two oscillators

[Lec 18)

Recall we have discovered eigen values of J^2 and J are $\hbar^2 j(j+1)$ and $m\hbar$ respectively.

$[J^2 \doteq \hbar^2 j(j+1) \downarrow$ eigen values (and **not** the operator)]

$$J^2 \doteq \hbar^2 j(j+1) \quad j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$$

$$J_3 \doteq \hbar m \quad m = -j, -j-1, \dots, 0, j-2, j-1, j = \{(2j+1) \text{ values}\}$$

where

$$j = \frac{(n_a + n_b)}{2}, \quad m = \frac{n_a - n_b}{2} \quad \Rightarrow \quad \begin{cases} n_a = j + m \\ n_b = j - m \end{cases}$$

$$m = \frac{0-4}{2}, \frac{1-3}{2}, \frac{2-2}{2}, \frac{3-1}{2}, \frac{4-0}{2}$$

$$m = -2, -1, 0, 1, 2$$

[for given $j = 2$; $\frac{N_a + N_b}{2} = 2$; $N_a + N_b = 4$. $N_a + N_b = 4$ can be obtained from $(0, 4)$ $(4, 0)$ $(1, 3)$ $(3, 1)$ $(2, 2)$ which produce $m = -j, -j+1, \dots, 0, j-1, j]$

Now we have two oscillators, so the eigen state depends on both n_a & n_b .

$|n_a, n_b\rangle$ = Abstract state vector corresponding to $(\hat{n}_a)$ and $(\hat{n}_b)$ operators.

The states we are interested in are angular momentum states labelled by j, m .

$$|j, m\rangle \quad \text{or} \quad |n_a, n_b\rangle \quad \text{or} \quad |n_a = j + m, n_b = j - m\rangle$$

such that

$$J^2 |j, m\rangle = \hbar^2 j(j+1) |j, m\rangle$$

and

$$J_3 |j, m\rangle = \hbar m |j, m\rangle$$

$$|j, m\rangle = |n_a = j + m, n_b = j - m\rangle$$

{Abstract state in basis formed from $|j\rangle \otimes |m\rangle$ } ; {(abstract state in) basis formed from $|n_a\rangle \otimes |n_b\rangle$ }

Direct product of Hilbert spaces

$$|n_a, n_b\rangle = \frac{(a^\dagger)^{n_a}}{\sqrt{n_a!}} \frac{(b^\dagger)^{n_b}}{\sqrt{n_b!}} |n_a = 0, n_b = 0\rangle$$

{Recall that $|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}} |0\rangle$ }

▷ "what do we mean by two ket vectors $|n_a, n_b\rangle$?"

$|n_a, n_b\rangle$ is direct product of these two hilbert spaces.

The idea is following :-

$$\mathcal{H}_a : \{|n_a\rangle\}$$

$$\mathcal{H}_b : \{|n_b\rangle\}$$

$\exists$ for oscillators a , a hilbert space $\mathcal{H}_a$ which has states/vectors labelled by $\{|n_a\rangle\}$.

similarly for oscillator b , $\exists$ a hilbert space $\mathcal{H}_b$ defined by states $\{|n_b\rangle\}$

operator/Oscillator space basis set

$$a^\dagger, a \xrightarrow{\text{acts upon}} \mathcal{H}_a \quad \{|n_a\rangle\}$$

$$b^\dagger, b \xrightarrow{\text{acts upon}} \mathcal{H}_b \quad \{|n_b\rangle\}$$

(direct sum of Hilbert space)

$$\mathcal{H}_a \oplus \mathcal{H}_b = \mathcal{H}$$

$\hookrightarrow$ This space is span by following states/basis:

$$\{|n_a\rangle \otimes |n_b\rangle\}$$

or $\{|n_a, n_b\rangle\}$ short hand notation

So,

$$|n_a, n_b\rangle = \frac{(a^\dagger)^{n_a}}{\sqrt{n_a!}} |n_a = 0\rangle \otimes \frac{(b^\dagger)^{n_b}}{\sqrt{n_b!}} |n_b = 0\rangle$$

$\hookrightarrow$ direct product / Tensor product / cartesian product

It is Just like plane is direct product of $R \otimes R = R^2$;

$$x \in R$$

$$y \in R$$

$$(x, y) \in R^2 = R \otimes R$$

Note ; If $\mathcal{H}_a$ has 10 states, $\mathcal{H}_b$ has 15 states then $\mathcal{H} \{|n_a\rangle \otimes |n_b\rangle\}$ has $10 \times 15 = 150$ states.

let us find $|j, m\rangle$?

$$|n_a, n_b\rangle = |j, m\rangle = \frac{(a^\dagger)^{j+m} (b^\dagger)^{j-m}}{\sqrt{(j+m)! (j-m)!}} |n_a = 0, n_b = 0\rangle$$

$$\left\{ |j, m\rangle = \frac{(a^\dagger)^{j+m} (b^\dagger)^{j-m}}{\sqrt{(j+m)! (j-m)!}} \left| \frac{0+0}{2}, \frac{0-0}{2} \right\rangle \right\}$$

If $|0, 0\rangle$ is normalized to unity, it is guarantee that $|j, m\rangle$ is normalized to unity because of $\sqrt{(j+m)!(j-m)!}$

So, in space of angular momentum states $|j, m\rangle$ The orthonormality condition becomes

$$\langle j, m | j', m' \rangle = \delta_{jj'} \delta_{mm'}.$$

for a system whose total angular quantum no (j) is specified, then the angular momentum states are $(2j + 1)$ in total. So the direct product space will be $(2j + 1)$ dimensional.

$\hookrightarrow$ so dimension of operators (angular momentum) will be $(2j + 1) \times (2j + 1)$.

So we can represent them in form of finite-dimensional matrices

once it is finite dimensional matrix, we know how to find basis/natural basis.

For a given 'j'

let's assume $|j, m\rangle$ is angular momentum state in which j is specified.

what is $J_+ |j, m\rangle = ? \longrightarrow$ state in $|j\rangle$ & $|m\rangle$ basis

$$J_+ = J_1 + iJ_2 = (a^\dagger b)\hbar$$

$\{J_+ = (a^\dagger b)\hbar$ in harmonic oscillator basis $\{|n_a\rangle\} \{|n_b\rangle\}$

$$J_+ |j, m\rangle = \hbar(a^\dagger b) \left| \frac{n_a + n_b}{2}, \frac{n_a - n_b}{2} \right\rangle$$

$|j, m\rangle$ & $|n_a, n_b\rangle$ Both represents angular momentum states, let's first find $J_+ |n_a, n_b\rangle = ?$ in harmonic oscillator basis spanned by $\{|n_a\rangle \otimes |n_b\rangle\}$

first let us work in $|n_a, n_b\rangle$ basis.

$$\begin{aligned} J_+ |j, m\rangle &= \hbar(a^\dagger b) |n_a, n_b\rangle = a^\dagger b |j + m, j - m\rangle \\ J_+ |n_a, n_b\rangle &= \hbar a^\dagger |j + m\rangle \otimes b |j - m\rangle \\ &= \hbar \sqrt{j + m + 1} |j + m + 1\rangle \otimes \sqrt{j - m} |j - m - 1\rangle \end{aligned}$$

or

$$J_+ |n_a, n_b\rangle = \hbar \sqrt{(j + m + 1)(j - m)} \left| \underbrace{j + m + 1}_{n_a}, \underbrace{j - m - 1}_{n_b} \right\rangle$$

In $|j, m\rangle$ basis

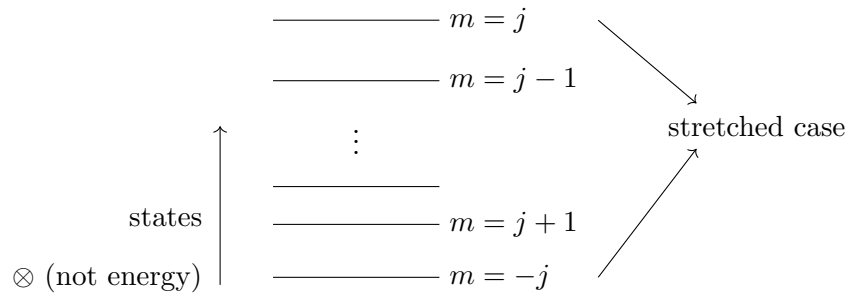
$$\begin{aligned} J_+ |j, m\rangle &= \hbar \sqrt{(j + m + 1)(j - m)} \left| \frac{(j + m + 1) + (j - m - 1)}{2}, \frac{j + m + 1 - j + m + 1}{2} \right\rangle \\ J_+ |j, m\rangle &= \hbar \sqrt{(j + m + 1)(j - m)} |j, m + 1\rangle \end{aligned}$$

{for $m = j \Rightarrow J_+ |j, j\rangle = 0$; $J_- |j, -j\rangle = 0$

similarly,

$$J_- |j, m\rangle = \hbar \sqrt{(j + m)(j - m + 1)} |j, m - 1\rangle$$

For a given j , there are $2j + 1$ eigen states of angular momentum.



[✓ states ⊗ (not energy)]

So J_+ operators increase the eigen value of J_3 by 1.

$J_{\pm}$; for example $j = 1$

$$m = -1, 0, 1$$

total angular momentum states are 3.

$$|1, -1\rangle, |1, 0\rangle, |1, 1\rangle$$

$$J_+|1, -1\rangle = \hbar\sqrt{2}|1, 0\rangle$$

$$J_+|1, 0\rangle = \hbar\sqrt{2}|1, 1\rangle$$

But since, there are no states above $|1, 1\rangle$

$$J_+|1, 1\rangle = 0 \quad \text{or} \quad \boxed{J_+|j, j\rangle = 0}$$

similarly there are no states below $m = -j$

$$\boxed{J_-|j, -j\rangle = 0}$$

Therefore in this space of $\{|j, m\rangle\}$ we can represent J_+ operator as,

$$\boxed{(J_{\pm})^{2j+1} = 0}$$

$\hookrightarrow$ nil potent matrix. $M^n = 0$.

Components of 'J' transform like a vector under rotation :-

J_+ & J_- are not hermitian, but are hermitian conjugate of each other.

(watch from 25:00 to 45:00)

▷ We know that J_1, J_2, J_3 are operators which can be expressed as matrices (for fixed j , $(2j+1) \times (2j+1)$ dimensional matrices), but here we call J_1, J_2, J_3 act as if they are vectors. (calling collection of 3 operators as component of a vector J). (Though ' J ' is also an operator).

How that is justified.?

" J_1, J_2, J_3 transform like component of a vector under physical rotations of coordinate system."

Rotation transformation of states and operators :- (Rotation representation)

If we have a transformation under which a system is invariant, then physical quantities don't change under this transformation.

It implies $\langle\psi|A|\psi\rangle$ must remain invariant under a transformation. Suppose transformation acts on states themselves

so, $|\Psi\rangle \longrightarrow U|\psi\rangle$; U = unitary operator

$$\langle\psi|A|\psi\rangle \longrightarrow \langle\psi|U^\dagger AU|\psi\rangle$$

we can see use heisenberg picture suggesting

$$\left\{ A \longrightarrow U^\dagger AU \right.$$

[U' depends on object it acts upon it.]

Since ' U ' is unitary $U^\dagger = U^{-1}$

$$\boxed{A \longrightarrow U^{-1}AU} \quad \Rightarrow \quad \text{rotation(transformation).}$$

[$|\Psi\rangle$ depends on coordinate system. $|\Psi\rangle$ changes to $|\psi\rangle$ for change of coordinate system and under rotation $|\Psi'\rangle = U|\Psi\rangle$; U = unitary operator representing rotation Transformation]

Just like a vector under rotation changes transforms from $v \rightarrow v'$ under rotation about z axis.

$$v'_1 = v_1 \cos \alpha + v_2 \sin \alpha$$

$$v'_2 = -v_1 \sin \alpha + v_2 \cos \alpha$$

$$v'_3 = v_3$$

$\{v_1, v_2, v_3 \text{ can be matrices also}\}$ so operators component can be transform like a vector.

Here v' is a vector which can be written as linear combinations of v_1 and v_2 .

Note :-

$$|\psi(t)\rangle = e^{-\frac{iHt}{\hbar}}|\psi(0)\rangle$$

$\leftarrow$ state vectors transform like -this and operators transform like $\longrightarrow (e^{iHt/\hbar} A(0) e^{-iHt/\hbar})$ under rotation

Similarly

If $[A, B] \neq 0$

$$\left\{ e^{\lambda A} B e^{-\lambda A} = B + \lambda[A, B] + \frac{\lambda^2}{2!}[A, [A, B]] + \frac{\lambda^3}{3!}[A, [A, [A, B]]] \right\} + \dots$$

Transformation of operators :-

$$\begin{aligned} 1) \quad e^{i\gamma J_3} J_1 e^{-i\gamma J_3} &= J'_1 = J_1 + \lambda i\hbar J_2 + \frac{\lambda^2}{2!}(-J_1 i\hbar) + \frac{\lambda^3}{3!}(-i\hbar J_2) \\ &= J_1 \cos \gamma + J_2 \sin \gamma \end{aligned}$$

$$\begin{aligned} 2) \quad e^{i\gamma J_3} J_2 e^{-i\gamma J_3} &= J'_2 = -J_1 \sin \gamma + J_2 \cos \gamma \\ 3) \quad e^{i\gamma J_3} J_3 e^{-i\gamma J_3} &= J_3 = J'_3 = J_3 \end{aligned}$$

linear combination of J_1 & J_2

The abstract representation of rotation of operators about an axis $\hat{n}$ through an angle ' θ ' is

$$R(\theta, \hat{n}) = e^{i\vec{J} \cdot \hat{n} \theta}$$

unitary operator ; $\vec{J}$ = hermitian operator

So J_1, J_2, J_3 transforms like a component of a vector.

Representation of $R(\theta, \hat{n})$ depends on the state it acts upon.

Note :- orbital angular momentum $\vec{L}$ can only take integer eigen values $(0, 1, 2, \dots)$; this result arrived from the requirement of wave function to be single-valued.

Note :- the representation of physical transformation (translation, parity, rotation etc of coordinate system) would be represented by an abstract operator ; the actual representation of this unitary operator would depend upon the object it acts upon.

$[R(\theta, \hat{n}) = e^{i\vec{J} \cdot \hat{n} \theta}$; we will do several representations of rotation group, one of them for angular momentum operator acting on physical wavefunctions $\psi(R, \theta, \phi)]$

j = 1/2 (very important case)

How does angular momentum operators look like for $j = \frac{1}{2}$;

[The experiments suggested that the value of spin (intrinsic) angular momentum corresponds to $j = \frac{1}{2}$; $\{|\frac{1}{2}, m\rangle\}$ = set of eigen states of spin angular mom. op^r]

we are looking at case where

$$J^2 \doteq \hbar^2 \frac{1}{2} \left(\frac{1}{2} + 1 \right)$$

are eigen values of J^2 operator.

It turns out that particles like e^- , p , n have a property called intrinsic angular momentum, even in the rest frame of the particle there is an angular momentum.

For point particles like e^- , you still have associated an intrinsic angular momentum, even in the frame where linear momentum of e^- is zero. It is like charge / mass of e^- . This property is intrinsic property.

and experiments have shown us that $j = \frac{1}{2}$ for e^- spin.

$$j = \frac{1}{2} \quad \Rightarrow \quad (2j + 1 = 2)$$

So every operator can be represented with $(2j + 1) \times (2j + 1)$ i.e. 2×2 matrices.

$$\begin{aligned} J^2 &\doteq \hbar^2 \frac{1}{2} \left(\frac{1}{2} + 1 \right) \\ J_3 &\doteq \pm \frac{\hbar}{2} \quad m = -\frac{1}{2}, \frac{1}{2} \end{aligned}$$

"we shouldn't picture spin as spinning e^- , or proton, first of all e^- is elementary particle, it does not have structure. P^+ do have structure, but however the origin of spin is in relativistic Q.M. It's not a non relativistic concept nor and not a classical concept at all"

It turns out, every elementary particle have intrinsic property called spin. (just like rest mass). so ($\vec{L} \neq \vec{r} \times \vec{p}$)

This property is dictated by requirement of lorentz invariance. (Invariance of laws of physics under lorentz transformations). No mechanical model is possible to represent spin angular momentum.

Note :- The photons also have spin, but since, photon has zero rest mass, the idea of spin (the origin and meaning of spin) for zero rest mass particles is slightly different from what it is for non-zero rest mass particles.

spin quantum no: (s) or (j)

$$\begin{aligned} s &= \frac{1}{2} \quad \text{for } e^-, p, n, \nu_e \\ &= 1 \quad \text{for } \gamma, W^\pm, Z^0 \\ &= 0 \quad \text{for } \pi^0, \pi^\pm, K^0 \end{aligned}$$

[these are elementary particles]

So these quantum particles have intrinsic angular momentum even in rest frame.

since J^2 commutes with J_1, J_2, J_3

$$J^2 = \hbar^2 \left(\frac{3}{4}\right) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$\rightarrow$ (schur's lemma theorem) $\leftrightarrow$ unit matrix

[even when $\vec{p} = 0$ so it can not be of form $\vec{r} \times \vec{p}$ but this operator transforms like an vector / angular momentum]

States :-

$$\begin{aligned} \{|j, m\rangle\} &= \left\{ \left| \frac{1}{2}, m \right\rangle \right\} \\ &= \left\{ \left| \frac{1}{2}, \frac{1}{2} \right\rangle \text{ \& } \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \right\} \\ &\text{or } \{ |\uparrow\rangle, |\downarrow\rangle \} \end{aligned}$$

$|\uparrow\rangle$ spin up ; $|\downarrow\rangle$ spin down

$$J_3 \doteq \pm \hbar/2 \quad (\text{eigen values of } J_3)$$

$$J_3 |\uparrow\rangle = \frac{\hbar}{2} |\uparrow\rangle \quad \text{or} \quad J_3 \left| \frac{1}{2}, \frac{1}{2} \right\rangle = \frac{\hbar}{2} \left| \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$\{|1\rangle = |\uparrow\rangle ; |2\rangle = |\downarrow\rangle \text{ [?]}\}$$

$\leftrightarrow$ eigen value

$$J_3 |\downarrow\rangle = -\frac{\hbar}{2} |\downarrow\rangle$$

$$\text{or } J_3 \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = -\frac{\hbar}{2} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\{\langle \uparrow | \uparrow \rangle = 1 = \langle \downarrow | \downarrow \rangle ; \langle \uparrow | \downarrow \rangle = 0$$

$$\boxed{\hat{J}_3 |j, m\rangle = m\hbar |j, m\rangle}$$

→ meaning of eigen states

$$| \uparrow \rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad | \downarrow \rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

representation in natural basis.

$$J_{\ell m} = \langle \ell | J_3 | m \rangle$$

$$\begin{Bmatrix} \langle 1 | J_3 | 1 \rangle & \langle 1 | J_3 | 2 \rangle \\ \langle 2 | J_3 | 1 \rangle & \langle 2 | J_3 | 2 \rangle \end{Bmatrix}$$

so J_3 should be : $\propto \sigma_3$ (pauli matrix)

$$J_3 = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

[| $\uparrow$ & | $\downarrow$ > states implies magnetic moment which either points along the direction of ext. $\vec{B}$ or opposite to ext $\vec{B}$.]

such that

$$J_3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$J_3 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{\hbar}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

schur's lemma theorem :-

(If we fix ' j ' and work in ' $2j + 1$ ' space)

Any operator in a subspace $(2j + 1)$ dimensionality which commutes with all the operators in subspace, must be some multiple of unit matrix.

$$[J^2, \vec{J} \cdot \hat{n}] = 0$$

$$J^2 = (\text{no.}) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

J_1, J_2, J_3 forms lie algebra of rotation group ; & J^2 commutes with all J_1, J_2, J_3

Any operator in lie algebra which commutes with its generators of Lie algebra is called Casimir operator. So, J^2 is casimir operator.

no. of casimir operators is called the rank of the algebra.

In this case it is one. (which is J^2 only). there is no other operator which commutes with its generators (J_1, J_2, J_3) .

[$J^2 = J_1^2 + J_2^2 + J_3^2$ is only one casimir operator]

(Q) *Can be J_1 and J_2 are also diagonal matrices ?* (just like J_3)

No, if they were diagonal matrix they could have commute with each other. so J_1 & J_2 can't be diagonal.

$$[J_1, J_2] \neq 0 \neq [J_2, J_3] \neq [J_1, J_3] \neq 0$$

of course we could have chose different basis such that any one of J_1, J_2, J_3 is diagonal matrix and rest is not.

18 Lecture 19

Spin operator for an electron

[Lec 19)

The spin operator for an e^- ($\vec{S}$) has three components S_x, S_y, S_z is represented by

$$\vec{S}_i = \frac{\hbar}{2} \vec{\sigma}_i$$

with spin quantum no ($s = \frac{1}{2}$)

which obey angular momentum algebra

for e^- , $s = \frac{1}{2} \Rightarrow j = \frac{1}{2}$

j or s

$$\begin{cases} |\frac{1}{2}, \frac{1}{2}\rangle = |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ |\frac{1}{2}, -\frac{1}{2}\rangle = |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{cases}$$

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \sigma_i^2 = I$$

[Notice only one operator is diagonal matrix]

↓ (representation in natural basis)

$$\langle \uparrow | \uparrow \rangle = \langle \downarrow | \downarrow \rangle = 1$$

$$\langle \uparrow | \downarrow \rangle = \langle \downarrow | \uparrow \rangle = 0$$

} orthonormality.

" $\vec{S}$ is simply J for $j = \frac{1}{2}$ (e^- , p , n , ν_e)

$$\vec{J}_{j=1/2} = \vec{S}."$$

How do we detect / find the spin angular momentum ?

"If we took case of a stone $m = 1$ kg performing circular motion with radius $r = 1$ m, has angular momentum

$$L = I\omega = mr^2\omega = \omega$$

but this ' L ' should be $\sim \hbar l$

$\hookrightarrow$ some angular momentum quantum no.

$$\hbar l \sim 1$$

(for stone)

$$(l \sim 10^{34}) \quad \text{so as } l \text{ is very enormous } (l = j).$$

The discreteness of angular momentum completely washed out. (as values of m varies from $-j$ to $+j$ in steps of 1.) That is why we see states as continuous rather than discrete. (-10^{34} , $-10^{34}+1$, $---$, $10^{34}-1$, 10^{34}).

(Q) So question arises how to detect such small $s = 1/2$

"How do we detect / find the spin angular momentum". ?

The spin angular momentum also implies magnetic moment for the e^- , (μ_e), and that μ_e (magnetic moment) would couple to applied magnetic fields, and then we can manipulate the spin of e^- using the magnetic field.

$$\vec{\mu}_e = \frac{g_e e}{2m_e} \vec{S}$$

[remember $\vec{S}$ is simply the J for $j = 1/2$ / $s = 1/2$]

$$\begin{aligned} &= \frac{-(2)|e|\hbar}{2m_e} \cdot \frac{1}{2} \vec{\sigma} \\ &= - \left(\frac{|e|\hbar}{2m_e} \right) \sigma \end{aligned}$$

$\longrightarrow$ Bohr magneton (μ_B)

$$\mu_B = \frac{|e|\hbar}{2m_e}$$

is natural unit of magnetic moment for charge $|e|$ and mass m_e .

$$\boxed{\vec{\mu}_e = -\mu_B \vec{\sigma}}$$

If we switch on $\vec{B}$. There will be extra term in $\hat{H}$ of charge particle

$$\left(\hat{H} = \frac{(p-eA)^2}{2m} + e\phi + \hat{H}_{\text{magnetic}} \right)$$

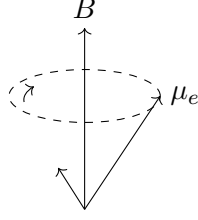
$$H_{\text{magnetic}} = -\vec{\mu}_e \cdot \vec{B}$$

$\hookrightarrow$ P.E of magnetic moment in applied $\vec{B}$.

$$H_{\text{magnetic}} = +\mu_B \vec{\sigma} \cdot \vec{B}$$

(our expectations :-)

Classically) If we have a magnetic dipole moment placed in an external magnetic field at non zero angle θ . then the dipole moment vector precesses around the $\vec{B}_{\text{ext}}$.



[”precession frequency is called larmor frequency. If the $\vec{B}$ is sufficiently non uniform / strong then we know that dipole moment also experience force along with torque. And such $\vec{B}_{\text{ext}}$ if strong can separate out magnetic dipole moment states. It is what done in Stern Gerlach exp. which is root of Quantum computing & quantum information”]

$$\left(\frac{dL}{dt} = \tau ; \frac{d\mu}{dt} = \kappa \tau \right)$$

(classical magnetic moment)

$$\frac{d\mu_e}{dt} = \kappa(\tau) = \kappa(\mu_e \times B)$$

$\hookrightarrow$ constant

$L \propto \mu_e$ so,

$$\frac{d\mu_e}{dt} = (\kappa) \mu_e \times B$$

$\hookrightarrow$ provides larmor frequency

Quantum mechanically :-

From Ernfest theorem, the expectation value of any operator obey classical equation. So

$$\frac{d\langle \mu \rangle}{dt} = () \cdot \langle \mu_e \rangle \times B$$

} should behave quantum mechanically

Spin states in a constant magnetic field

let us ask what happens to spin states in const $\vec{B}$. as time passes. :-

Since $|\uparrow\rangle$ and $|\downarrow\rangle$ form basis, we can write $|\Psi(t)\rangle$ as linear combination of those basis. ($|\Psi(t)\rangle$ = angular momentum state).

$$|\Psi(t)\rangle = a(t)|\uparrow\rangle + b(t)|\downarrow\rangle$$

$$\{\langle\uparrow|\downarrow\rangle = 0 ; \langle\uparrow|\uparrow\rangle = 1 ; \langle\downarrow|\downarrow\rangle = 1$$

$$|a(t)|^2 + |b(t)|^2 = 1 \quad \text{for all time}$$

let us look at simplest system.

at $t = 0$ we only have $|\uparrow\rangle$ state (or prepare $|\uparrow\rangle$ states)

means,

$$\begin{cases} a(0) = 1 \\ b(0) = 0 \end{cases}$$

$$|\psi(0)\rangle = |\uparrow\rangle$$

and what happens if we switch on the hamiltonian.

If $B = B \hat{e}_z$

$$(\sigma \cdot B = \sigma_1 B_x + \sigma_2 B_y + \sigma_3 B_z)$$

$$H_{\text{magnetic}} = \mu_B \sigma_3 B$$

$$(\vec{\sigma} \cdot \vec{B} = \sigma_3 \cdot B).$$

$$\begin{aligned} |\psi(t)\rangle &= e^{\frac{-iHt}{\hbar}} |\psi(0)\rangle \\ &= e^{\frac{-iB\mu_B\sigma_3 t}{\hbar}} |\psi(0)\rangle \\ &= e^{\frac{-iB\mu_B\sigma_3 t}{\hbar}} |\uparrow\rangle \\ &= e^{\frac{-iB\mu_B t}{\hbar}} |\uparrow\rangle \end{aligned}$$

$$[e^{-i\alpha\sigma_3} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = e^{-i\alpha} \begin{pmatrix} 1 \\ 0 \end{pmatrix}]$$

$$[\sigma_3 |\uparrow\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}]$$

$$[J_3 |S_3\rangle = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} (\text{sign})]$$

[so nothing happens upon applying ext. $\vec{B}$. $H|\Psi(0)\rangle = E|\Psi(0)\rangle]$

Even if we start from arbitrary state $|\downarrow\rangle$

$$|\psi(t)\rangle = e^{\frac{+iB\mu_B t}{\hbar}} |\downarrow\rangle$$

$$[e^{-i\alpha\sigma_3} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = e^{i\alpha} \begin{pmatrix} 0 \\ 1 \end{pmatrix} ; (\alpha = \frac{B\mu_B t}{\hbar})]$$

$[\downarrow\rangle$ remains in same state with eigen value $\frac{\hbar}{2} \left(\frac{iB\mu_B t}{\hbar} \right)$ [?]]

Interesting things occur if we start with arbitrary wave function

$$|\psi(0)\rangle = a(0)|\uparrow\rangle + b(0)|\downarrow\rangle$$

for $B = B \hat{e}_x$ let $B = b_x \hat{e}_x$ (in x direction only)

$$|\psi(t)\rangle = e^{-\frac{iHt}{\hbar}} |\psi(0)\rangle$$

$$H = \mu_B \vec{\sigma} \cdot \vec{B} = \mu_B (\sigma_1) \cdot B = +(B) \frac{|e|\hbar}{2m} \cdot \sigma_1$$

$$\begin{aligned} |\psi(t)\rangle &= e^{-\frac{i}{\hbar} \frac{\mu_B B |e|\hbar}{2m} \sigma_1 t} |\psi(0)\rangle \\ &= e^{-\frac{i}{2} \frac{|e|B}{m} \sigma_1 t} |\psi(0)\rangle \end{aligned}$$

$\hookrightarrow \omega_c$ (cyclotron frequency)

$$= e^{-\frac{i\omega_c t}{2} \sigma_1} |\psi(0)\rangle$$

let, if $|\psi(0)\rangle = |\uparrow\rangle \begin{pmatrix} b(0) = 0 \\ a(0) = 1 \end{pmatrix}$

So,

$$|\psi(t)\rangle = e^{-\frac{i\omega_c t}{2} \sigma_1} |\uparrow\rangle$$

here, $e^{-\frac{i\omega_c t}{2} \sigma_1}$ is not a phase factor as $|\uparrow\rangle$ is not eigen state of σ_1

also let's write $e^{-\frac{i\omega_c t}{2} \sigma_1}$ in matrix form.

$$[e^{\alpha x} = 1 + \frac{(\alpha x)}{1!} + \frac{(\alpha x)^2}{2!} + \dots]$$

$$e^{-i\vec{a} \cdot \vec{\sigma}} = (\cos a) I - i \left(\frac{\vec{a} \cdot \vec{\sigma}}{|\vec{a}|} \right) \sin(a)$$

$$\left\{ a = \frac{\omega_c t}{2} ; \frac{\vec{a} \cdot \vec{\sigma}}{|\vec{a}|} = \sigma_1 \right.$$

$\hookrightarrow$ generalization of $e^{i\theta} = \cos \theta + i \sin \theta$

$$\begin{aligned} |\Psi(t)\rangle &= \left[\left(\cos \frac{\omega_c t}{2} \right) I - i \sigma_1 \left(\sin \frac{\omega_c t}{2} \right) \right] |\uparrow\rangle \\ I &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} \cos \frac{\omega_c t}{2} & -i \sin \frac{\omega_c t}{2} \\ -i \sin \frac{\omega_c t}{2} & \cos \frac{\omega_c t}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \cos \frac{\omega_c t}{2} \\ -i \sin \frac{\omega_c t}{2} \end{pmatrix} = \cos \frac{\omega_c t}{2} |\uparrow\rangle + \left(-i \sin \frac{\omega_c t}{2} \right) |\downarrow\rangle \\ \text{or } &\left(\cos \frac{\omega_c t}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \left(-i \sin \frac{\omega_c t}{2} \right) \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\ &\begin{cases} a(t) = \cos \frac{\omega_c t}{2} \\ b(t) = -i \sin \frac{\omega_c t}{2} \end{cases} \quad (|a(t)|^2 + |b(t)|^2 = 1) \end{aligned}$$

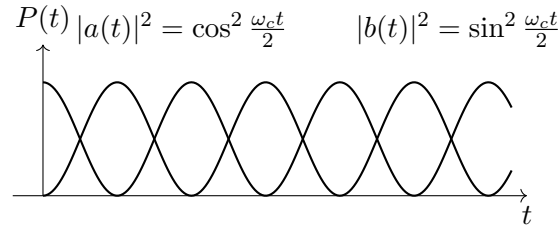
also, we can see that $|\Psi(t)\rangle$ remains normalized.

Q.) Now, what is significance of these coefficients ?

$|a(t)|^2$ = probability that at time 't' the state is $|\uparrow\rangle$

$|b(t)|^2$ = Probability that the state is down $|\downarrow\rangle$.

Such that $|a|^2 + |b|^2 = 1$



$$\{|b(t)|^2 = 1 ; \frac{\omega_c t}{2} = n\pi$$

So, quantum mechanically, this H ($B = B\hat{e}_x$) plays the role of spin flip operator. It flips the spin from up to down and down to up repeatedly.

$$|\uparrow\rangle \longrightarrow |\downarrow\rangle$$

$$|\downarrow\rangle \longrightarrow |\uparrow\rangle$$

so we can't say unlike classically, to that the system is in $|\uparrow\rangle$ state or $|\downarrow\rangle$ state. It exist in superposition of states. except those instances where $a(0) = 0$ or $b(0) = 0$.

for only $\left(t = \frac{2n\pi}{\omega_c}\right)$ system is in $|\downarrow\rangle$ state, at for $t = \frac{2\pi(n+\frac{1}{2})}{\omega_c}$ system is in $|\uparrow\rangle$ state).

suppose we have eigen state of σ_1 $|\rightarrow\rangle$

$$|\rightarrow\rangle = a|\uparrow\rangle + b|\downarrow\rangle$$

(thus any spin-state in this hilbert space can be written as superposition of $|\uparrow\rangle$ and $|\downarrow\rangle$ states.)

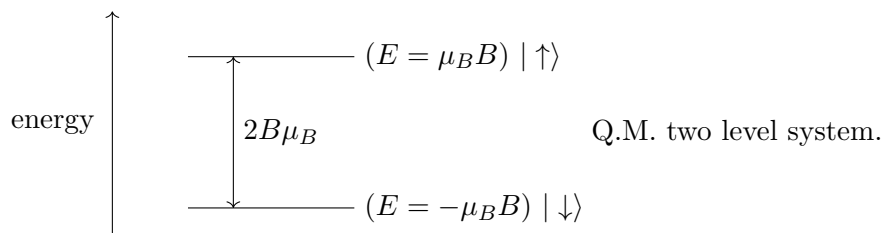
For the case $B = B\hat{e}_z$

$$H = \mu_B \vec{\sigma} \cdot \vec{B} = \mu_B \sigma_3 B \quad \Rightarrow \quad \begin{aligned} H|\uparrow\rangle &= E_1|\uparrow\rangle \\ H|\downarrow\rangle &= E_2|\downarrow\rangle \end{aligned}$$

the eigen states

since σ_3 has two eigen values $+1, -1$.

$$E_1 = \mu_B B \quad E_2 = -\mu_B B$$



"any state can be written as superposition, $\exists$ infinite no. of superpositions possible. ($a, b \in$ any number).

we start with two level system but we can have ∞ -no of possibilities, this is called qubit."

so it is superposition which leads to all the interesting phenomena.

Entanglement

Suppose we have such 2 e^- each with two level systems. we look at spin states of 2 electrons together.

[we are here not interested in coulomb interaction, other d.of, not interested in P.E, not interested in K.E due to motion]

we will look only at spin-states of system.

what is hilbert space looks like

$$\begin{aligned}\mathcal{H}_1 & \quad \{|\uparrow\rangle_1, |\downarrow\rangle_1\} \\ \mathcal{H}_2 & \quad \{|\uparrow\rangle_2, |\downarrow\rangle_2\}\end{aligned}$$

what are possible states of full / combined system of e^- s.

There are four possible states to start with.

$$|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle.$$

▷) How to add angular momenta in Q.M ?

let J_1 and J_2 be two total angular momentum operators for e_1^- and e_2^-

components of J_1 commutes with components of J_2

also, $[J_1, J_2] = 0$. i.e. $[J_{1i}, J_{2j}] = 0$

[for e_1 : $J_1 \rightarrow J_1^2, J_{11}, J_{12}, J_{13}$; e_2 : $J_2 \rightarrow J_2^2, J_{21}, J_{22}, J_{23}$]

Theory of addition of angular momenta :-

If

$$\vec{J}_1 + \vec{J}_2 = \vec{J}$$

→ Total angular momentum.

So let us ask

(Q) what is/are possible eigen values of J/J^2 given the eigen values of J_1/J_1^2 or J_2/J_2^2 .

Theory of addⁿ of angular momenta :-

$$J_{\text{e.value}}^2 = j(j+1)\hbar^2$$

$$J_{1\text{e.v.}} = j_1(j_1+1)\hbar^2$$

$$J_{2\text{e.v.}} = j_2(j_2+1)\hbar^2$$

Then the rule is, for total quantum no ' j ' is :-

$$j = |j_1 - j_2| \text{ to } (j_1 + j_2)$$

$\hookrightarrow$ quantum no of $\vec{J}$.

[J is an operator so it does not add up like vectors, operators have their own peculiarity.]

But, any component adds up linearly

for $(\vec{J}_1 + \vec{J}_2 = \vec{J})$

(example :-)

$$J_{13} = J_1^z = m_1 \hbar$$

$$J_{23} = J_2^z = m_2 \hbar$$

$$[m_1 = -j_1, -j_1 + 1, \dots, j_1 - 1, j_1 ; m_2 = -j_2, -j_2 + 1, \dots, j_2 - 1, j_2]$$

$$J^z = (m_1 + m_2) \hbar \quad m_1 + m_2 = -j, -j + 1, \dots, j - 1, j$$

$\rightarrow$ *what if we add spins of two e^- :-*

what happens if we solve for $j_1 = \frac{1}{2}$ & $j_2 = \frac{1}{2}$ ($j = \frac{1}{2}$ for both e^-)

so J is represented as $\vec{S}$.

$$\vec{S} = \vec{S}_1 + \vec{S}_2 = \vec{S}$$

possible values for total spin quantum no. $S = ?$)

$$\because S_1 = \frac{1}{2}, S_2 = \frac{1}{2}$$

$$s = |j_1 - j_2| \text{ to } j_1 + j_2 = 0 \text{ to } 1 \text{ in steps of } 1$$

$$(s = 0 \text{ or } 1)$$

' s' is total spin quantum no.

$$[s_1 = \frac{1}{2} ; m_1 = -j \text{ to } +j ; s_2 = \frac{1}{2} ; m_2 = -j \text{ to } j ; m_2 = -\frac{1}{2}, \frac{1}{2}]$$

$$[\text{so 's' runs from } |s_1 - s_2| \text{ to } (s_1 + s_2) \Rightarrow 0 \text{ to } 1 ; (|\frac{1}{2} - \frac{1}{2}| \text{ to } |\frac{1}{2} + \frac{1}{2}|)]$$

$\Rightarrow$ So, if we take two fermions we put them together, we now create a boson (integral spin quantum no (s) 0, 1, 2 ...).

"This is how super conductivity occurs"

Q). what are the possible states?

$$|\uparrow\rangle_1 \otimes |\uparrow\rangle_2 = |\uparrow\uparrow\rangle$$

$\rightarrow$ representation.

$$[\text{where } |\uparrow\rangle_1 = |\frac{1}{2}, \frac{1}{2}\rangle_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} ; |\downarrow\rangle_1 = |\frac{1}{2}, -\frac{1}{2}\rangle_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}]$$

So we have 4 - states.

$$|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle.$$

[rough work in a box on this page :

$$2 \sin^2 \theta \Rightarrow 1 - \cos 2\theta$$

$$1 - \cos(\theta + \theta)$$

$$1 - [\cos^2 \theta - \sin^2 \theta] = 1 - (1 - \sin^2 \theta - \sin^2 \theta) = 1 - 1 + 2 \sin^2 \theta$$

]

⇒ So they form a basis in full Hilbert space.

So every spin-state (of combined system of 2 e^-) can be written as superposition of these above mentioned four states.

$$S = 0 \quad \text{or} \quad 1$$

$$(m_1 + m_2) = S^z = 0 \quad \text{or} \quad -1, 0, 1$$

$$[|\Psi\rangle = a|\uparrow\uparrow\rangle + b|\uparrow\downarrow\rangle + c|\downarrow\uparrow\rangle + d|\downarrow\downarrow\rangle]$$

[a signature "Vishal." is boxed in the middle of this page]

$$S = 0, \quad S^z = 0 \quad \} \text{ singlet state}$$

($-s$ to $+s$ in steps of 1)

$$S = 1, \quad S^z = \begin{matrix} -1 \\ 0 \\ 1 \end{matrix} \quad \} \text{ triplet state}$$

($m_s = -s$ to $+s$ in steps of 1)

(let us write states) :-

$$|s, s_z\rangle \quad \text{or} \quad |j, m\rangle$$

spin singlet state

$$|0, 0\rangle = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$$

→ anti-symmetric under exchange of 2 e^- .

spin triplet states are symmetric under exchange

$$|1, 1\rangle = |\uparrow\uparrow\rangle \quad \text{symmetric}$$

$$|1, -1\rangle = |\downarrow\downarrow\rangle \quad \text{symmetric}$$

$$|1, 0\rangle = (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)/\sqrt{2} \quad (\text{normalized state})$$

} symmetric under the exchange of 2 electrons.

$$\langle s_1, s_1^z | s_2, s_2^z \rangle = \delta_{s_1 s_2} \delta_{s_1^z s_2^z}$$

} orthogonal states.

pauli exclusion principle :-

[this page is written over a printed sudoku puzzle; the block below is the highlighted / coloured annotation – colour ignored]

If we have two fermions which are absolutely identical then no matter what states these fermions are in, total state must be anti-symmetric under exchange.

But this is only spin part of wave function. There is also a special part, if e^- are moving around constitutes $\psi(r, t)$ then Total wave function $\Psi = \psi(r, t) * \psi(\text{spin})$, and that $|\Psi\rangle$ should be anti-symmetric.

So, in singlet state special part of wavefunction must be symmetric and in triplet states special part of state must be anti-symmetric.

$|1, 0\rangle$ should be symmetric under exchange so, it would be superposition of $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$ state.
all states should be orthogonal with each other so

$$|0, 0\rangle = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$$

after exchange

$$|0, 0\rangle = (|\downarrow\uparrow\rangle - |\uparrow\downarrow\rangle)/\sqrt{2}$$

anti-symmetric under exchange

overview — [⌂](#) [prev](#)